

Supplemental Appendix to “Measuring Uncertainty and its Impact on the Economy”

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May 2017

Abstract

This appendix provides additional details, on priors, the estimation algorithm, estimation results, and robustness.

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1 Priors and algorithm details

1.1 Priors

For the VAR coefficients contained in Π , we use a Minnesota-type prior. With the variables of interest transformed for stationarity, we set the prior mean of all the VAR coefficients to 0. We make the prior variance-covariance matrix $\underline{\Omega}_\Pi$ diagonal. The variances are specified to make the prior on the $\log m_t$ and $\log f_t$ terms uninformative and the prior on the lags of y_t take a Minnesota-type form. Specifically, for the $\log m_t$, and $\log f_t$ terms of equation i , the prior variance is $\theta_3^2 \sigma_i^2$. For lag l of variable j in equation i , the prior variance is $\frac{\theta_1^2}{l^2}$ for $i = j$ and $\frac{\theta_1^2 \theta_2^2}{l^2} \frac{\sigma_i^2}{\sigma_j^2}$ otherwise. In line with common settings, we set overall shrinkage $\theta_1 = 0.1$, cross-variable shrinkage $\theta_2 = 0.5$, and factor coefficient shrinkage $\theta_3 = 1000$. With these settings, we have deliberately made the prior on the uncertainty terms in the VAR uninformative (making the prior less uninformative by setting $\theta_3 = 10$ yielded results very similar to the baseline estimates). Finally, consistent with common settings, the scale parameters σ_i^2 take the values of residual variances from $\text{AR}(p)$ models from the estimation sample.

We now turn to describing the priors attached to the volatility-related components of the model. Since the estimation of common uncertainties is a key new feature of our model, we have checked sensitivity of our results to the priors attached to the volatility related parameters, including the factor loadings. Below we describe the baseline prior setup used in the paper. After specifying the prior for each parameter group we also briefly report some alternative prior specifications we have experimented with. In particular we used a looser version of the prior and a tighter one. Our results are generally unaffected by changes of the priors described below, with more diffuse priors typically yielding the same qualitative pattern for impulse responses, only with slightly less pronounced effects for some variables and wider credible sets. The more tight priors lead to similar median impulse responses and slightly narrower credible sets. Additional results are provided below in Section 3.5.

For the rows a_j of the matrix A , we follow Cogley and Sargent (2005) and make the prior fairly uninformative, with prior means of 0 and variances of 10 for all coefficients. The prior variance of 10 is large enough for this prior to be considered uninformative. Increasing it to 1000 does not change the posterior estimates in any significant way.

For the coefficients $(\gamma_{i,0}, \gamma_{i,1})$ (intercept, slope) of the idiosyncratic processes of equation i , $i = 1, \dots, n$, the prior mean is $(\log \sigma_i^2, 0.0)$, where σ_i^2 is the residual variance of an $\text{AR}(p)$ model over the estimation

sample. The prior standard deviations (assuming 0 covariance) are $(2^{0.5}, 0.4)$. In the alternative prior specifications we have changed these standard deviations to $(4^{0.5}, 0.4)$.

For the factor loadings β_j , $j = 2, \dots, n$, we use a prior mean of 1 and a standard deviation of 0.4. This is a rather diffuse prior, and indeed increasing the prior standard deviation from 0.4 to 1 does not lead to major differences in the posterior estimates.

For the coefficients of the VAR process of the factors, the prior means are zero, except that the first-order lag coefficients of each factor has a mean of 0.8. The prior standard deviations are set to 0.2 for the elements of $D(L)$ and 0.4 for those of δ_m and δ_f . In the alternative prior specifications we have used a looser prior featuring the same prior means but prior standard deviations of 1 for all the elements of $D(L)$ and those of δ_m and δ_f .

For the innovations to the idiosyncratic components of volatility $(\phi_1, \dots, \phi_n)$, we use a mean of 0.03, with 10 degrees of freedom for each. For the variance-covariance matrix of innovations to the factor processes (Φ_u) , we use a mean of 0.01 times an identity matrix, with 10 degrees of freedom. In the alternative prior specifications we used 5 or 25 degrees of freedom (rather than 10) for all these coefficients.

For the period 0 values of $\log m_t$ and $\log f_t$, and $\log h_{i,t}$, we set the mean at 0 and in each draw use the variance implied by the VAR representation of the factors (treating the δ_m and δ_f coefficients as 0) and the draws of the coefficients and error variance matrix. Finally, for the period 0 values of $\log h_{i,t}$, we set the mean and variance at $\log \sigma_i^2$ and 2.0, respectively. In the alternative prior specifications we used a prior variance of 4.0.

1.2 Triangularization for estimation

In this subsection we briefly summarize the VAR triangularization which is needed to handle a large system with asymmetric priors and time varying volatilities, such as the model used here.¹ More details can be found in Carriero, Clark and Marcellino (2016b). With the triangularization, the estimation algorithm will block the conditional posterior distribution of the system of VAR coefficients in n different blocks. In the step of the typical Gibbs sampler that involves drawing the set of VAR coefficients Π , all of the remaining

¹Since the triangularization obtains computational gains of order n^2 , the cross-sectional dimension of the system can be extremely large, and indeed Carriero, Clark and Marcellino (2016b) present results for a VAR with 125 variables.

model coefficients are given. Consider again the reduced form residuals:

$$\begin{bmatrix} v_{1,t} \\ v_{2,t} \\ \dots \\ v_{n,t} \end{bmatrix} = \begin{bmatrix} 1 & 0 & \dots & 0 \\ a_{2,1}^* & 1 & & \dots \\ \dots & & 1 & 0 \\ a_{n,1}^* & \dots & a_{n,n-1}^* & 1 \end{bmatrix} \begin{bmatrix} \lambda_{1,t}^{0.5} & 0 & \dots & 0 \\ 0 & \lambda_{2,t}^{0.5} & & \dots \\ \dots & & \dots & 0 \\ 0 & \dots & 0 & \lambda_{n,t}^{0.5} \end{bmatrix} \begin{bmatrix} \epsilon_{1,t} \\ \epsilon_{2,t} \\ \dots \\ \epsilon_{n,t} \end{bmatrix}, \quad (1)$$

where $a_{j,i}^*$ denotes the generic element of the matrix A^{-1} which is available under knowledge of A . The VAR can be written as:

$$\begin{aligned} y_{1,t} &= \sum_{i=1}^n \sum_{l=1}^p \pi_{1,l}^{(i)} y_{i,t-l} + \sum_{l=0}^{p_m} \pi_{1,l}^{(m)} \ln m_{t-l} + \sum_{l=0}^{p_f} \pi_{1,l}^{(f)} \ln f_{t-l} + \lambda_{1,t}^{0.5} \epsilon_{1,t} \\ y_{2,t} &= \sum_{i=1}^n \sum_{l=1}^p \pi_{2,l}^{(i)} y_{i,t-l} + \sum_{l=0}^{p_m} \pi_{2,l}^{(m)} \ln m_{t-l} + \sum_{l=0}^{p_f} \pi_{2,l}^{(f)} \ln f_{t-l} + a_{2,1}^* \lambda_{1,t}^{0.5} \epsilon_{1,t} + \lambda_{2,t}^{0.5} \epsilon_{2,t} \\ &\dots \\ y_{n,t} &= \sum_{i=1}^n \sum_{l=1}^p \pi_{n,l}^{(i)} y_{i,t-l} + \sum_{l=0}^{p_m} \pi_{n,l}^{(m)} \ln m_{t-l} + \sum_{l=0}^{p_f} \pi_{n,l}^{(f)} \ln f_{t-l} + a_{n,1}^* \lambda_{1,t}^{0.5} \epsilon_{1,t} + \dots \\ &\dots + a_{n,n-1}^* \lambda_{n-1,t}^{0.5} \epsilon_{n-1,t} + \lambda_{n,t}^{0.5} \epsilon_{n,t}, \end{aligned}$$

with the generic equation for variable j :

$$\begin{aligned} &y_{j,t} - (a_{j,1}^* \lambda_{1,t}^{0.5} \epsilon_{1,t} + \dots + a_{j,j-1}^* \lambda_{j-1,t}^{0.5} \epsilon_{j-1,t}) \\ &= \sum_{i=1}^n \sum_{l=1}^p \pi_{j,l}^{(i)} y_{i,t-l} + \sum_{l=0}^{p_m} \pi_{j,l}^{(m)} \ln m_{t-l} + \sum_{l=0}^{p_f} \pi_{j,l}^{(f)} \ln f_{t-l} + \lambda_{j,t} \epsilon_{j,t}. \end{aligned} \quad (2)$$

Consider estimating these equations in order from $j = 1$ to $j = n$. When estimating the generic equation j the term of the left hand side in (2) is known, since it is given by the difference between the dependent variable of that equation and the estimated residuals of all the previous $j - 1$ equations. Therefore we can define:

$$y_{j,t}^* = y_{j,t} - (a_{j,1}^* \lambda_{1,t}^{0.5} \epsilon_{1,t} + \dots + a_{j,j-1}^* \lambda_{j-1,t}^{0.5} \epsilon_{j-1,t}), \quad (3)$$

and equation (2) becomes a standard generalized linear regression model for the variable in equation (3) with Gaussian disturbances with mean 0 and variance $\lambda_{j,t}$.

Accordingly, drawing on results detailed in Carriero, Clark and Marcellino (2016b), the posterior

distribution of the VAR coefficients can be factorized as:

$$\begin{aligned}
p(\Pi|A, \beta, f_{1:T}, m_{1:T}, h_{1:T}, y_{1:T}) &= p(\pi^{(n)}|\pi^{(n-1)}, \pi^{(n-2)}, \dots, \pi^{(1)}, A, \beta, f_{1:T}, m_{1:T}, h_{1:T}, y_{1:T}) \\
&\quad \times p(\pi^{(n-1)}|\pi^{(n-2)}, \dots, \pi^{(1)}, A, \beta, f_{1:T}, m_{1:T}, h_{1:T}, y_{1:T}) \\
&\quad \times \dots \times p(\pi^{(1)}|A, \beta, f_{1:T}, m_{1:T}, h_{1:T}, y_{1:T}),
\end{aligned} \tag{4}$$

where the vector β collects the loadings of the uncertainty factors and $f_{1:T}, m_{1:T}, h_{1:T} = (h_{1,T}, \dots, h_{n,T})$, and $y_{1:T}$ denote the history of the states and data up to time T .² As a result, we are able to estimate the coefficients of the VAR on an equation-by-equation basis. This greatly speeds estimation and permits us to consider much larger systems than we would otherwise be able to consider.

Importantly, although the expression (1) and the following triangular system are based on a Cholesky-type decomposition of the variance Σ_t , the decomposition is simply used as an estimation device, not as a way to identify structural shocks. The ordering of the variables in the system does not change the joint (conditional) posterior of the reduced form coefficients, so changing the order of the variables is inconsequential to the results.³ Moreover, since a shock to uncertainty is uncorrelated with shocks to the conditional mean of the variables, the ordering of the variables in the system has no influence on the shape of impulse responses in our application.

1.3 Drawing the loadings

Finally, we note that in drawing the loadings, we make use of the information in the observable $\ln(\tilde{v}_{j,t}^2)$, with the following transformation of the observation equations:

$$\ln(\tilde{v}_{j,t}^2 + \bar{c}) - \ln h_{j,t} = \begin{cases} \beta_{m,j} \ln m_t + \ln \epsilon_{j,t}^2, & j = 1, \dots, n_m \\ \beta_{f,j} \ln f_t + \ln \epsilon_{j,t}^2, & j = n_m + 1, \dots, n. \end{cases},$$

²Note we have implicitly used the fact that the matrix Ω_{Π}^{-1} is block diagonal, which is the case in our application, as our prior on the conditional mean coefficients is independent across equations, with a Minnesota-style form.

³This statement refers to drawing from the conditional posterior of the conditional mean parameters, when Σ_t belongs to the conditioning set. One needs also to keep in mind that the joint distribution of the system might be affected by the ordering of the variables in the system due to an entirely different reason: the diagonalization typically used for the error variance Σ_t in stochastic volatility models. Since priors are elicited separately for A and Λ_t , the implied prior of Σ_t will change if one changes the equation ordering, and therefore different orderings would result in different prior specifications and then potentially different joint posteriors. This problem is not a feature of our triangular algorithm, but rather it is inherent to all models using the diagonalization of Σ_t . As noted by Sims and Zha (1998) and Primiceri (2005), this problem will be mitigated in the case (as the one considered in this paper) in which the covariances A do not vary with time, because the likelihood information will soon dominate the prior.

with the conditioning on $h_{1:T}$ and $s_{1:T}$ in the posterior for β , we use this equation, along with the mixture mean and variance associated with the draw of $s_{1:T}$, for sampling the factor loadings with a conditionally normal posterior with mean and variance represented in a GLS form.

2 Additional results from baseline model estimation

Table A.1 below reports the inefficiency factors computed for the retained draws from the posterior distribution. On balance, the inefficiency factors indicate, that in terms of efficiency and mixing, the performance of the MCMC algorithm is reasonably good.

Table A.2 below provides posterior estimates of the factor loadings — the β_m and β_f coefficients of equations (2) and (3). All of the posterior mean estimates are positive and clustered around a value of 1 (as noted above, the loadings on employment and S&P returns are fixed at 1). Some loadings are somewhat below 1 (e.g., orders for durable goods), and others are modestly above 1 (e.g., industrial production). In all cases, the loadings appear to be estimated with reasonable precision. However, the loadings on the finance factor have posterior standard deviations that are noticeably lower than those for the loadings on the macro factor (despite a prior that is the same for the macro and finance factors).

Figures A1 through A10 provide some additional detail on the estimates reported in the paper, including the correlations of our uncertainty estimates with those of JLN and LMN; reduced form and idiosyncratic volatility estimates for all variables in the model; and impulse responses, historical decompositions (including a color version of the paper’s Figure 5), and predictive distributions for all variables in the model.

Table A.1. Inefficiency Factors of Baseline Model Estimates

parameter block	# of parameters	median IF	mean IF	min IF	max IF
Π	5580	1.10	1.40	0.42	53.73
A	435	1.82	2.22	0.64	15.31
β_m	17	8.74	9.97	3.30	25.61
β_f	11	6.99	7.15	4.87	9.26
γ_0	30	23.22	28.85	2.97	89.38
γ_1	30	23.85	29.67	4.66	89.91
$\delta^{(m)}$	34	12.16	15.25	5.56	70.24
$\delta^{(f)}$	34	5.45	6.30	1.44	16.41
Φ_v	30	24.87	27.02	5.37	63.87
Φ_u	3	42.61	39.76	31.64	45.03
$\ln m_{1:T}$	647	13.23	14.86	2.64	51.35
$\ln f_{1:T}$	647	3.09	3.58	0.95	15.96

Notes:

1. All results are based on a sample of 5,000 retained draws, obtained by sampling a total of 30,000 draws, discarding the first 5,000, and retaining every 5th draw of the post-burn sample.
2. For each individual parameter, the inefficiency factor is estimated as $1 + 2 \sum_{k=1}^{\infty} \rho_k$, where ρ_k is the k -th order autocorrelation of the chain of retained draws. The estimates use the Newey-West kernel and a bandwidth of 4 percent of the sample of retained draws.
3. Parameter blocks are denoted as follows: Π , coefficients of VAR; A , lower triangular matrix entering the equation for the reduced form errors; β_m , loadings of macroeconomic volatilities on the macroeconomic uncertainty factor; β_f , loadings of financial volatilities on the financial uncertainty factor; γ_0 , intercepts of idiosyncratic volatility processes; γ_1 , slope coefficients of intercepts of idiosyncratic volatility processes; $\delta^{(m)}$, coefficients of the augmented VAR's equation for log macroeconomic uncertainty; $\delta^{(f)}$, coefficients of the augmented VAR's equation for log financial uncertainty; Φ_v , vector of variances of innovations of idiosyncratic volatility processes; Φ_u , variance-covariance matrix of innovations to log uncertainty; $\ln m_{1:T}$, time series of log macroeconomic uncertainty; and $\ln f_{1:T}$, time series of log financial uncertainty.

Table A.2: Posterior estimates of factor loadings

Variable	Posterior median	st. dev.
<i>Loadings of macro variables on macro factor</i>		
Employment	1.000	0.000
Ind. prod.	1.230	0.277
Capacity utilization	0.725	0.284
Help wanted/unemployment	0.786	0.296
Unemployment rate	0.875	0.244
Real personal income	0.767	0.314
Weekly hours, goods	0.754	0.316
Housing starts	1.153	0.256
Housing permits	1.246	0.309
Real consumer spending	1.159	0.268
Real manuf. and trade sales	0.599	0.276
ISM index, new orders	0.740	0.249
Orders for durable goods	0.679	0.278
Avg. hourly earnings, goods	1.074	0.295
PPI, finished goods	1.209	0.287
PPI, commodities	0.771	0.316
PCE price index	1.126	0.264
Federal funds rate	1.346	0.310
<i>Loadings of finance variables on finance factor</i>		
S&P 500	1.000	0.000
Spread, Baa-10y Treasury	1.100	0.171
Excess return	0.986	0.134
SMB	0.928	0.116
HML	1.008	0.121
Momentum	1.508	0.148
R15-R11	0.771	0.141
Industry 1	0.823	0.125
Industry 2	0.769	0.154
Industry 3	0.816	0.125
Industry 4	0.793	0.150
Industry 5	0.902	0.159

3 Robustness

In this section we provide additional details on the robustness checks of our results.

3.1 Alternative samples

3.1.1 Quarterly data

First, we consider the robustness of our results to the use of a quarterly, rather than monthly, data frequency. Graphs for these results are provided in Figure A11 (estimated uncertainty measures), Figure A12 (estimated volatilities), Figure A13 (impulse responses to a macro uncertainty shock) and Figure A14 (impulse responses to a financial uncertainty shock). Qualitatively, these quarterly results are very similar to our baseline estimates with monthly data. There are some differences, such as the more significant response of macro uncertainty to finance uncertainty in the quarterly estimates than the monthly estimates, but our key results from the monthly frequency carry over to the quarterly case.

3.1.2 Financial crisis effects

A relevant question is whether the recent financial crisis had a substantial impact on the results we have obtained. For example, Alessandri and Mumtaz (2014), using a nonlinear VAR model, find that uncertainty has a much stronger (negative) impact on output in periods of financial stress than otherwise. To assess whether this is the case in our baseline results, we recompute the impulse responses using estimates of the model with data through just December 2007. Figures A15 and A16 provide the results. In the shorter sample, it continues to be the case that shocks to both macro uncertainty and financial uncertainty have significant effects on economic activity and some financial variables, with patterns generally similar to those obtained for the full sample of data. However, in the case of the shock to macro uncertainty, the effects are modestly smaller and less precisely estimated than in the baseline case. In the case of the shock to financial uncertainty, the effects for the 1960-2007 sample are quite similar to those for the 1960-2014 period. Overall, we judge that the crisis period has provided useful information for assessing the effects of changes in macro uncertainty but by no means drives it, whereas the crisis period has less effect on the measurement of financial uncertainty and its effects.

3.2 Nested specifications

Second, we consider comparisons with simpler models that can be nested within our specification. In particular we will consider: i) the use of the two step approach, ii) shutting off the effect of stochastic volatility on the conditional mean, and iii) shutting down the time variation in the idiosyncratic volatility in all the variables.

3.2.1 Two step approach

To check the effects of the use of the two-step approach, we have proceeded as follows. In a first step we have estimated a standard BVAR (using the same 30 variables used for the baseline model) with stochastic volatility. Then, by averaging the resulting estimated volatilities we have obtained two measures of macroeconomic and financial uncertainty. Finally, in a second step we have plugged in these measures of uncertainty in a conventional homoskedastic BVAR in the 30 variables (hence yielding a 32 variables BVAR) and performed standard structural analysis, ordering the uncertainty measures first in the system, and ordering macro uncertainty ahead of financial uncertainty.

Results for this experiment are reported in Figure A17 and Figure A18. The resulting effects of a macro shock are qualitatively similar to our baseline results, although there are some differences for a few variables. The effects of a shock to financial uncertainty are very similar to our baseline estimates. As to precision of estimates in the two-step approach compared to our baseline, the responses of the uncertainty series are clearly more precise in the two-step approach than our baseline. This is an artificial consequence of the two-step approach, which does not take into account the uncertainty around uncertainty. But for other variables, it is harder to see a clear and consistent difference in precision.

We have also computed the historical decompositions using this two-step approach, and results are displayed in Figure A19. In broad terms, these results are qualitatively similar to those we obtain with our baseline model, but there are two changes that seem most notable: (1) the effects of uncertainty shocks are a little larger in this two-step approach than our approach, and (2) shocks to the VAR's other variables play a smaller role in uncertainty moves in these estimates than in our one-step estimates. Hence this evidence suggests that the two-step approach might overestimate the total contribution of uncertainty to macroeconomic and financial fluctuations.

3.2.2 No volatility in the conditional mean

We have repeated the estimation of our model, but eliminating the feature that the common volatility factors also enter the conditional mean of the main VAR. The resulting estimated uncertainty measures are shown in Figure A20. From this figure, it is clear that while the measures computed under the two cases are very much correlated, the measures that do not include uncertainty in the conditional mean appear much more volatile. Note that since the factors do not appear in the conditional mean of the VAR, for this model it is not possible to compute IRFs to uncertainty shocks.

3.2.3 Constant idiosyncratic volatilities

We have repeated the estimation of our model eliminating time variation in the idiosyncratic components of volatility, restricting these components to be constant — i.e., modifying the paper’s equation (3) such that $h_{j,t} = \gamma_{j,0}$, for all variables $j = 1, \dots, n$. Note that this model is close to the one used by e.g. Alessandri and Mumtaz (2014), and it has the merit to allow for common variation across the volatilities, but has the drawback that does not allow individual volatilities to have idiosyncratic time variation.

Indeed, Figures A21 and A22 suggest that time variation in the idiosyncratic volatility components is important in model estimates with monthly data. Most importantly, the estimated reduced form volatilities are fairly different for those variables with relatively large idiosyncratic variation in the baseline estimates. In addition, the uncertainty measures estimated under the assumption of no idiosyncratic movements appear much more volatile. In the IRFs (not reported) the economic effects of a macro shock are weaker than in the baseline. Overall, we take this evidence as supportive of our specification with time variation in the idiosyncratic volatility components.

3.3 Time variation in error covariances

Primiceri (2005) argued that time variation in the covariances of the error variance of a VAR (i.e. the matrix A^{-1}) might be relevant in analyzing monetary policy, and concluded that this was not the case using U.S. data. In order to check that this is the case also in our application, we have considered a smaller version of our model that allows the matrix A^{-1} to be time-varying. A time varying A^{-1} matrix affects the joint posterior of the model, and therefore it can potentially affect the posterior of the volatility measures and the IRFs. However, note that the change in the posterior estimates is the only way in which the time variation of A^{-1} can

change the IRFs to an uncertainty shock. Indeed, this matrix models the simultaneous relationship among the variables in the VAR, i.e. the covariance structure of the shocks v_t , but these shocks are orthogonal to the uncertainty shocks by assumption.

We have considered a smaller model, with 4 macroeconomic variables and one uncertainty factor (which therefore represents macroeconomic uncertainty). We have used the more flexible variance covariance matrix specification of Primiceri (2005), i.e. we have allowed the coefficients of the matrix A^{-1} to change over time, with each row of the matrix following a random walk process as in Primiceri (2005). Figures A23 - A25 show results for this experiment. From the figures it is clear that the time variation in the matrix A^{-1} has a very limited impact on the estimated uncertainty measure and the resulting impulse responses. Also, using a cross-section of only 4 variables does not deliver any significant effect of the uncertainty shock. This seems to point to the conclusion that the use of a large cross section of data is key in order to identify uncertainty and its effects. Instead, the use of a time varying as opposed to a fixed A^{-1} has a very limited impact on the impulse responses, which is in line with the findings of Primiceri (2005).

In order to check that this latter result holds also in a larger model, we have repeated the exercise including 12 macroeconomic variables rather than 4.⁴ Results are presented in Figures A26-A28. Also in this case we find that the time variation in the coefficients of A is limited, and it does not matter much for the shape of impulse response functions.

3.4 Alternative variable blocking

In order to check the robustness to the variable groupings described in Section 3, we have tried some alternative blocking of the variables in the system. We have focussed on the S&P and credit spread, as these are some variables that arguably could be thought as belonging to either the macro or finance set. In one case we have also considered enlarging the set of financial variables. Specifically, we considered some alternative blockings of the variables involving: i) S&P500 and credit spread both moved to the macro set; ii) S&P500 moved to the macro set, credit spread kept in the financial set; iii) Credit spread moved to the macro set, S&P500 kept in the financial set, and iv) 36 variables, obtained by adding six sorted portfolios (from the

⁴Repeating the exercise with the whole 30-variable system is computationally very challenging. Indeed, while the triangularization of Carriero, Clark, and Marcellino (2016b) solves the bottleneck due to the size of the variance matrix of the conditional mean coefficients (which is the main bottleneck), in systems large as the one presented here the estimation of a time varying A^{-1} matrix is still challenging. The number of states present in this matrix is $(n * (n - 1))/2$, which with $n = 30$ equals 435, and of course this poses computational problems. In theory, the simulation of these states can be parallelized, but there is a problem related on how to calibrate the prior on so many state variables, since pre-sample data are very limited.

datasets of Kenneth French) as financial variables, with S&P500 kept in the finance set, and credit spread moved to the macro set. Results for these alternative orderings are presented in Figures A29-A40 (with three figures for each alternative blocking). It turns out that, overall, the effects of a macro uncertainty shock are not much affected by moving some of the financial variables from the finance class to the macro class. But the results on the effects of a financial uncertainty shock instead can be affected when we move the S&P500 from the financial set to the macro set. As a result, as a general matter, we acknowledge that, as is often the case, there is some model uncertainty surrounding our results. Based on the many robustness checks we have conducted, we are confident in our basic results on the effects of uncertainty shocks. However, there can be settings in which it is difficult to disentangle macroeconomic and financial uncertainty.

3.5 Alternative priors on volatility-related parts of the model

3.5.1 Broadly looser prior (Alternative prior 1)

The looser prior (referred to as Alternative prior 1 in the figures) modifies the baseline prior as follows, to make it broadly less informative:

- $(\gamma_{i,0}, \gamma_{i,1})$ (intercept, slope) of the idiosyncratic processes of equation i : mean = $(0, 0.0)$; standard deviations (assuming 0 covariance) = $(4^{0.5}, 1.0)$
- factor loadings β_j : mean = 1, standard deviation = 1
- VAR in log factors: means = 0 for all coefficients, standard deviation = 1 for all
- $\phi_1, \dots, \phi_n$ (innovations to the idiosyncratic components of volatility): mean = 0.03 (unchanged), with 5 degrees of freedom for each
- Φ_u (variance-covariance matrix of innovations to the factor processes) = 0.01 times an identity matrix (unchanged), with 5 degrees of freedom
- Period 0 values of $\log h_{i,t}$: mean = 0, variance = 4.0

Results for this choice of priors are reported in Figures A41 and A42. The effects of a macro uncertainty shock are similar to the baseline, although with weaker effects on a few variables (housing in particular). Financial uncertainty shock effects are instead somewhat weaker than in the baseline, with slightly to modestly smaller responses and wider credible sets.

3.5.2 Broadly tighter prior (Alternative prior 2)

The tighter prior (referred to as Alternative prior 2 in the figures) modifies the baseline prior as follows:

- $(\gamma_{i,0}, \gamma_{i,1})$ (intercept, slope) of the idiosyncratic processes of equation i : mean = $(\log \sigma_i^2, 0.0)$; standard deviations = $(4^{0.5}, 0.4)$.
- $\phi_1, \dots, \phi_n$ (innovations to the idiosyncratic components of volatility): mean = 0.03 (unchanged), with 25 degrees of freedom for each
- Φ_u (variance-covariance matrix of innovations to the factor processes) = 0.01 times an identity matrix (unchanged), with 25 degrees of freedom
- Period 0 values of $\log h_{i,t}$: mean = $\log \sigma_i^2$, variance = 4.0
- factor loadings β_j : mean = 1, standard deviation = 1

This prior is broadly tighter than the baseline specification in that the degrees of freedom in the priors for ϕ and Φ are higher than in the baseline case. In some other aspects, the prior is actually looser than the baseline specification.

Results for this choice of priors are reported in Figures A43 and A44, and they are broadly consistent with those for the baseline specification. Specifically, the effects of a macro uncertainty shock are slightly to modestly smaller and more precise than in the baseline, whereas the effects of a finance uncertainty shock tend to be a little larger and more precise than in the baseline.

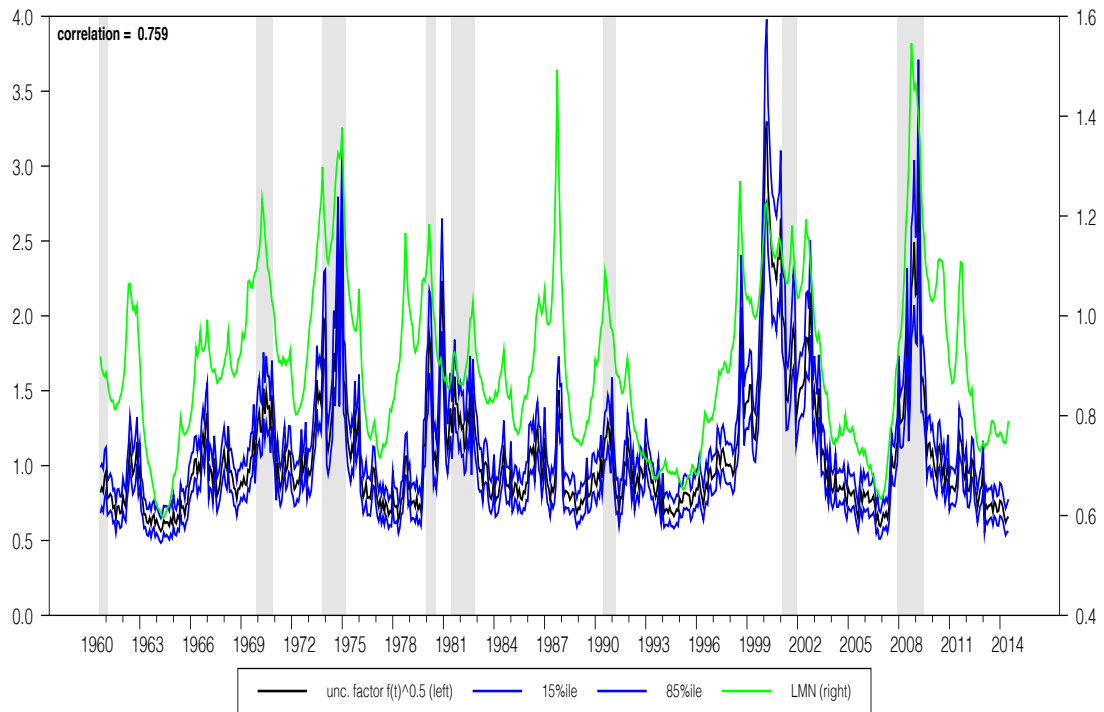
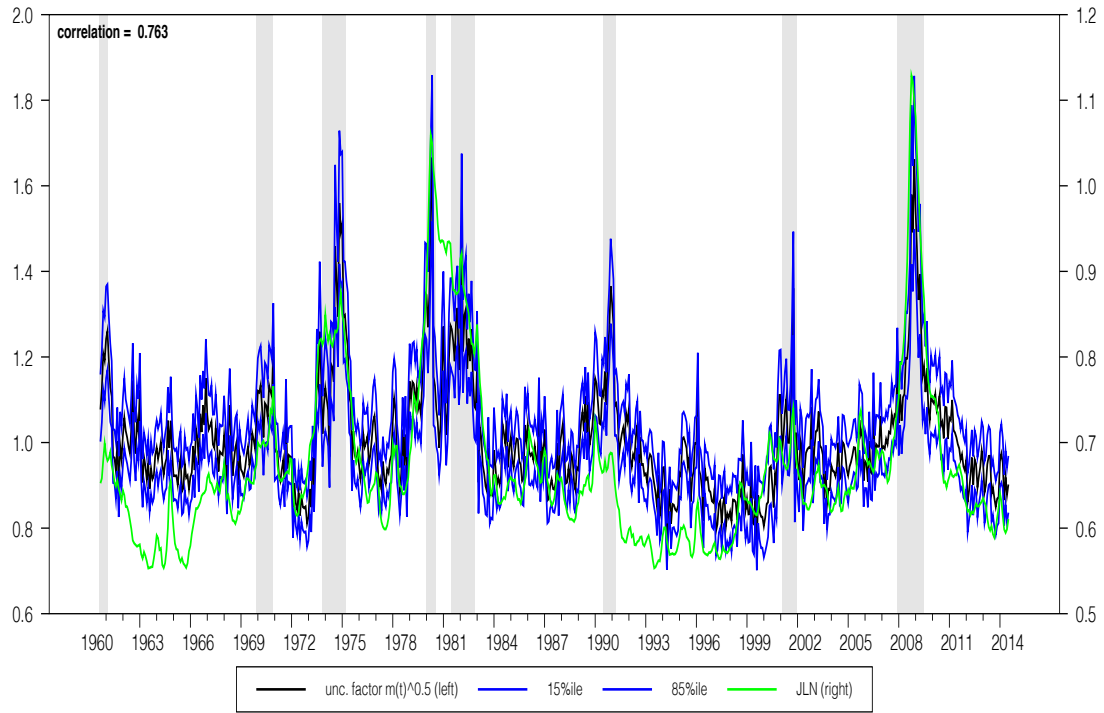


Figure A1: Baseline results with additional detail: uncertainty estimates. Posterior median (black line) and 15%/85% quantiles (blue lines), with macro uncertainty in the top panel and financial uncertainty in the bottom panel. The green line represents the corresponding estimates from JLN (top) and LMN (bottom). The gray bars indicate NBER recessions.

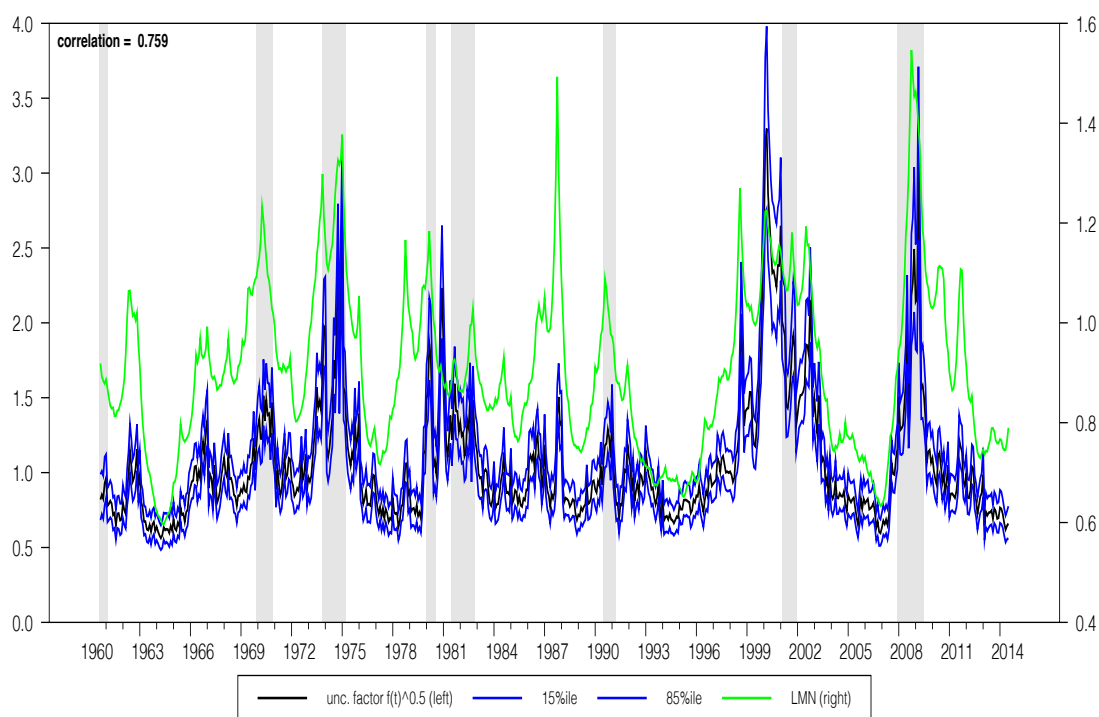
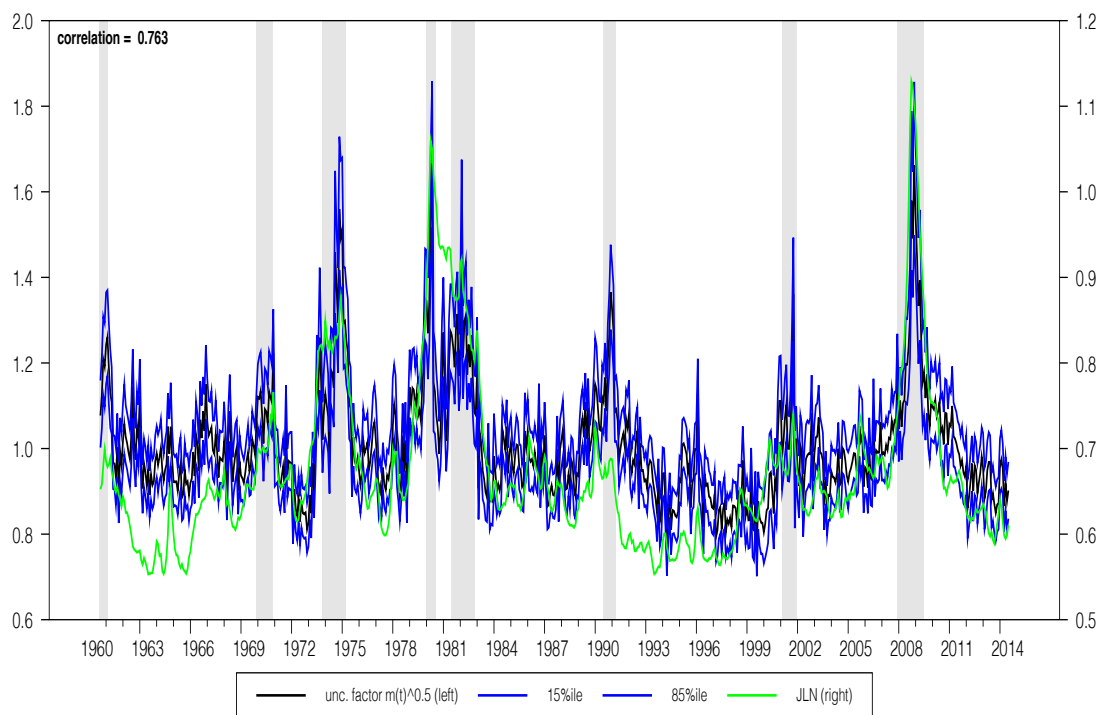


Figure A2: Baseline results with additional detail: uncertainty estimates. Posterior median (black line) and 15%/85% quantiles (blue lines), with macro uncertainty in the top panel and financial uncertainty in the bottom panel. The green line represents the corresponding estimates from JLN (top) and LMN (bottom). The gray bars indicate NBER recessions.

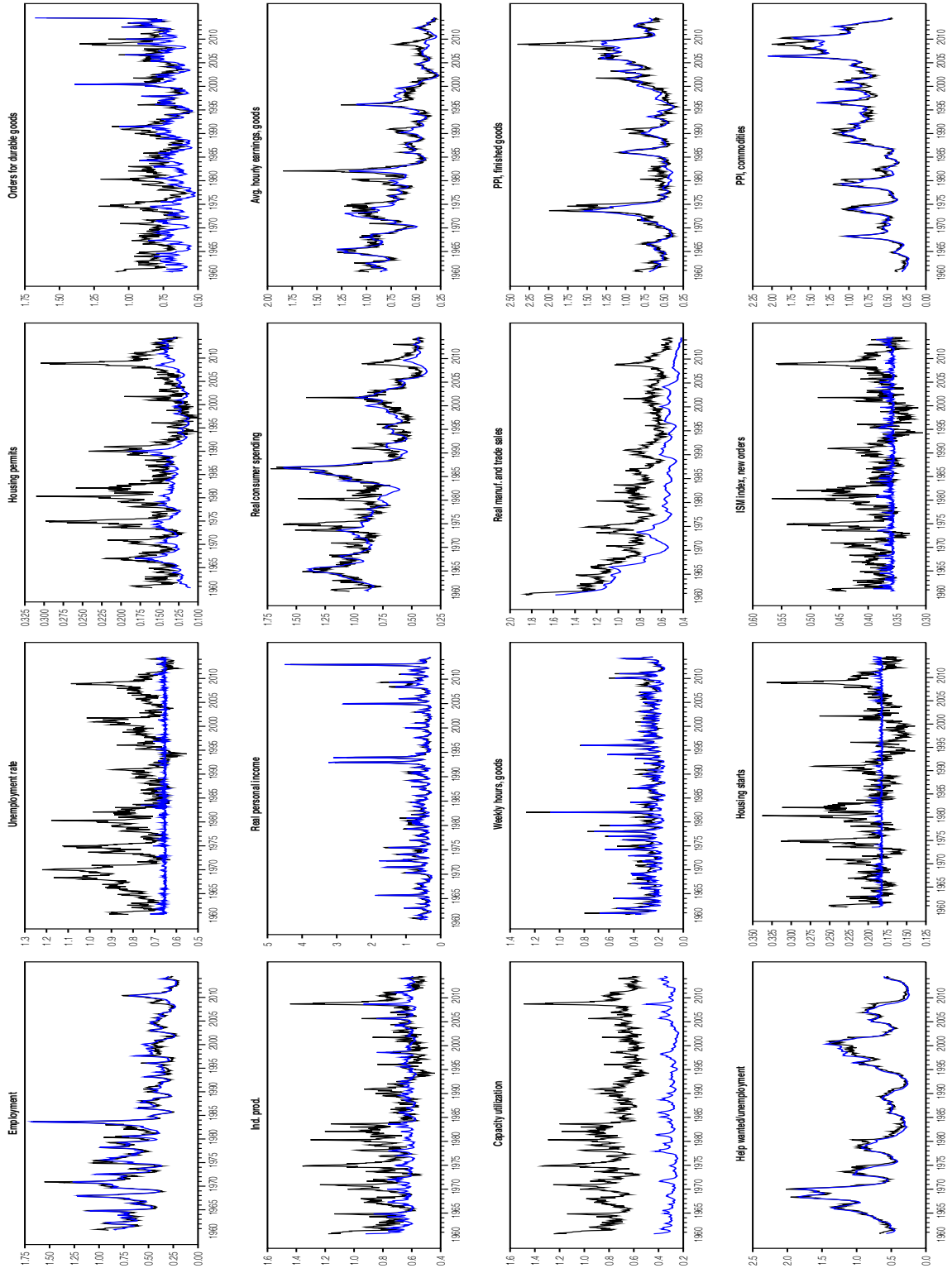


Figure A3: Baseline results with additional detail: reduced-form (black line) and idiosyncratic volatilities ($h_{i,t}^{0.5}$, blue line), posterior medians

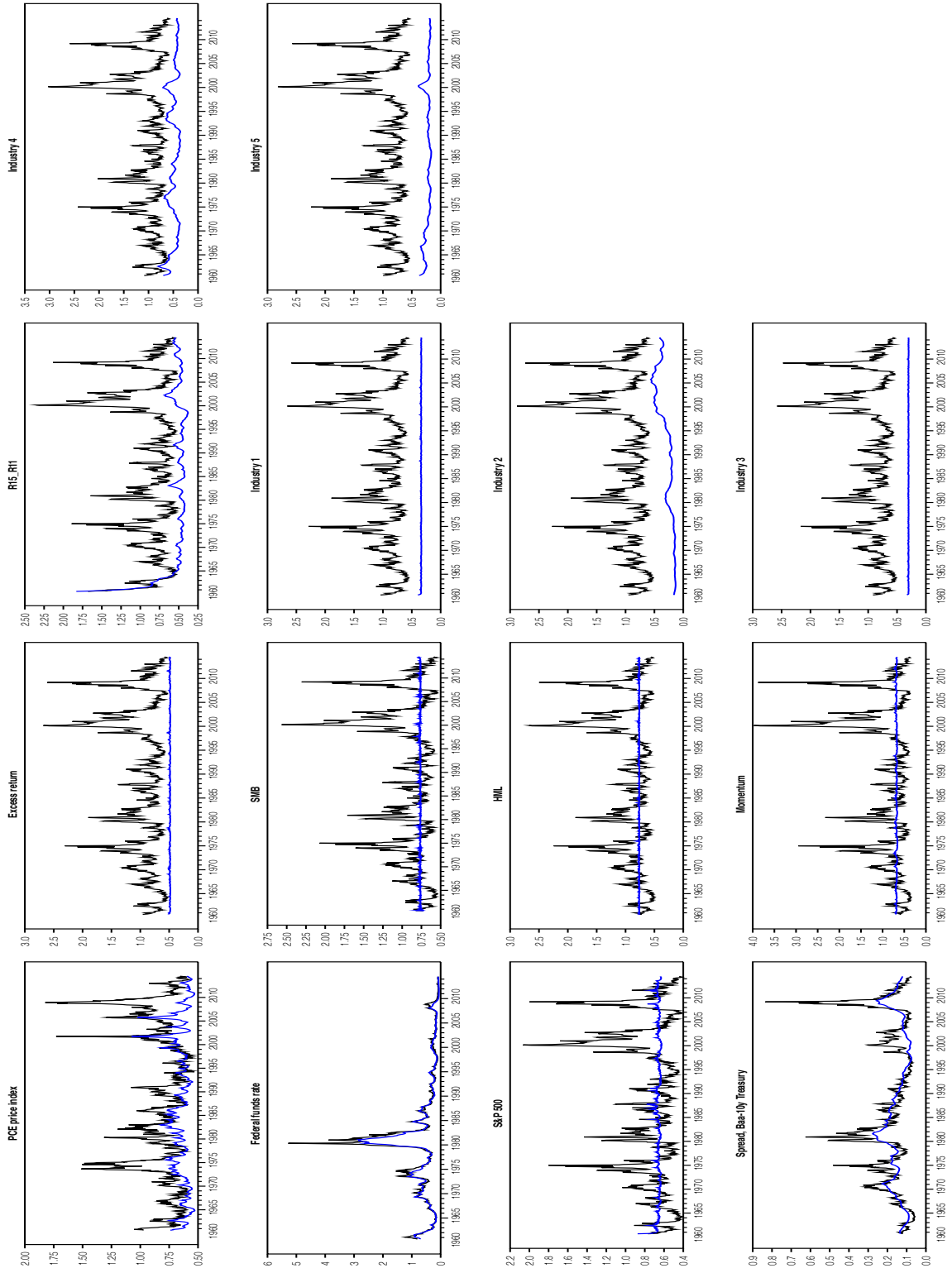


Figure A3: Continued, baseline results with additional detail: reduced-form (black line) and idiosyncratic volatilities ($h_{i,t}^{0.5}$, blue line), posterior medians

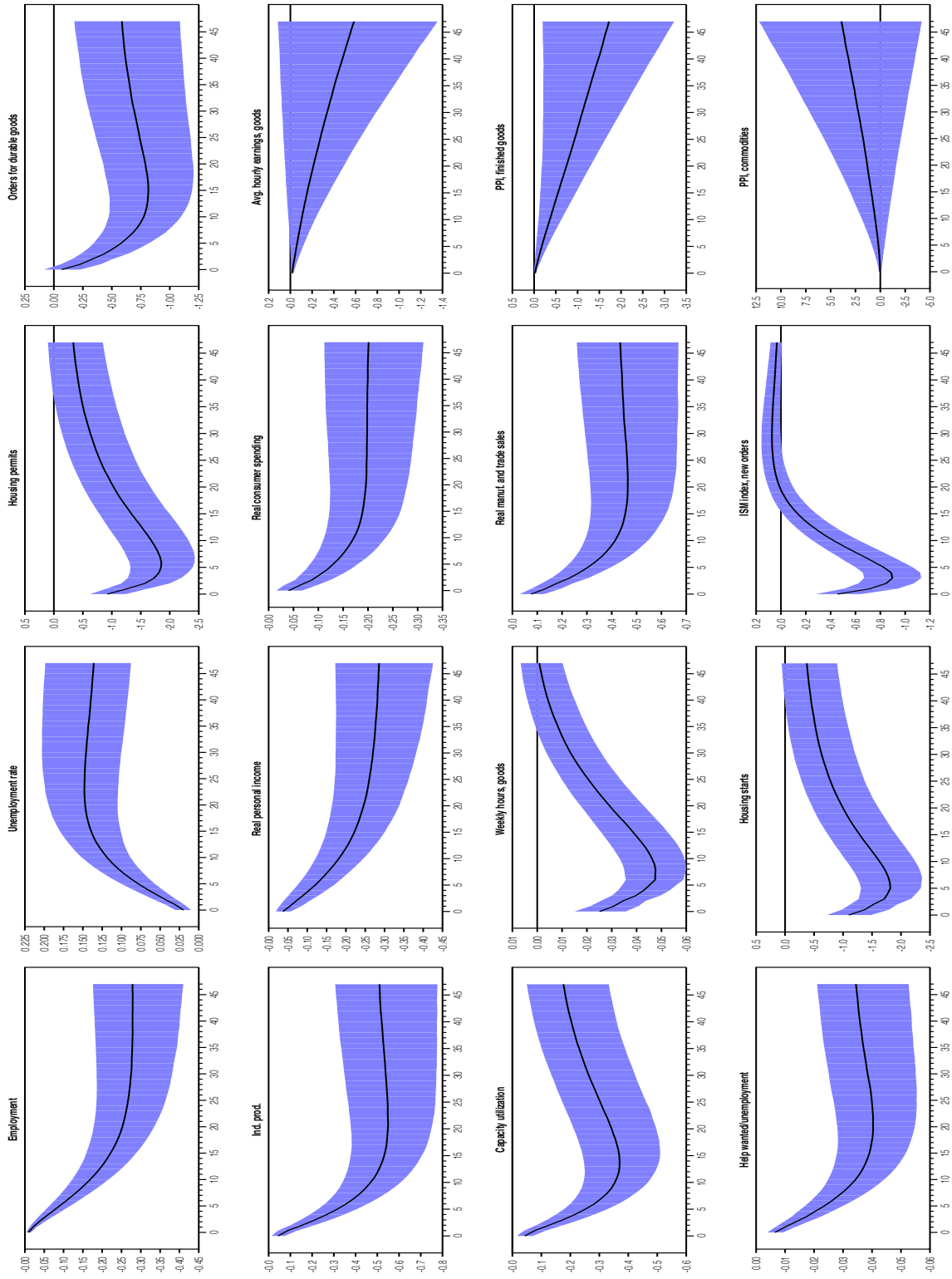


Figure A4: Baseline results with additional detail: impulse responses for shock to macro uncertainty. One standard deviation shock, posterior median (black line) and 15%/85% quantiles (blue shading)

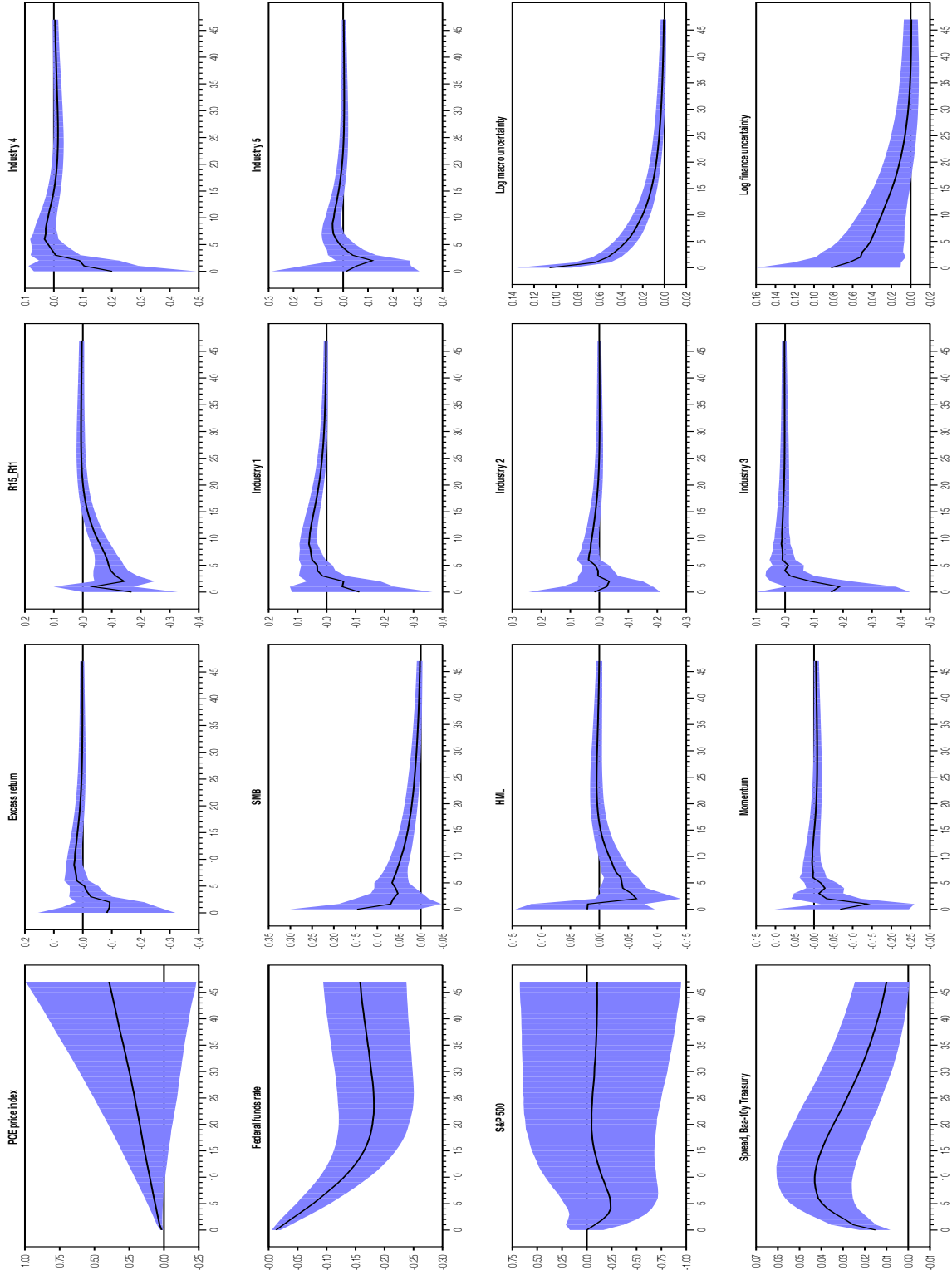


Figure A4: Continued, baseline results with additional detail: impulse responses for shock to macro uncertainty. One standard deviation shock, posterior median (black line) and 15%/85% quantiles (blue shading)

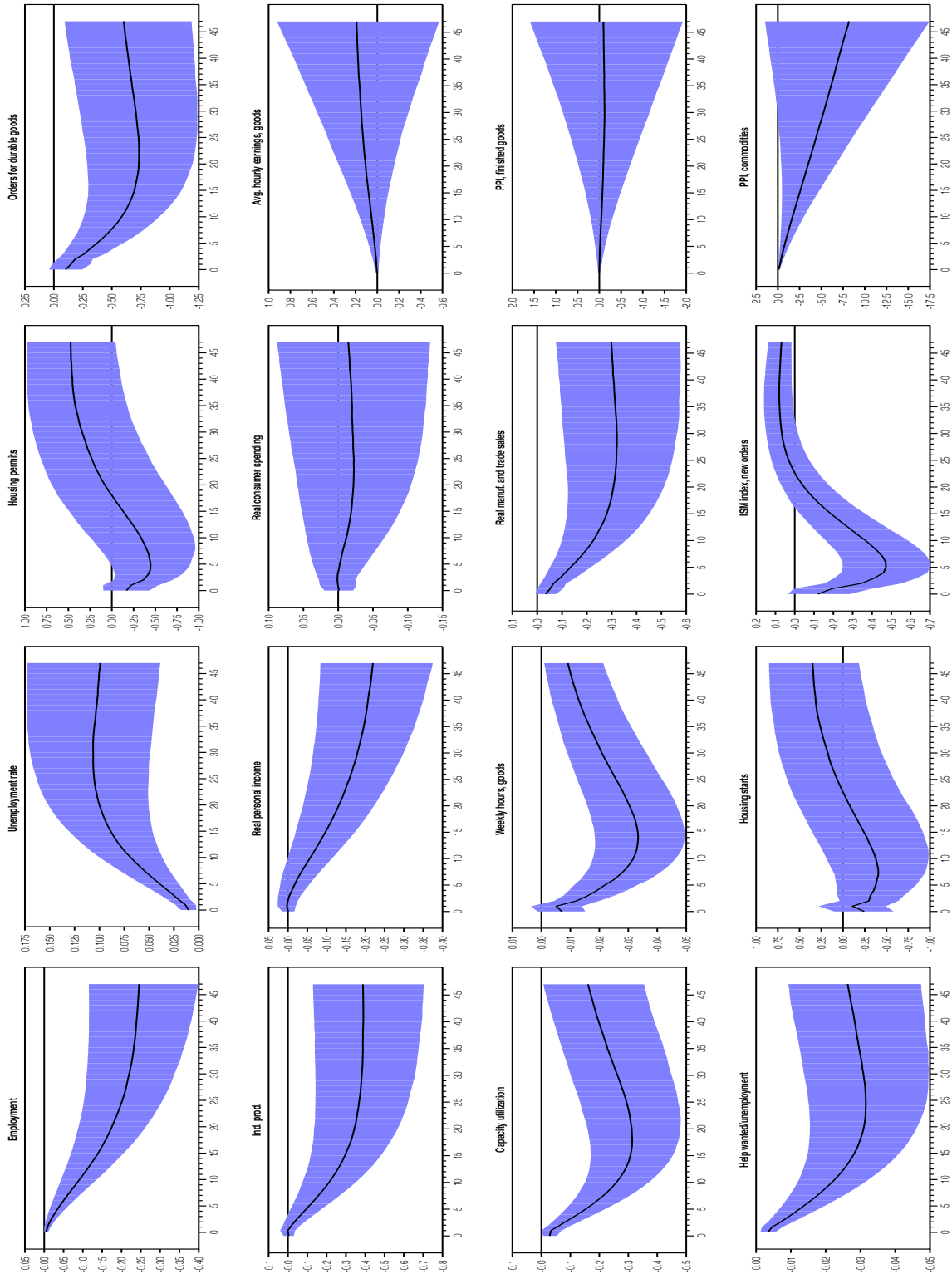


Figure A5: Baseline results with additional detail: impulse responses for shock to financial uncertainty. One standard deviation shock, posterior median (black line) and 15%/85% quantiles (blue shading)

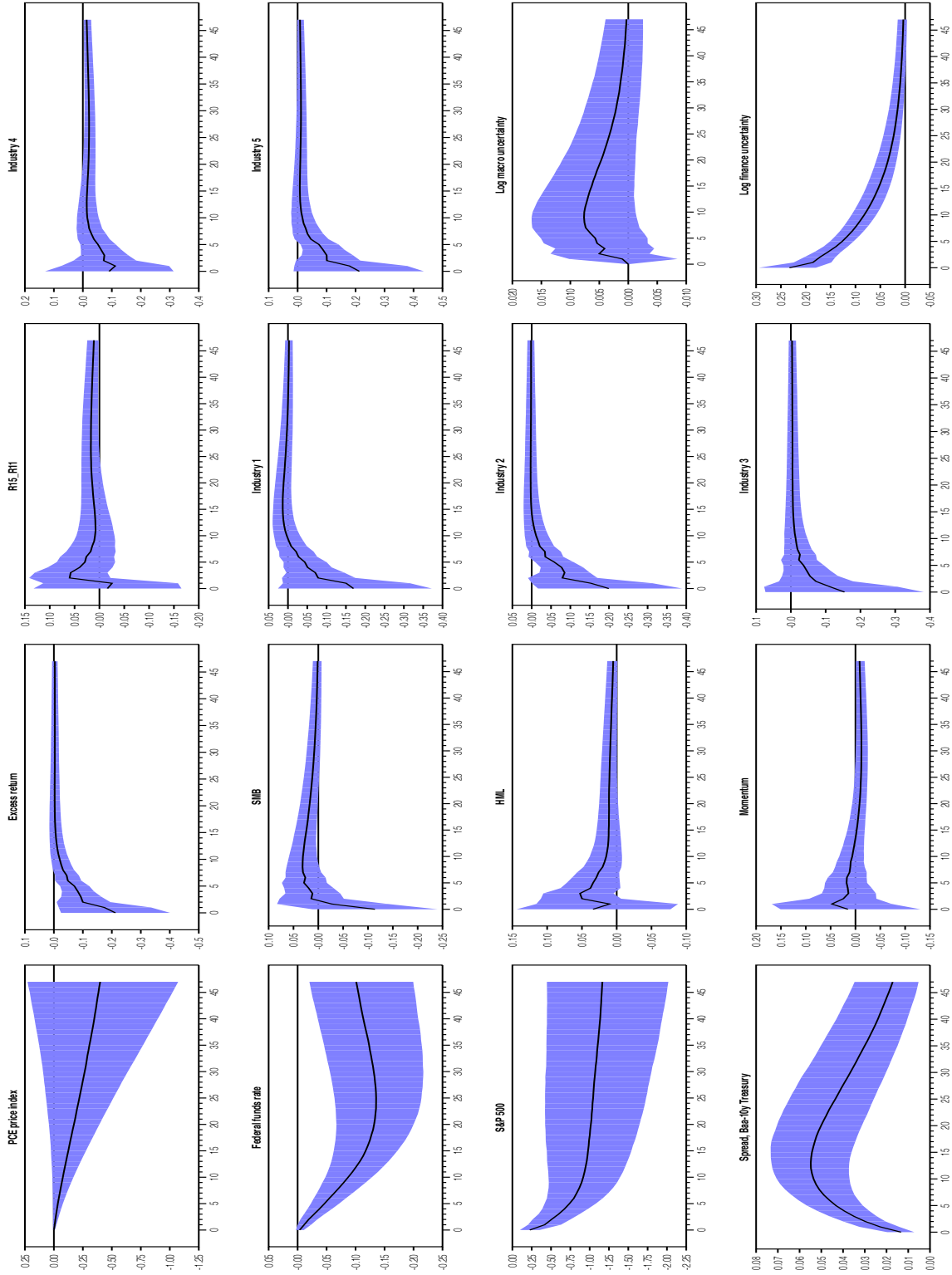


Figure A5: Continued, baseline results with additional detail: impulse responses for shock to financial uncertainty . One standard deviation shock, posterior median (black line) and 15%/85% quantiles (blue shading)

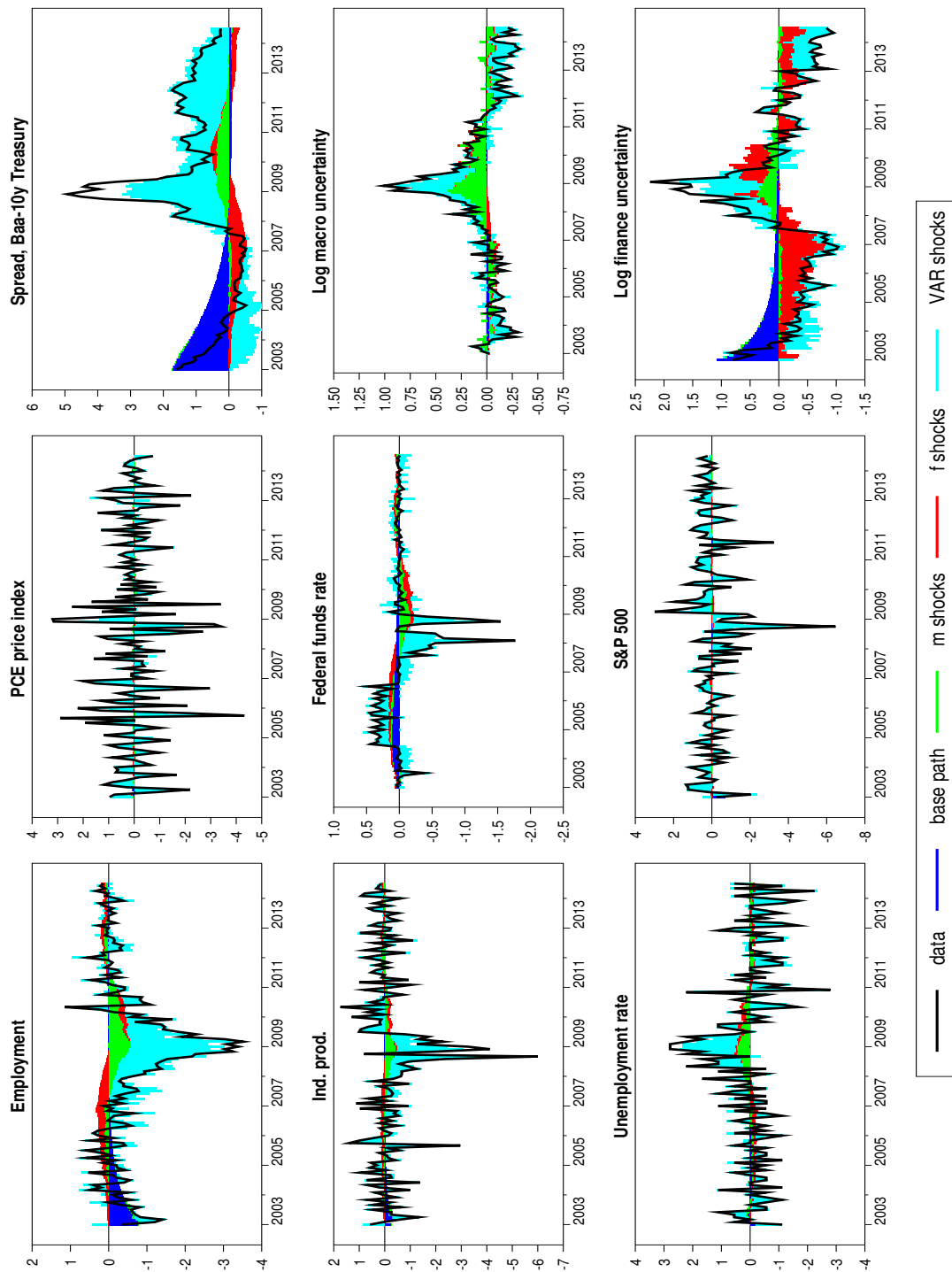


Figure A6: Baseline results with additional detail: historical decomposition in color, 2003-2014, selected variables

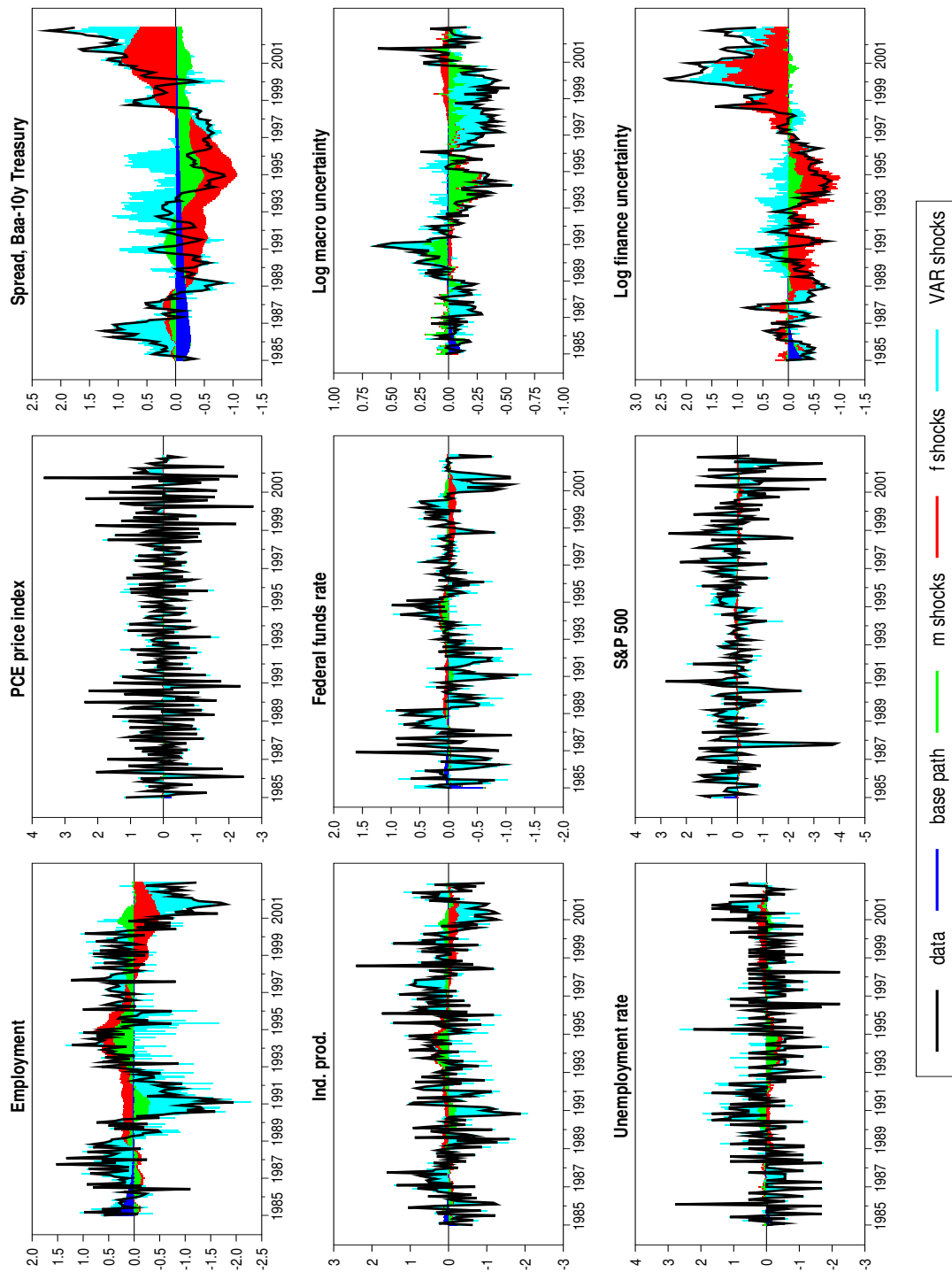


Figure A7: Baseline results with additional detail: historical decomposition in color, 1985-2002, selected variables

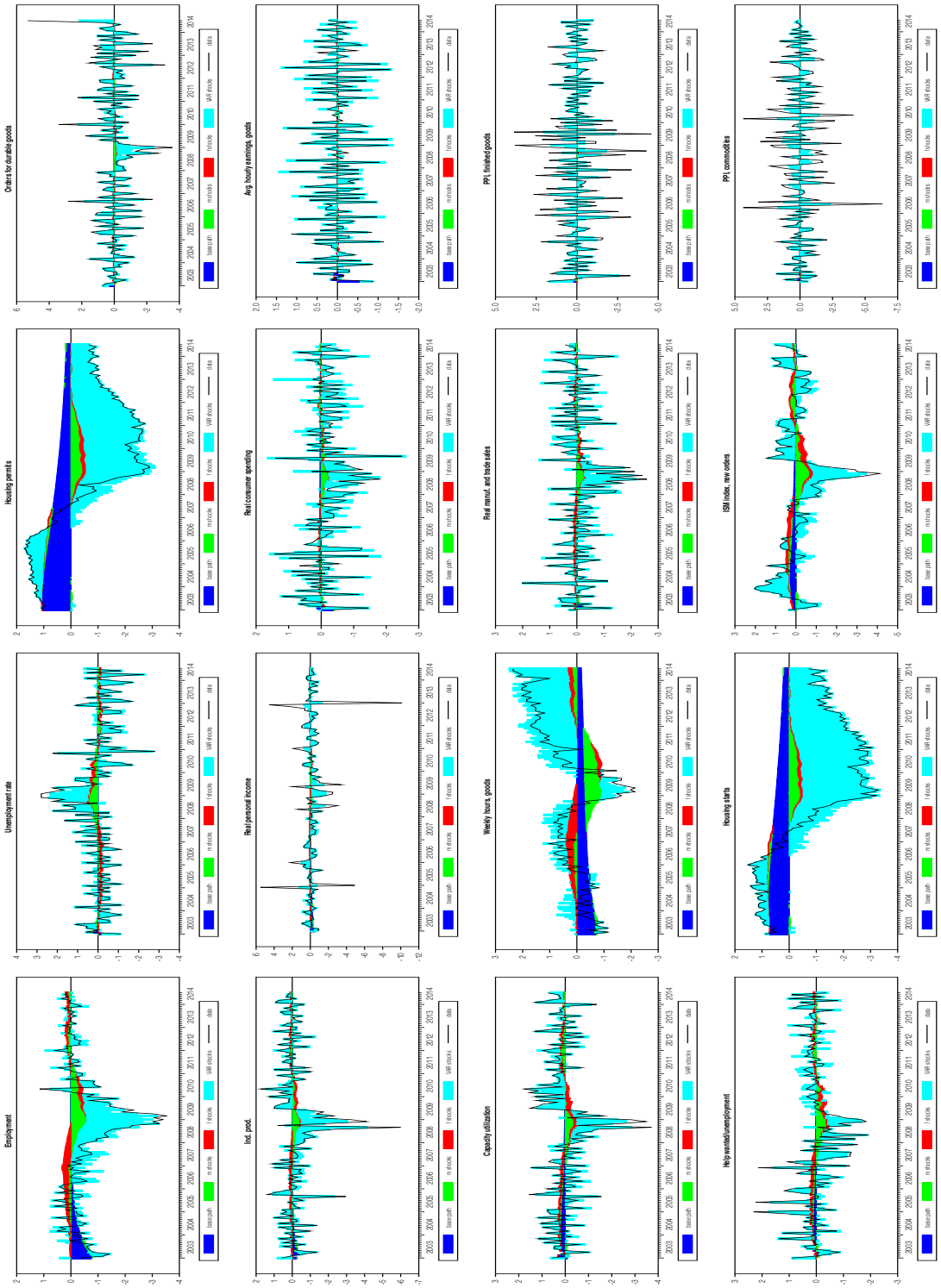


Figure A8: Baseline results with additional detail: historical decomposition, 2003-2014, all variables

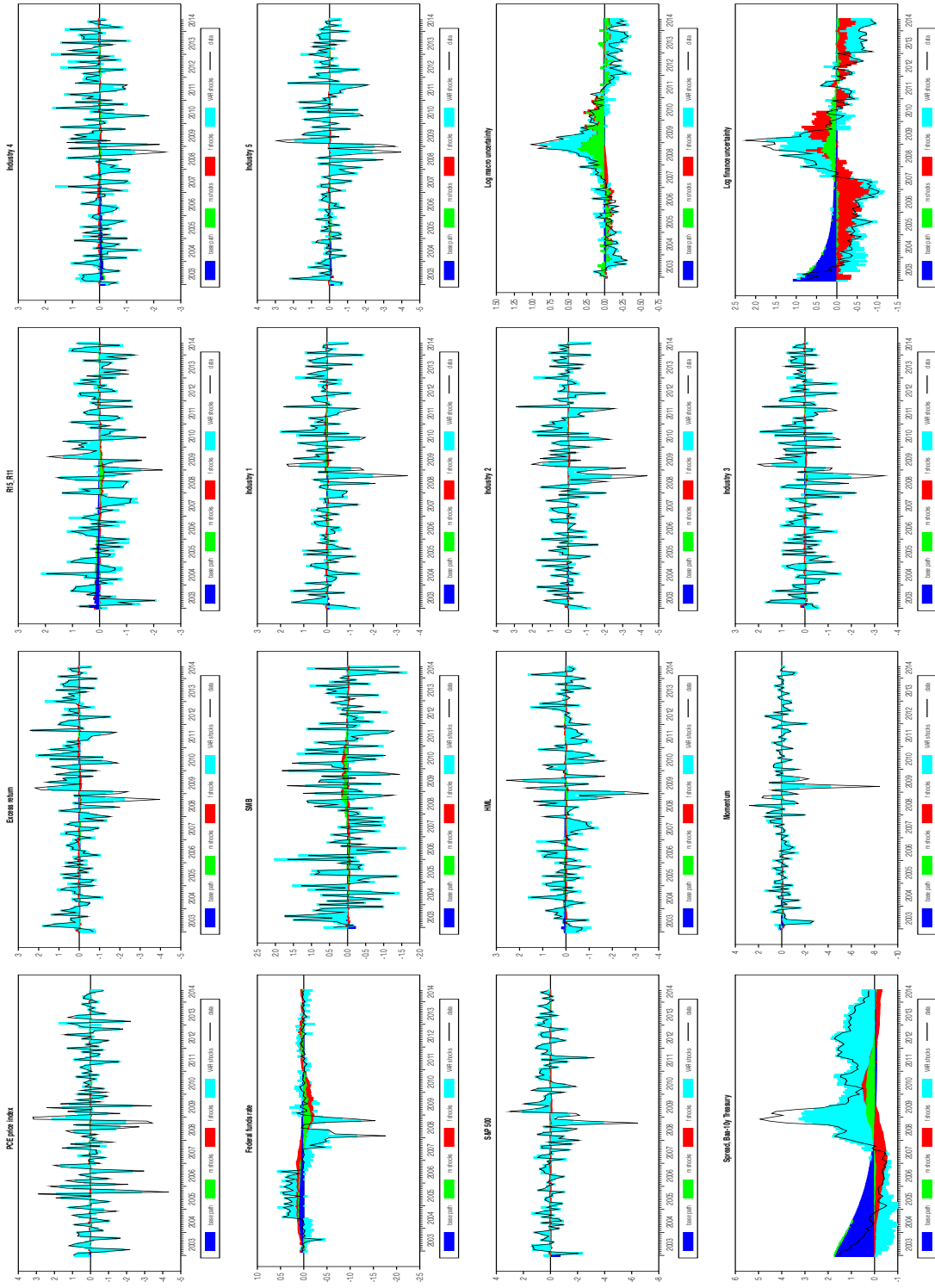


Figure A8: Continued, baseline results with additional detail: historical decomposition, 2003-2014, all variables

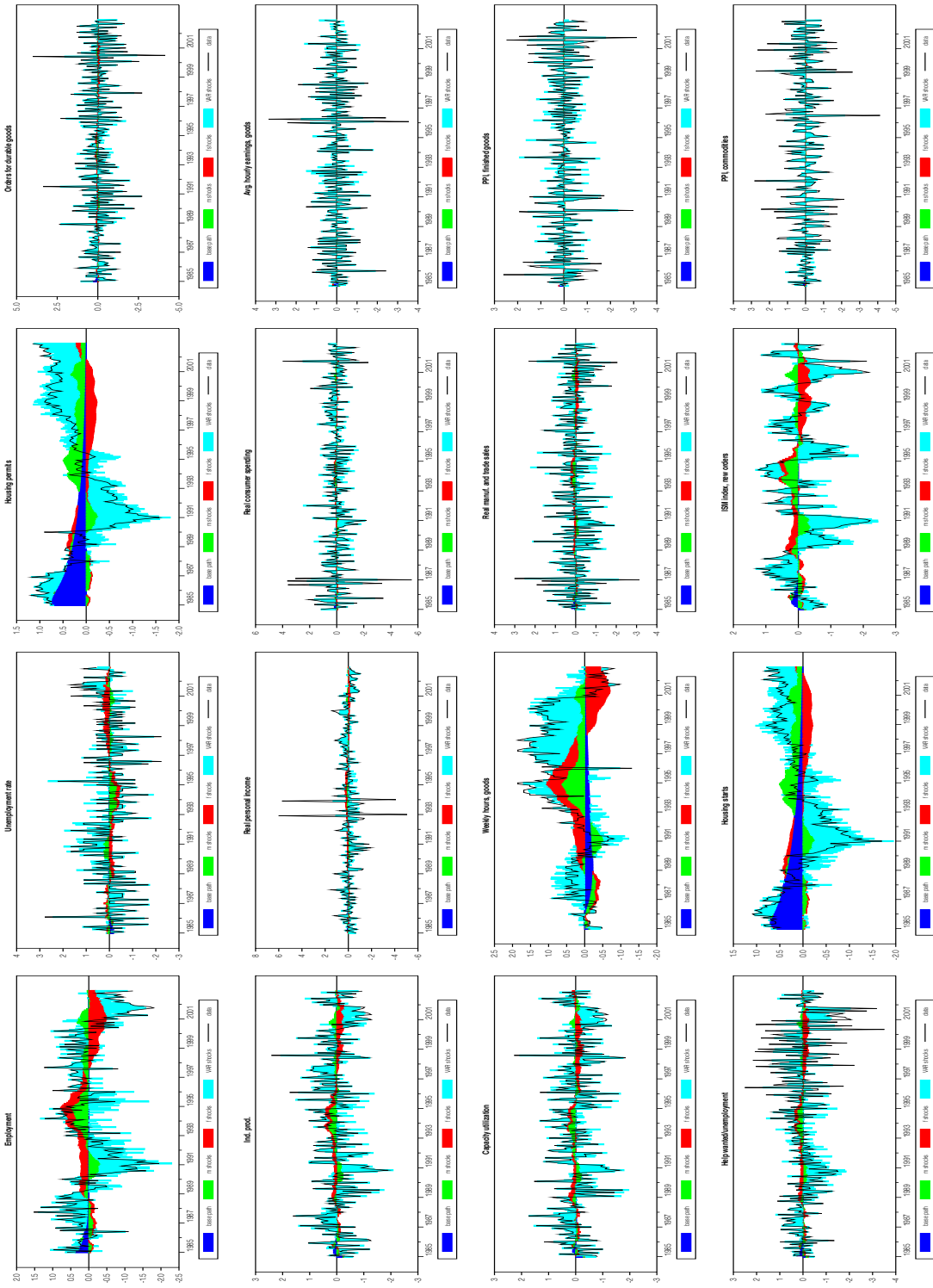


Figure A9: Baseline results with additional detail: historical decomposition, 1985-2002, all variables

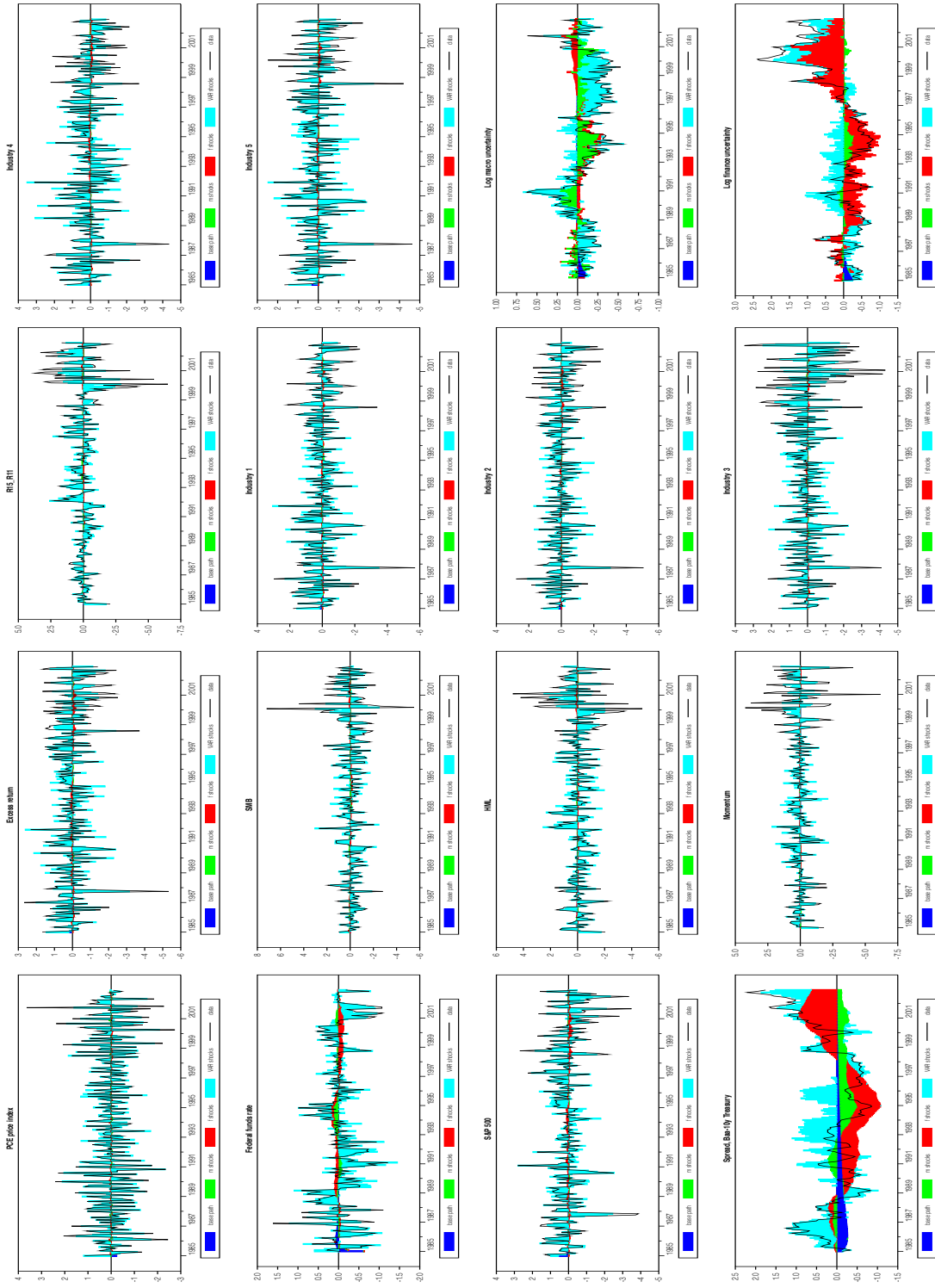


Figure A9: Continued, baseline results with additional detail: historical decomposition, 1985-2002, all variables

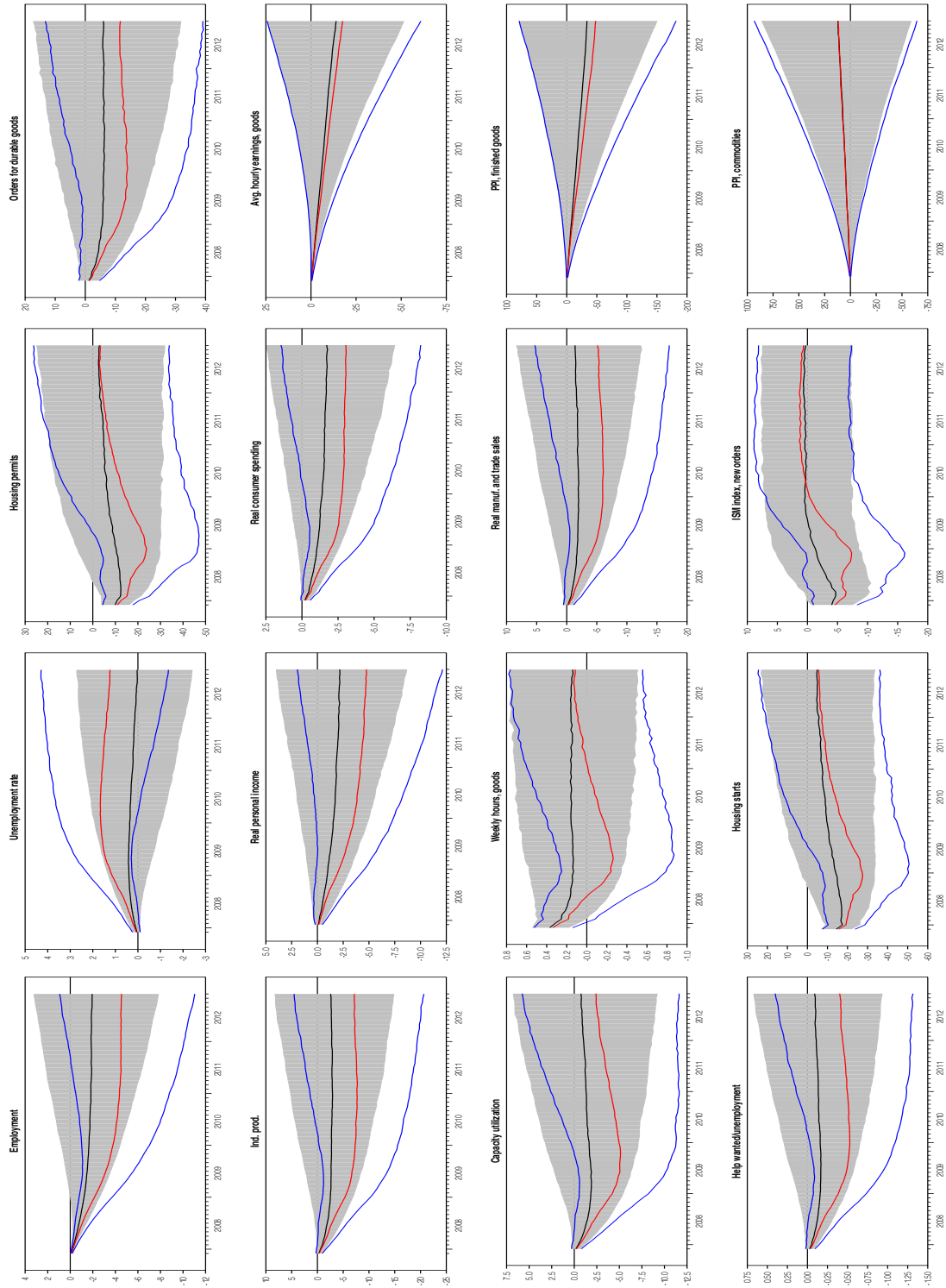


Figure A10: Baseline results with additional detail: effects of uncertainty shocks on predictive distributions, December 2007 through December 2012, all variables. The baseline path is reported as the black line (median) with gray shading (15%/85% quantiles). The path with the effects of the estimated uncertainty shocks over the period is reported as the red line (median) with blue lines (15%/85% quantiles).

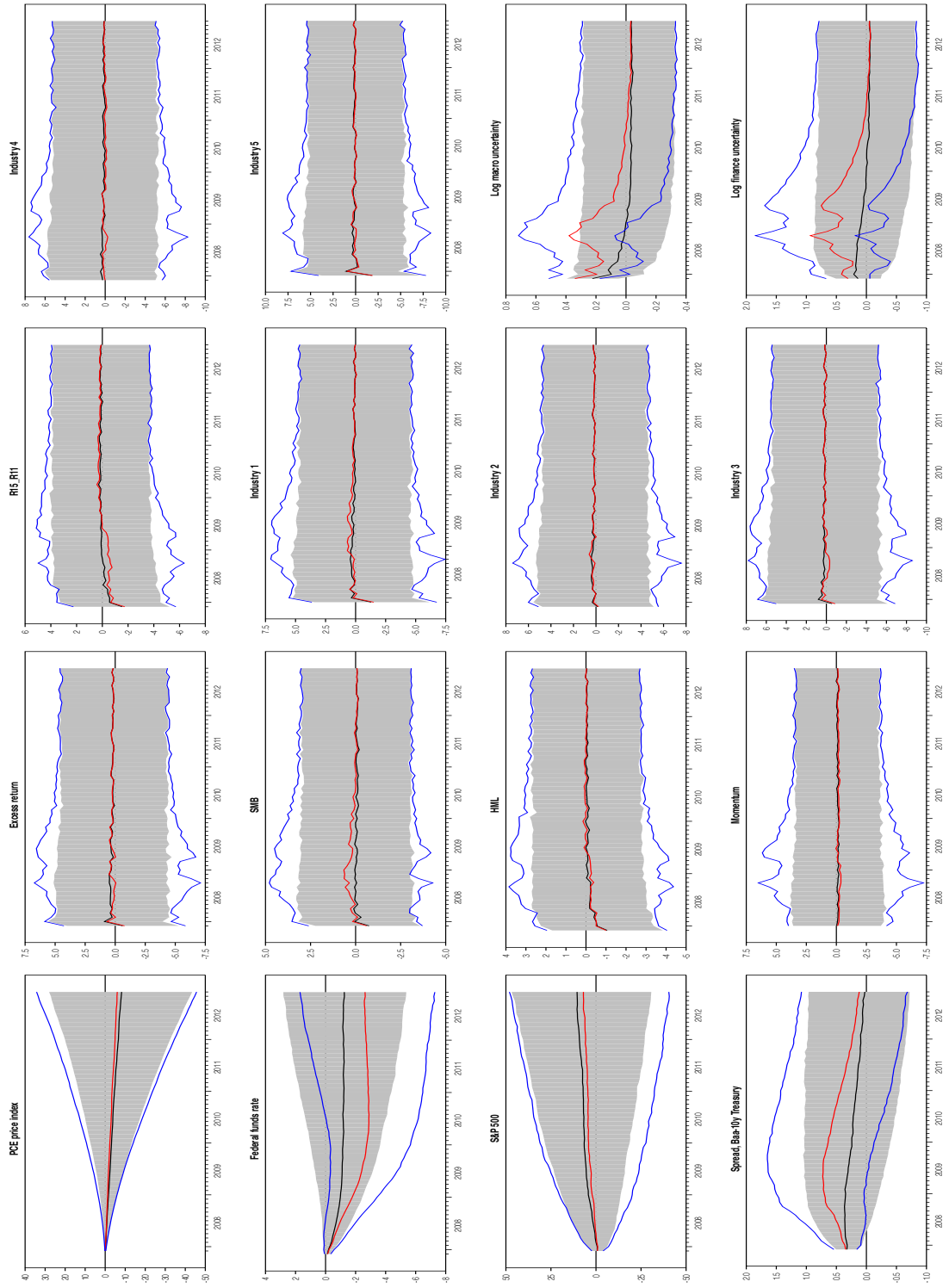


Figure A10: Continued, baseline results with additional detail: effects of uncertainty shocks on predictive distributions, December 2007 through December 2012, all variables. The baseline path is reported as the black line (median) with gray shading (15%/85% quantiles). The path with the effects of the estimated uncertainty shocks over the period is reported as the red line (median) with blue lines (15%/85% quantiles).

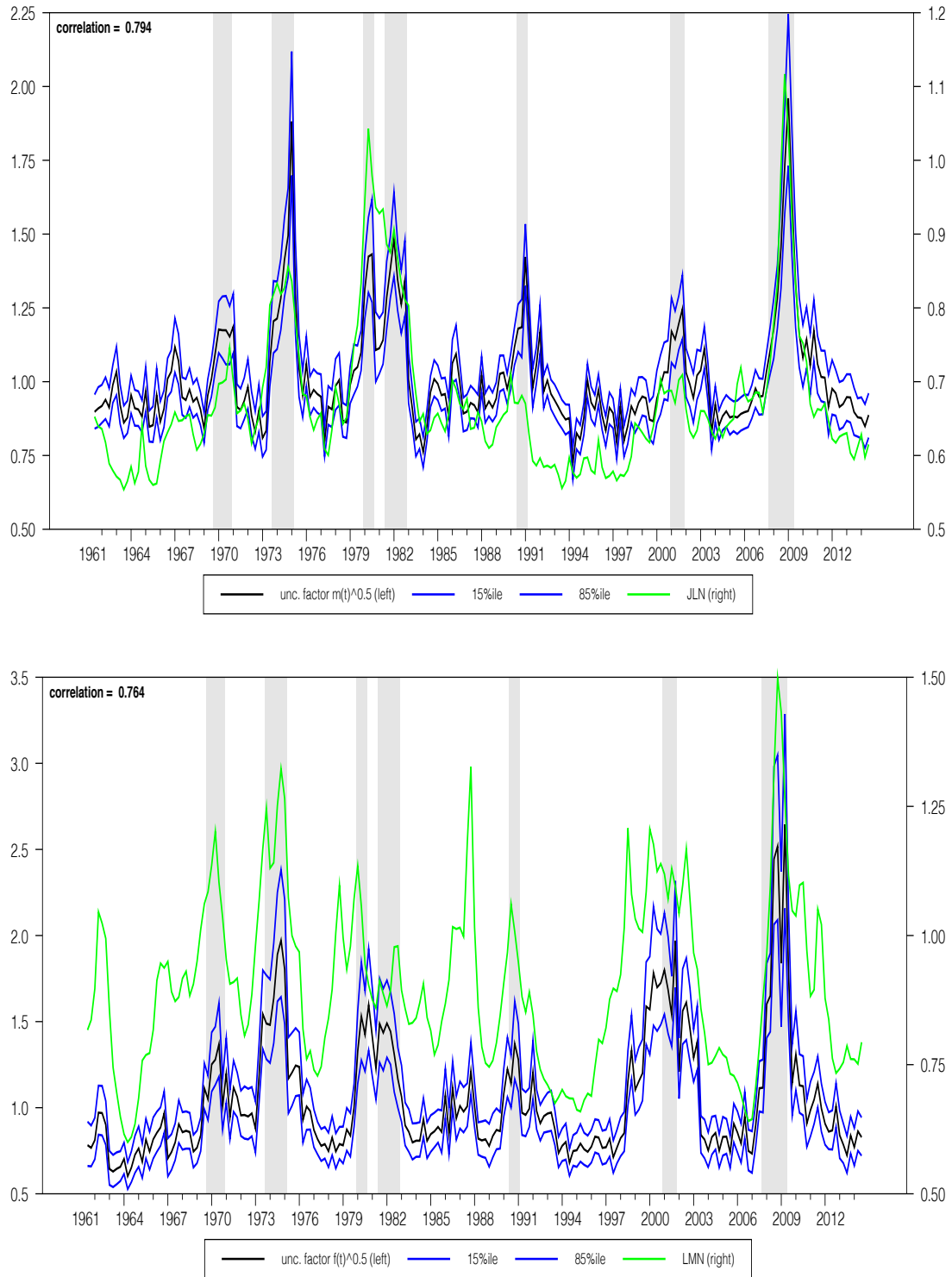


Figure A11: Results with quarterly data: uncertainty estimates. Posterior median (black line) and 15%/85% quantiles (blue lines), with macro uncertainty in the top panel and financial uncertainty in the bottom panel. The green line represents the corresponding estimates from JLN (top) and LMN (bottom). The gray bars indicate NBER recessions.

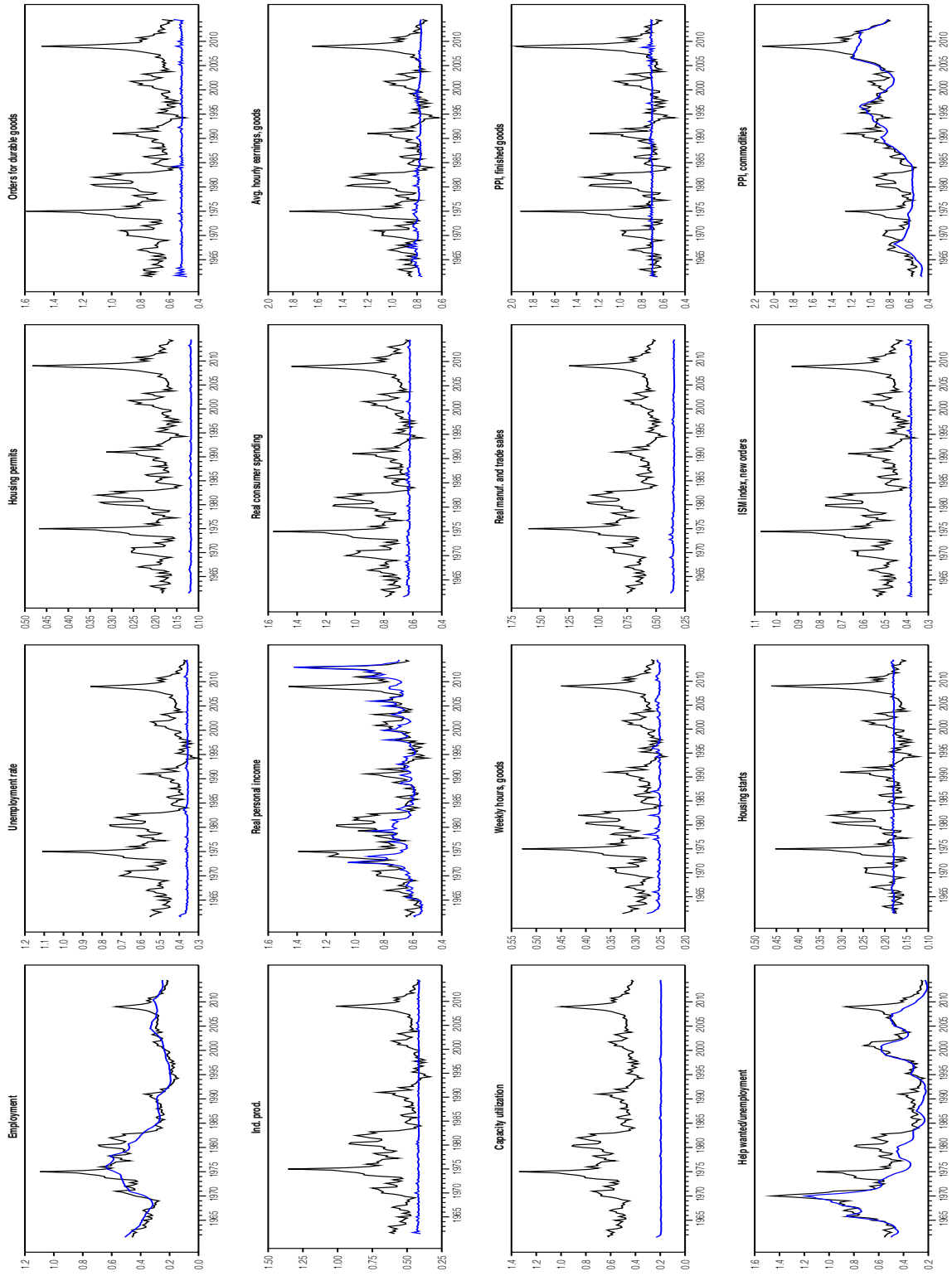


Figure A12: Results with quarterly data: reduced-form (black line) and idiosyncratic volatilities ($h_{i,t}^{0.5}$, blue line), posterior medians

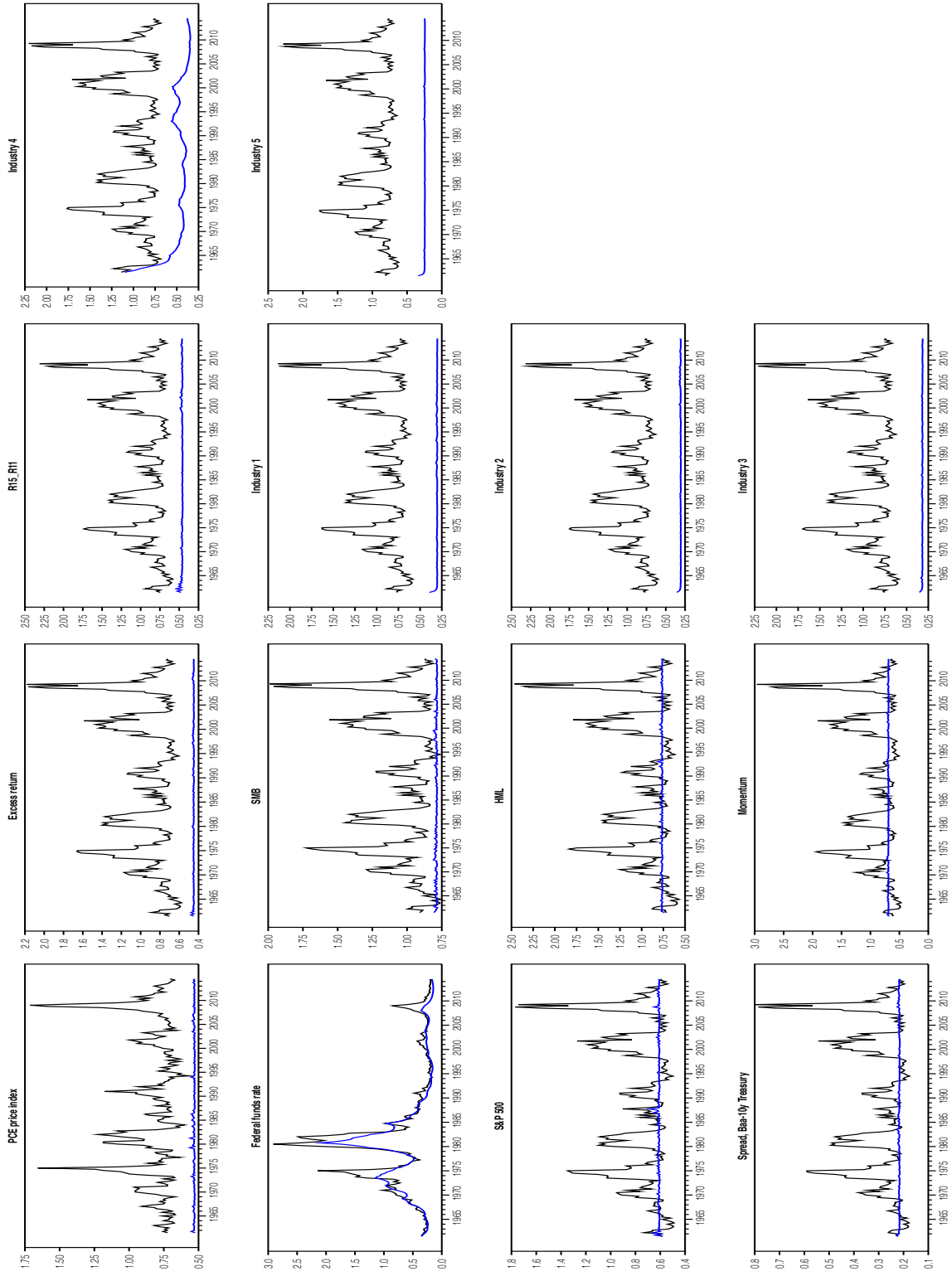


Figure A12: Continued, results with quarterly data: reduced-form (black line) and idiosyncratic volatilities ($h_{i,t}^{0.5}$, blue line), posterior medians

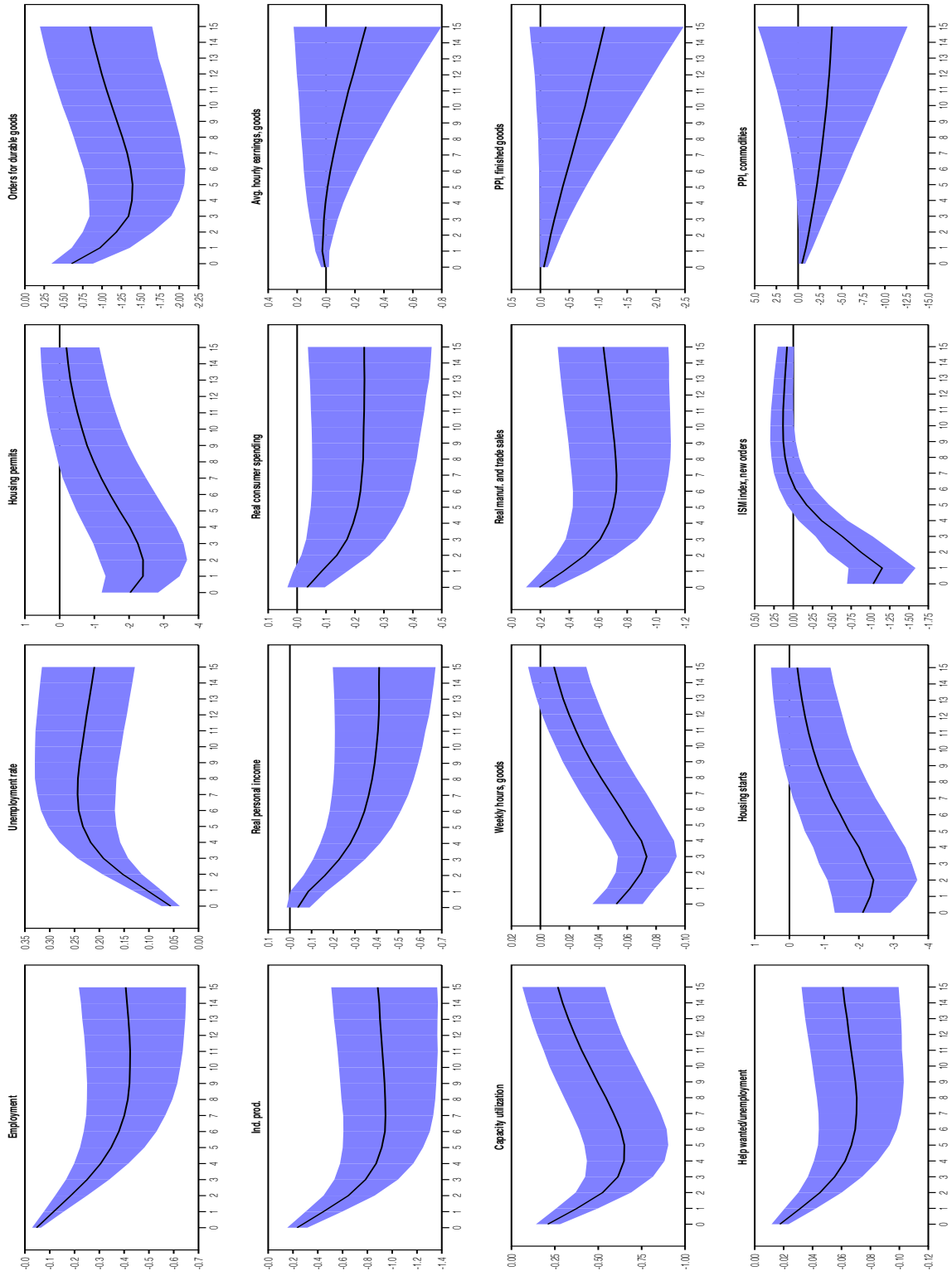


Figure A13: Results with quarterly data: impulse responses for shock to macro uncertainty. One standard deviation shock, posterior median (black line) and 15%/85% quantiles (blue shading)

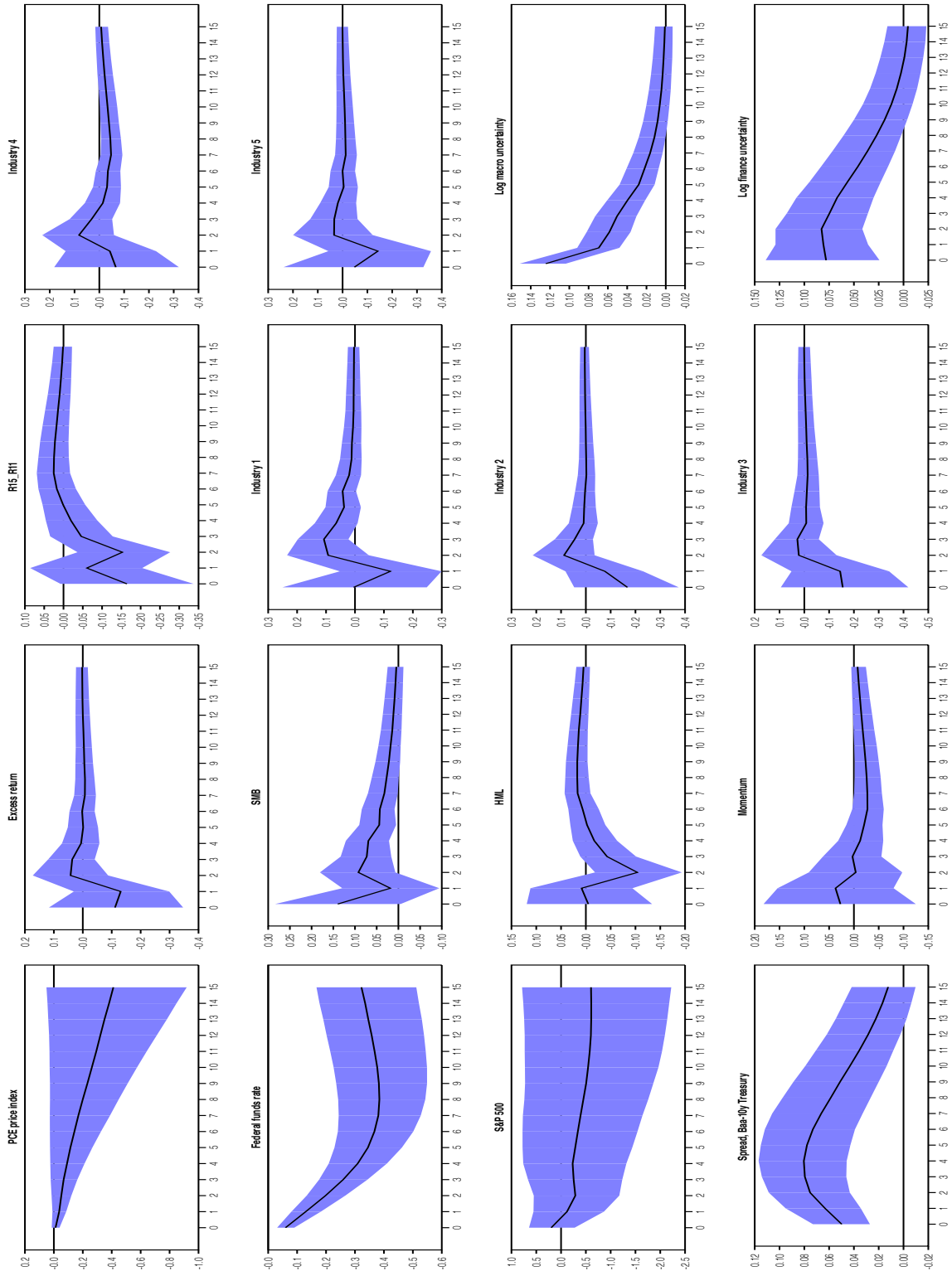


Figure A13: Continued, results with quarterly data: impulse responses for shock to macro uncertainty. One standard deviation shock, posterior median (black line) and 15%/85% quantiles

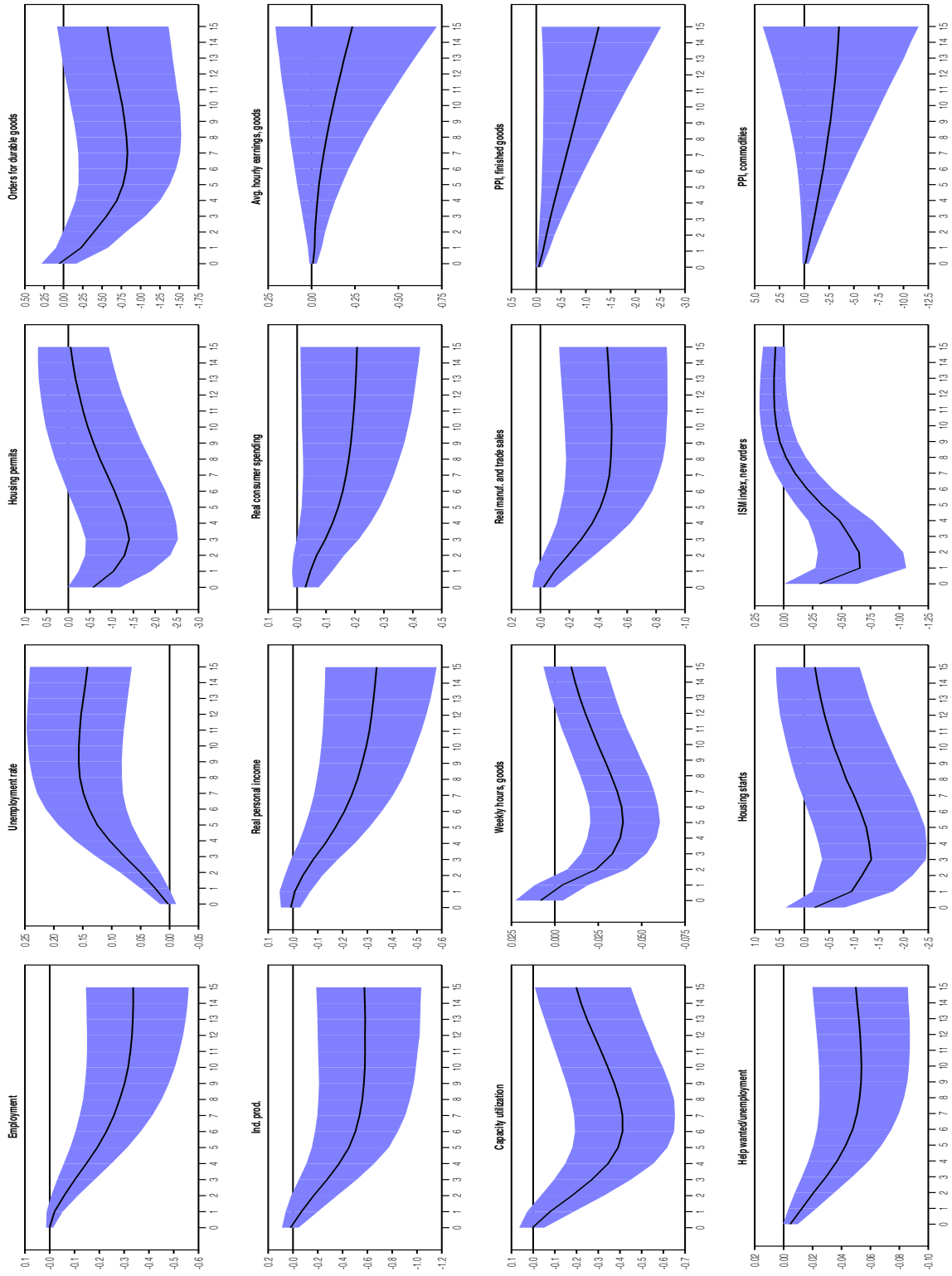


Figure A14: Results with quarterly data: impulse responses for shock to financial uncertainty. One standard deviation shock, posterior median (black line) and 15%/85% quantiles (blue shading)

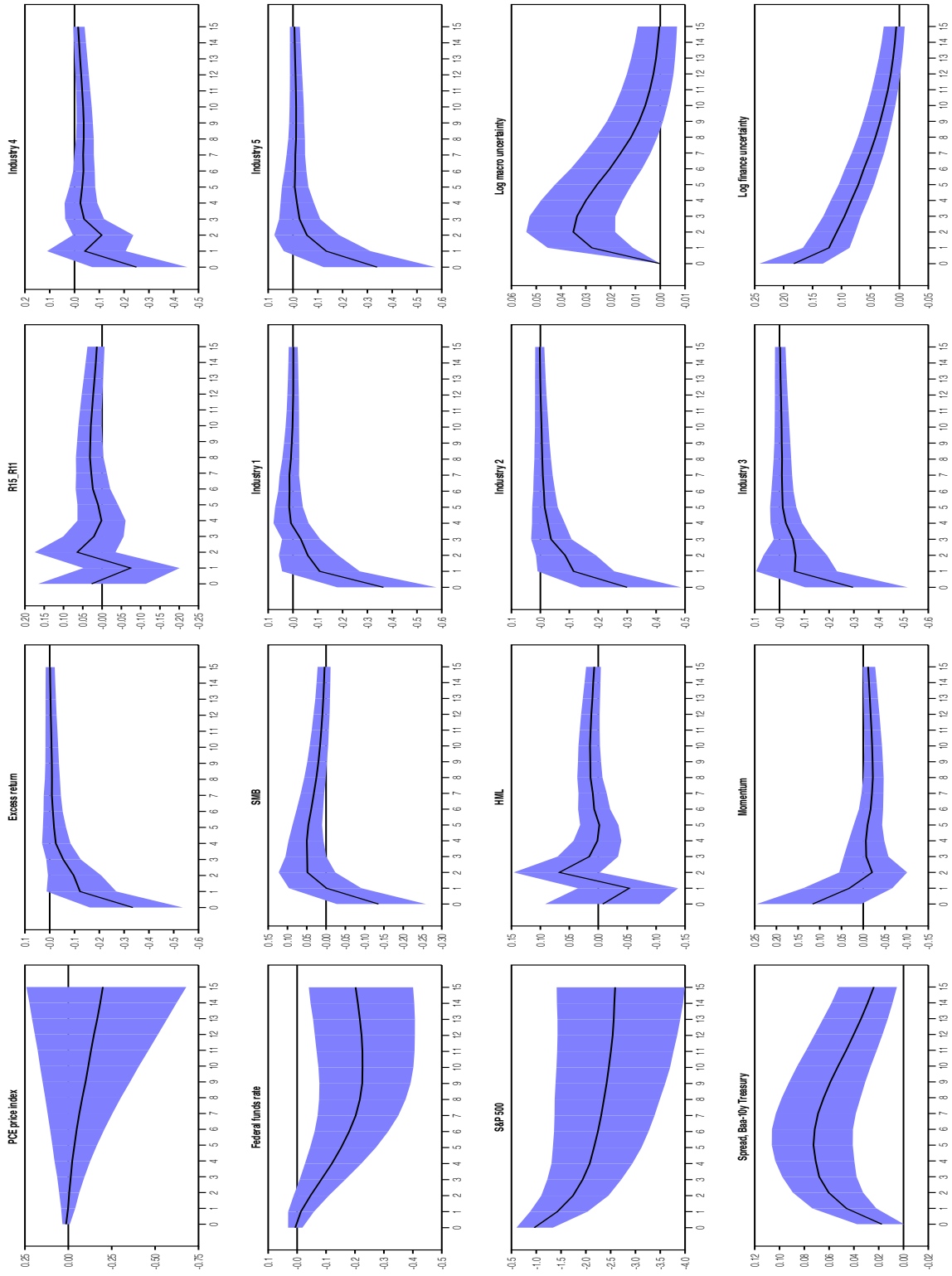


Figure A14: Continued, results with quarterly data: impulse responses for shock to financial uncertainty. One standard deviation shock, posterior median (black line) and 15%/85% quantiles

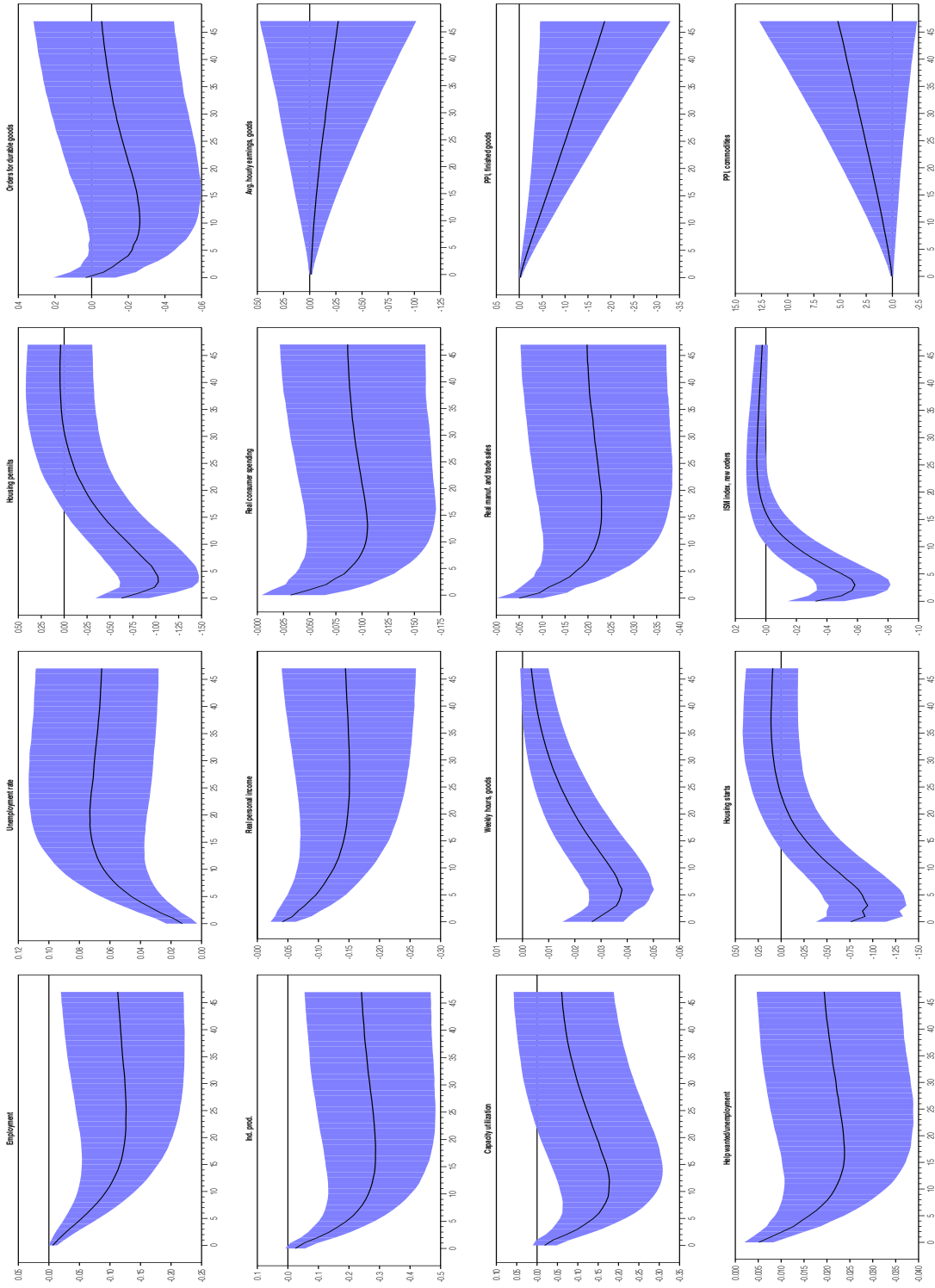


Figure A15: Results with sample ending in 2007: impulse responses for shock to macro uncertainty. One standard deviation shock, posterior median (black line) and 15%/85% quantiles (blue shading)

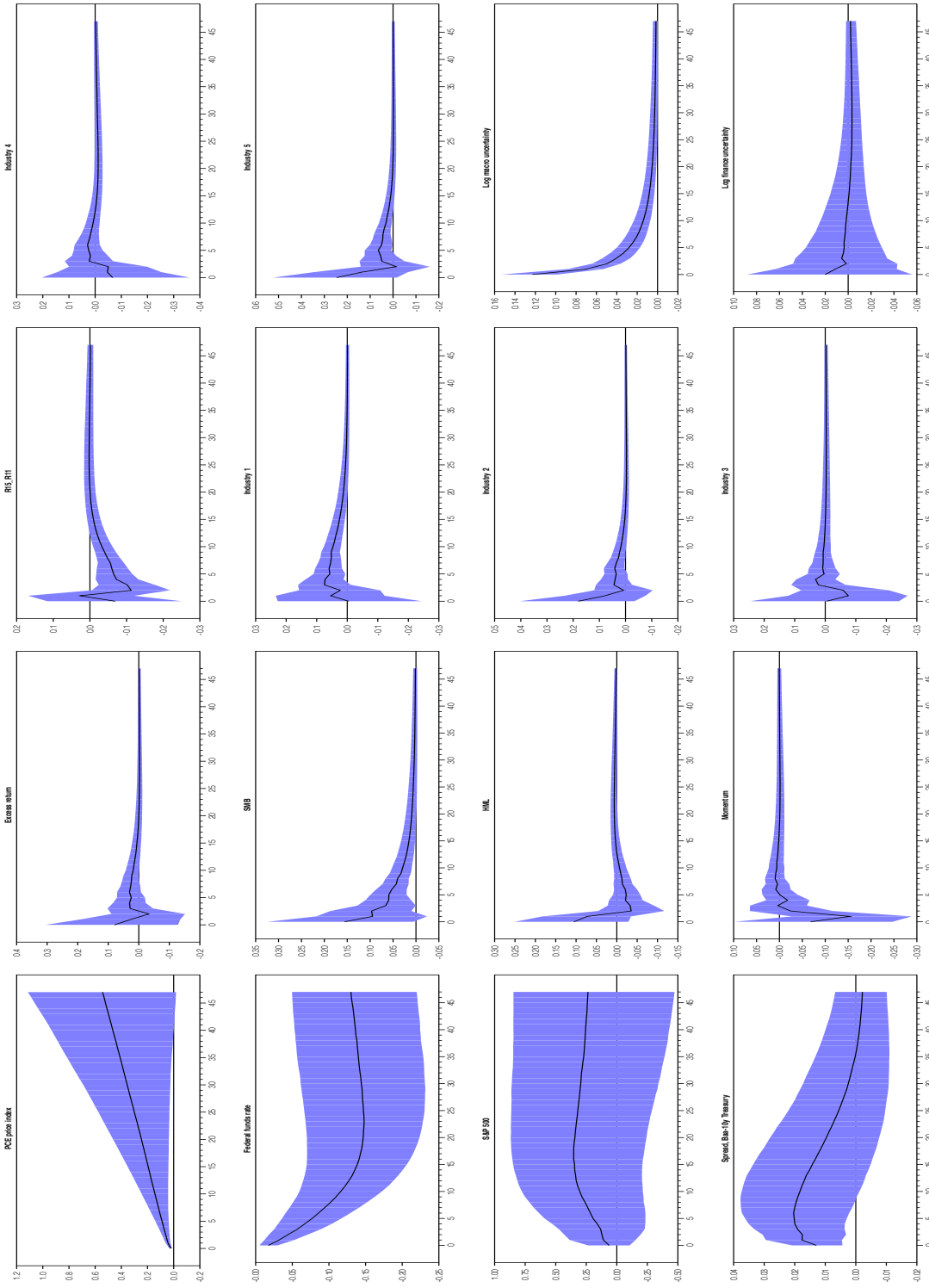


Figure A15: Continued, results with sample ending in 2007: impulse responses for shock to macro uncertainty. One standard deviation shock, posterior median (black line) and 15%/85% quantiles

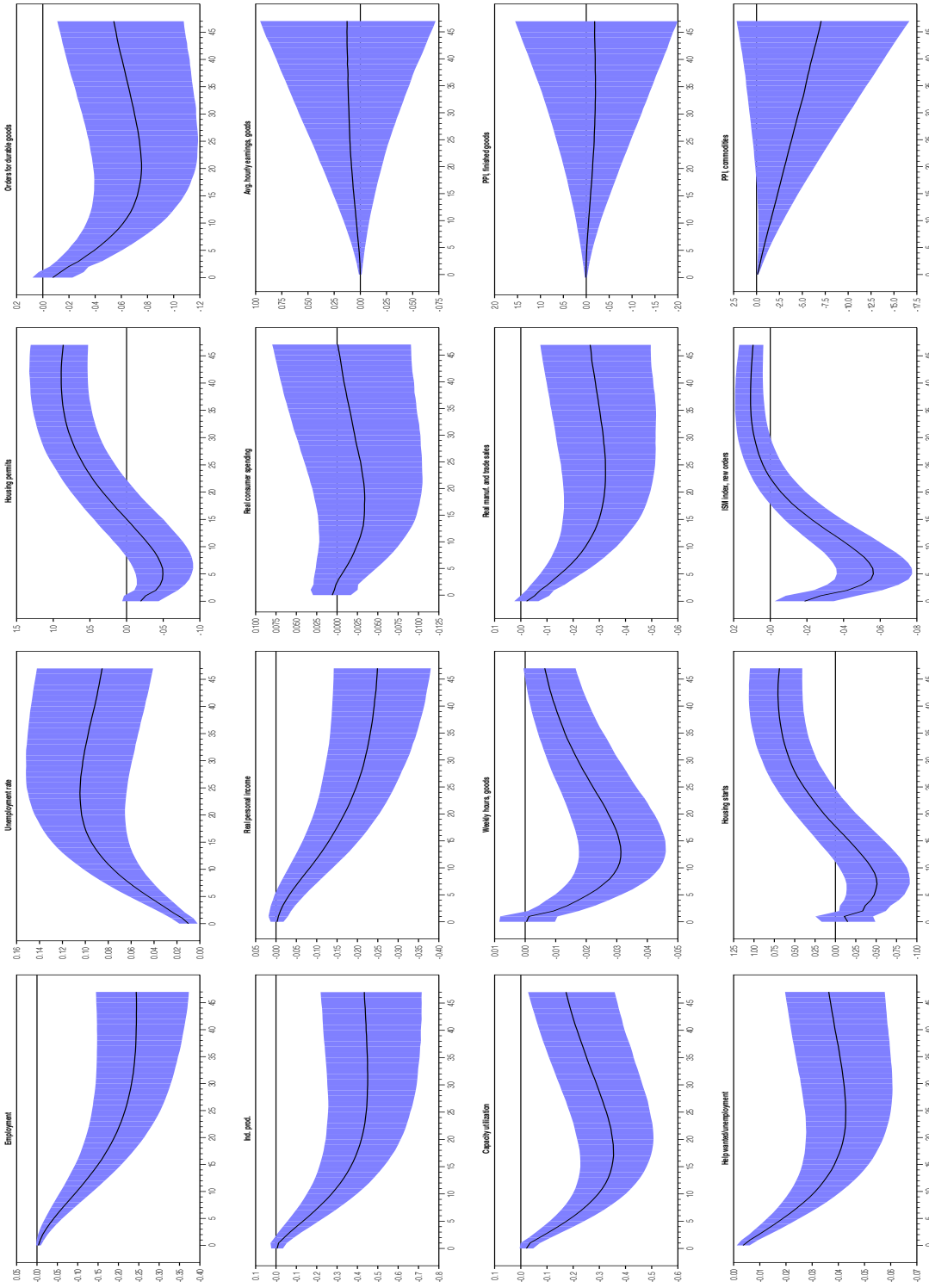


Figure A16: Results with sample ending in 2007: impulse responses for shock to financial uncertainty. One standard deviation shock, posterior median (black line) and 15%/85% quantiles (blue shading)

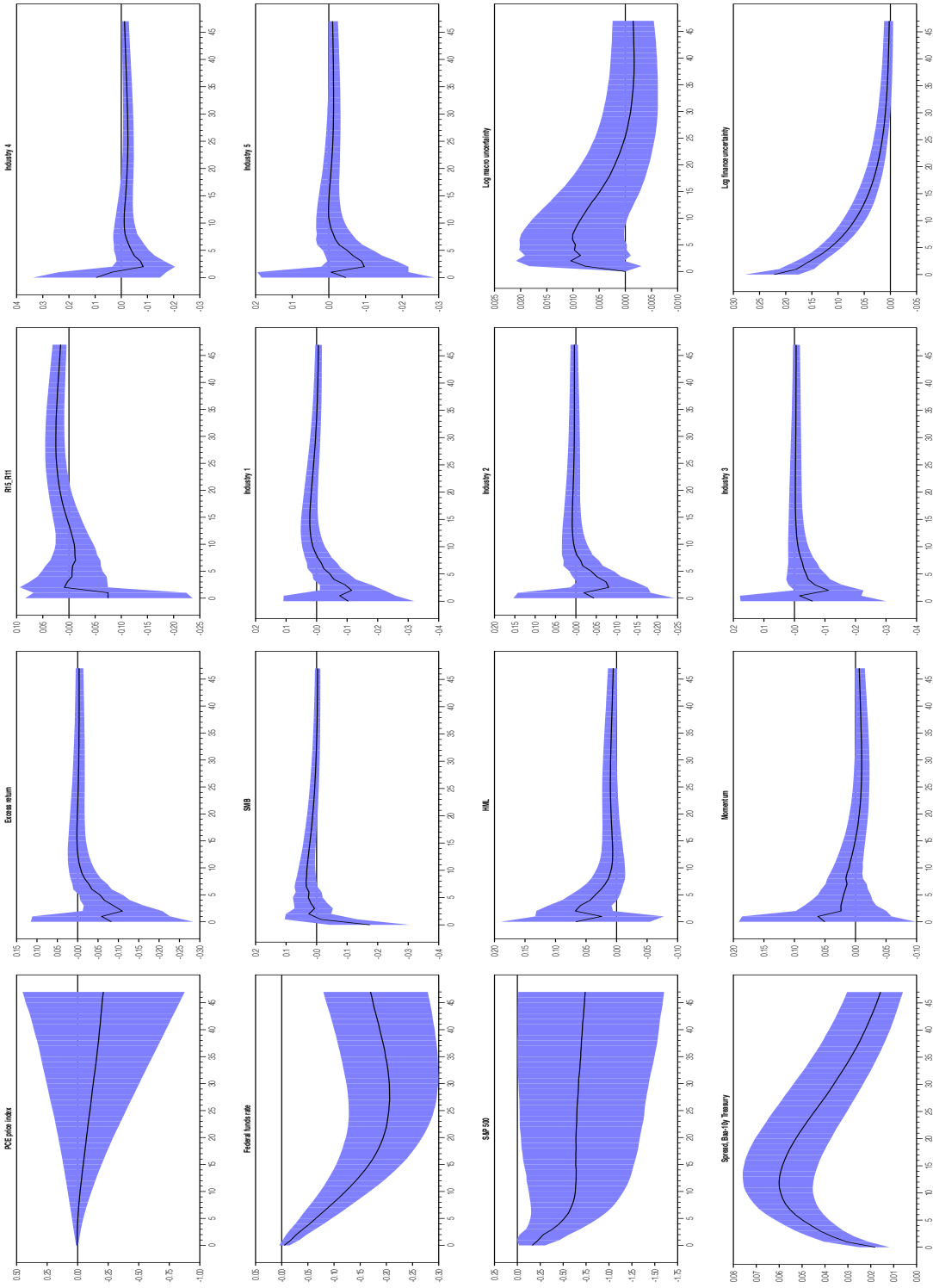


Figure A16: Continued, results with sample ending in 2007: impulse responses for shock to financial uncertainty. One standard deviation shock, posterior median (black line) and 15%/85% (blue shading)

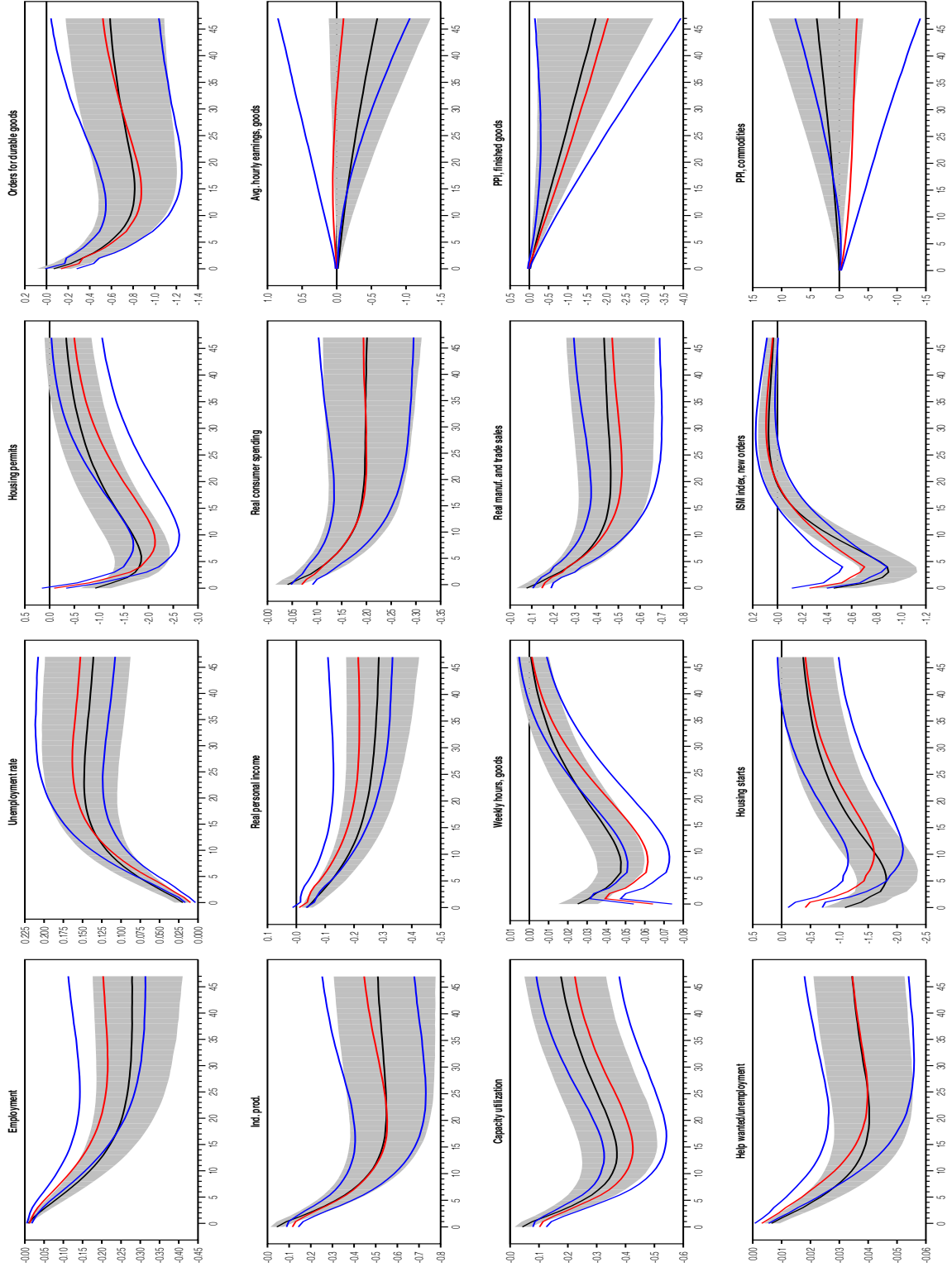


Figure A17: Comparison to two-step approach: impulse responses from shock to macroeconomic uncertainty. Baseline model estimates: black line (median) with gray fans (15%/85% quantiles). Alternative estimates, from two-step approach, with uncertainty defined as average of log volatilities, ordered first in VAR: red line (median) with blue lines (15%/85% quantiles).

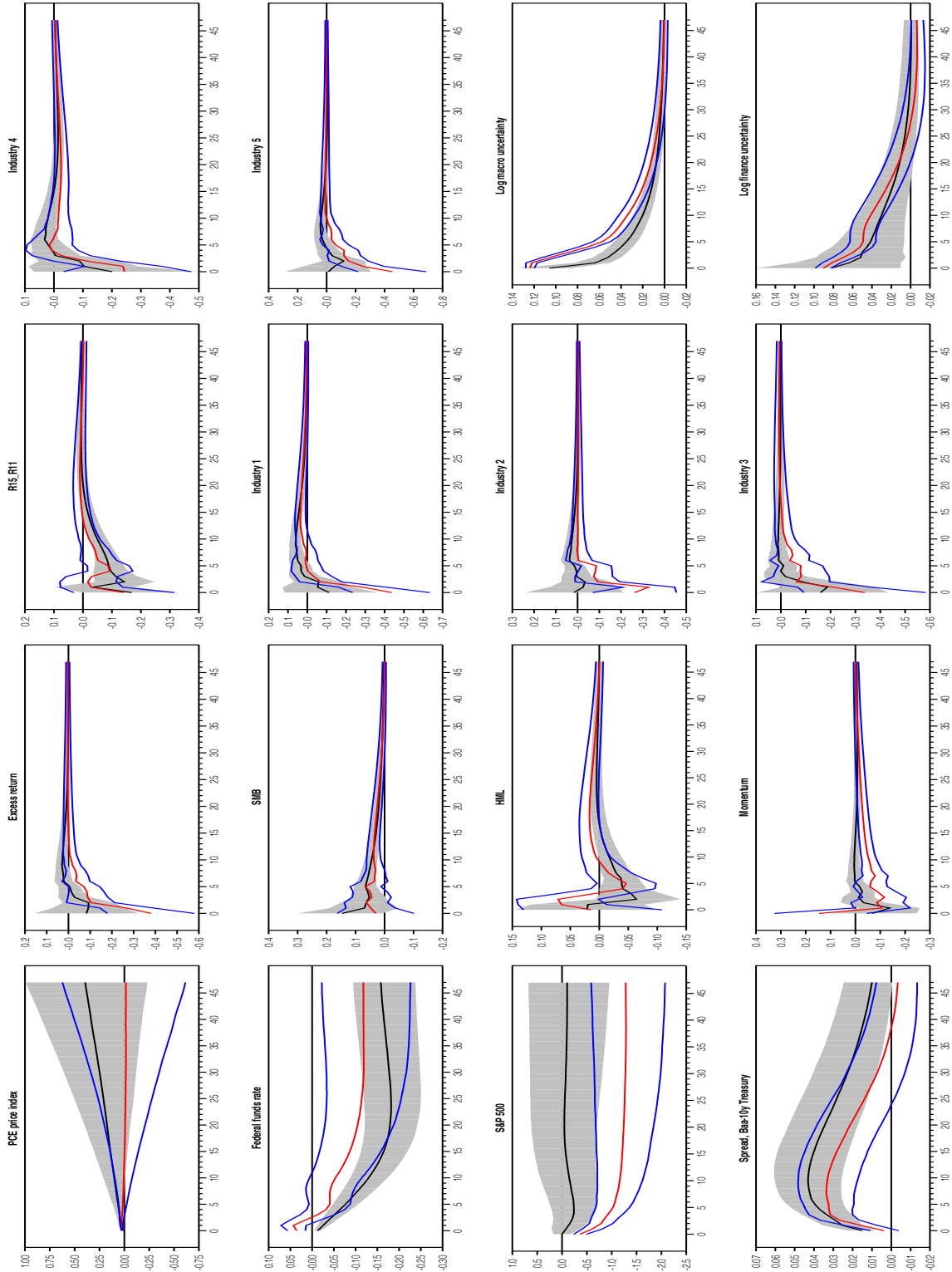


Figure A17: Continued, comparison to two-step approach: impulse responses from shock to macroeconomic uncertainty. Baseline model estimates: black line (median) with gray fans (15%/85% quantiles). Alternative estimates, from two-step approach, with uncertainty defined as average of log volatilities, ordered first in VAR: red line (median) with blue lines (15%/85% quantiles).

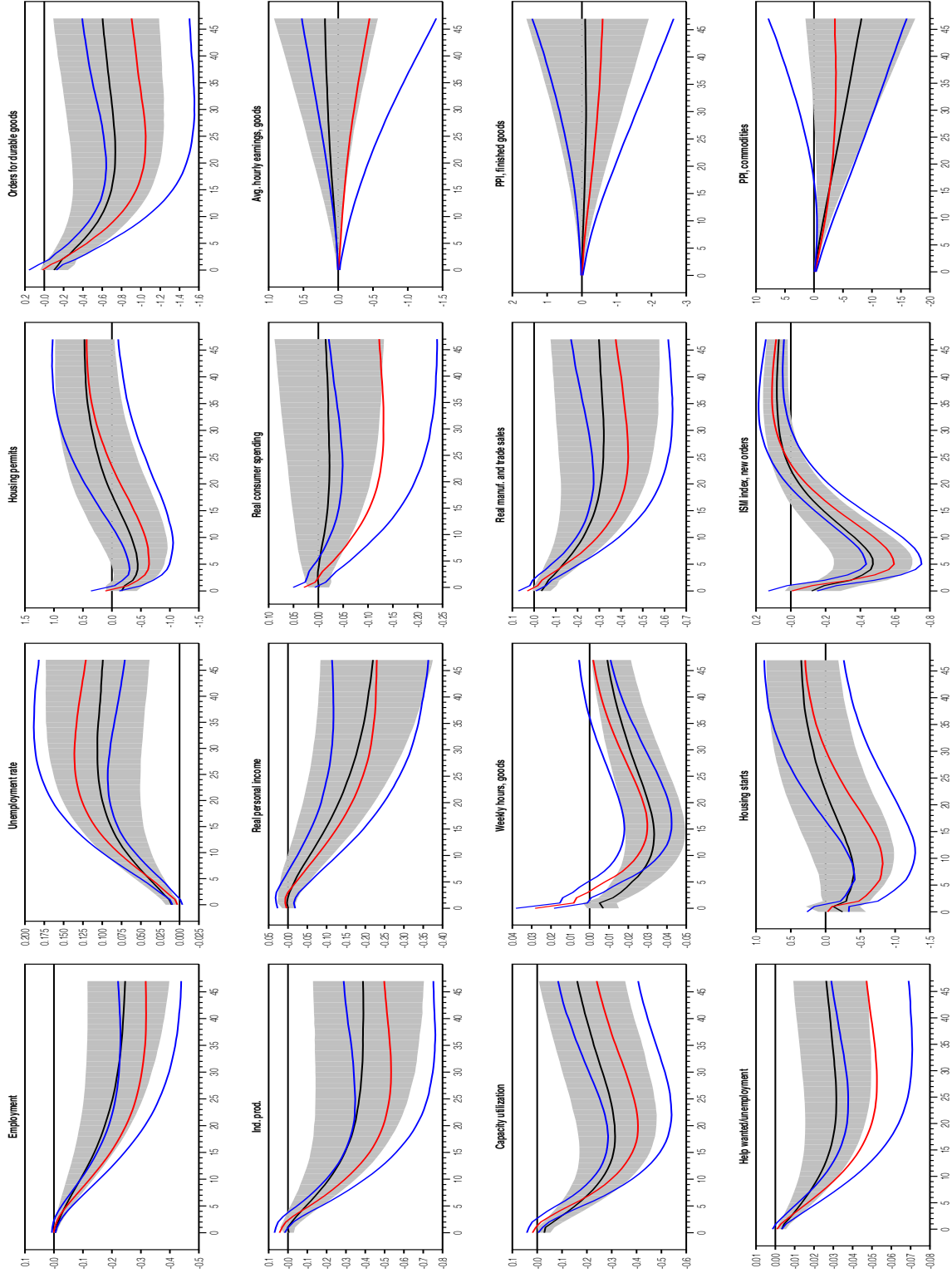


Figure A18: Comparison to two-step approach: impulse responses from shock to financial uncertainty. Baseline model estimates: black line (median) with gray fans (15%/85% quantiles). Alternative estimates, from two-step approach, with uncertainty defined as average of log volatilities, ordered first in VAR: red line (median) with blue lines (15%/85% quantiles).

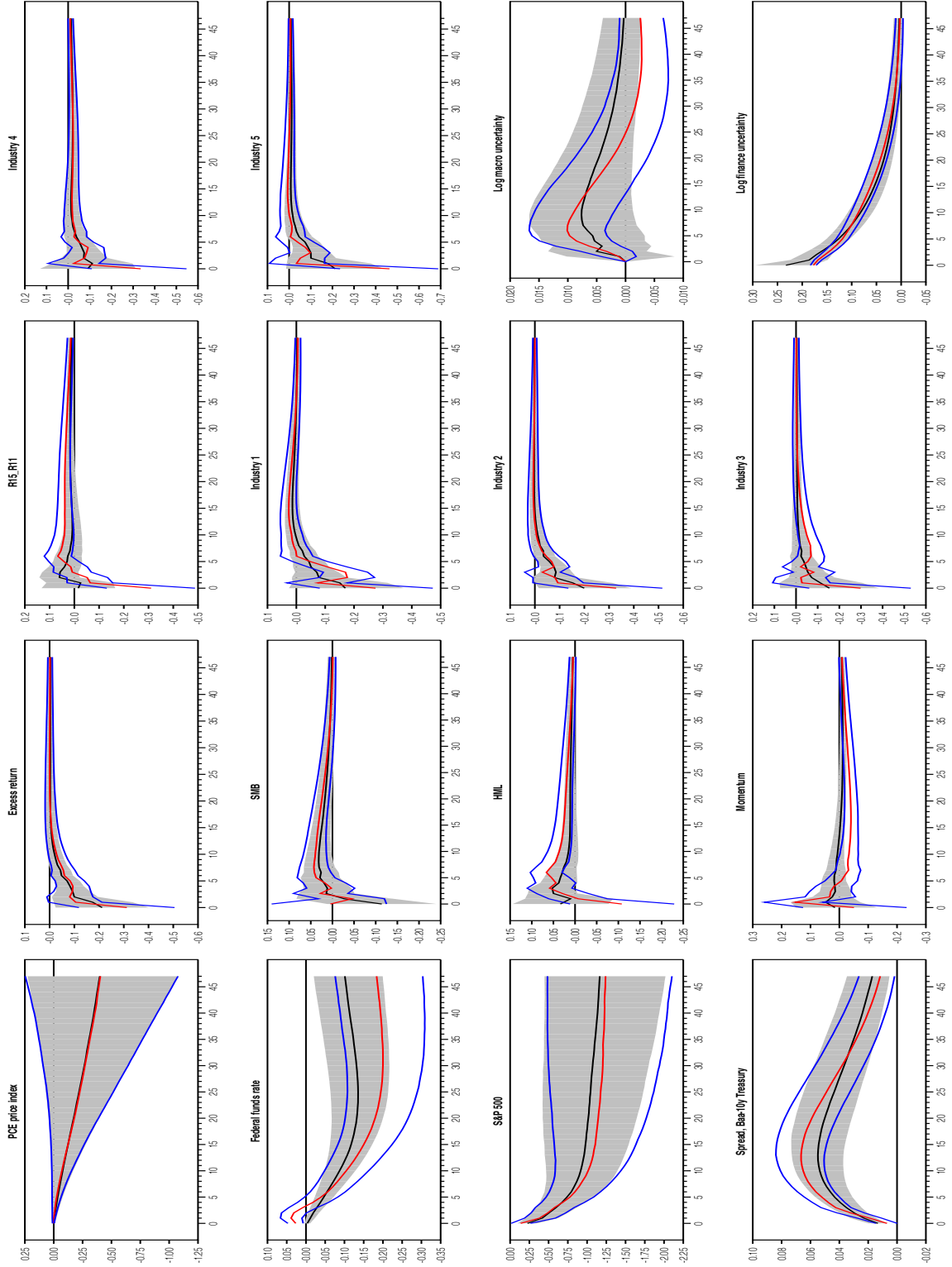


Figure A18: Continued, comparison to two-step approach: impulse responses from shock to financial uncertainty. Baseline model estimates: black line (median) with gray fans (15%/85% quantiles). Alternative estimates, from two-step approach, with uncertainty defined as average of log volatilities, ordered first in VAR: red line (median) with blue lines (15%/85% quantiles).

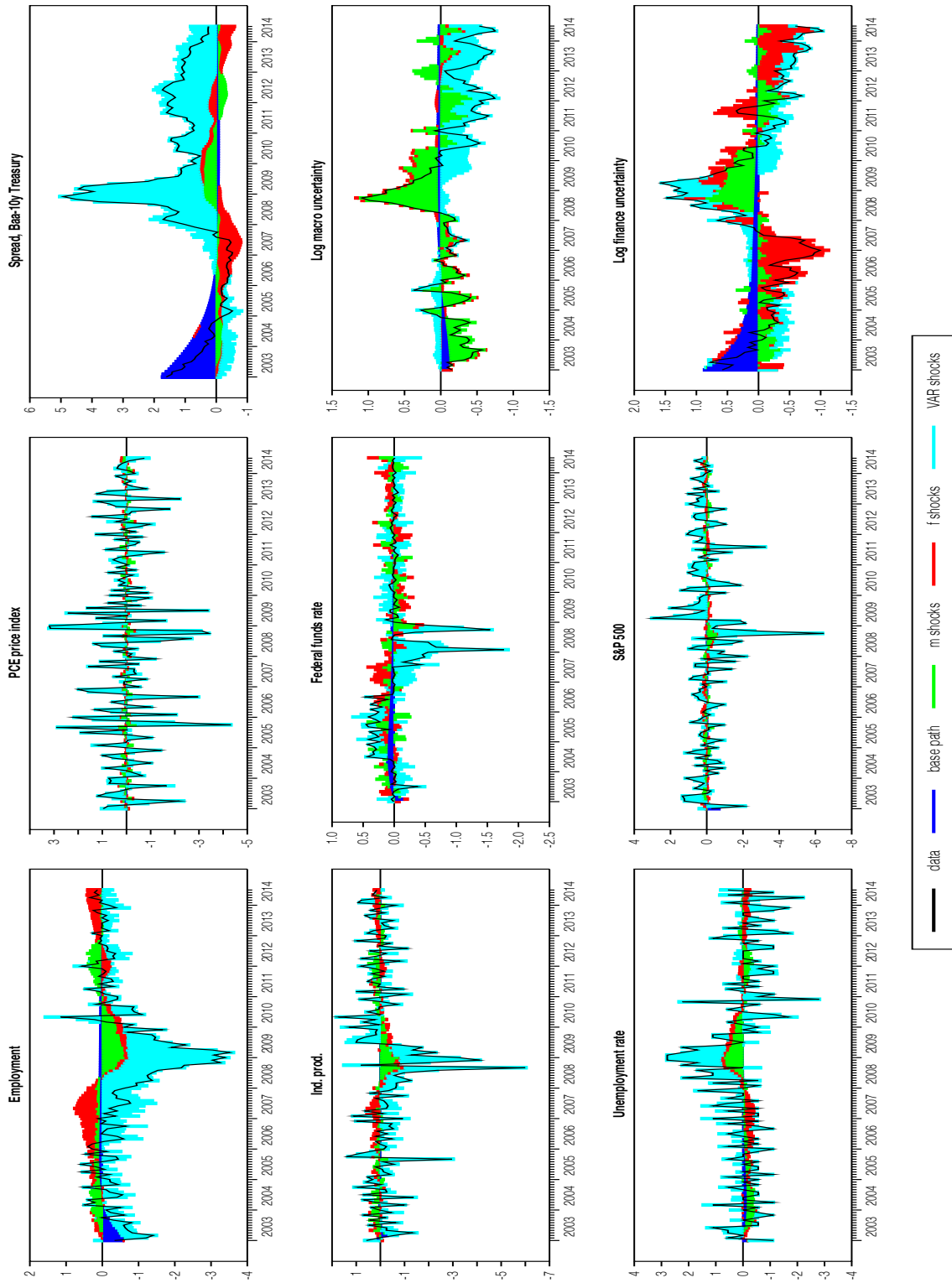


Figure A19: Comparison to two-step approach: historical decomposition, 2003-2014, obtained from two-step approach, with uncertainty defined as average of log volatilities, ordered first in VAR.

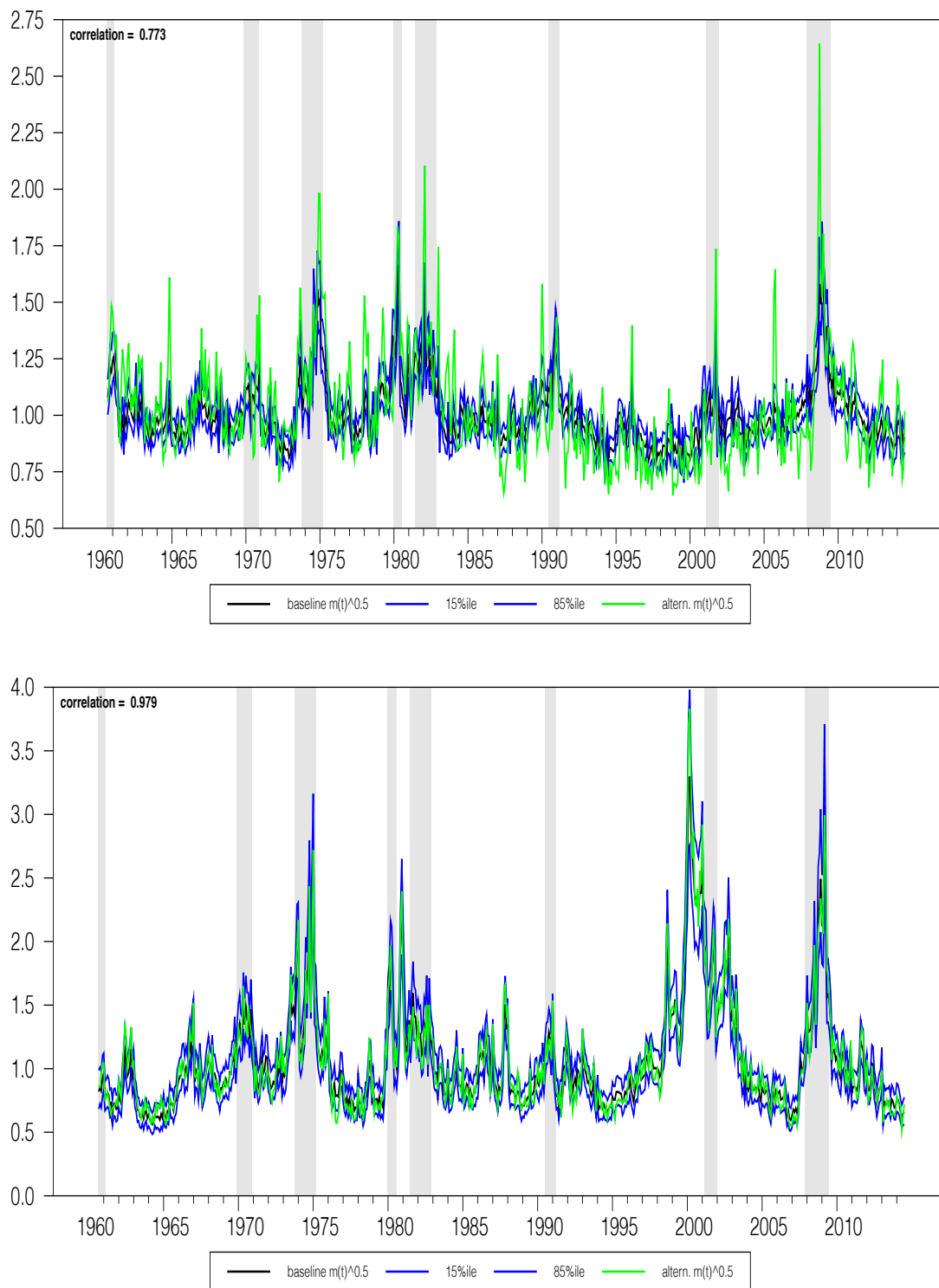


Figure A20: Uncertainty estimates from model without uncertainty in the VAR's conditional mean: posterior median (black line) and 15%/85% quantiles (blue lines), with macro uncertainty in the top panel and financial uncertainty in the bottom panel. The green line represents the corresponding posterior median estimate from a version of the model restricted such that uncertainty does not enter the conditional mean of the VAR. The gray bars indicate NBER recessions.

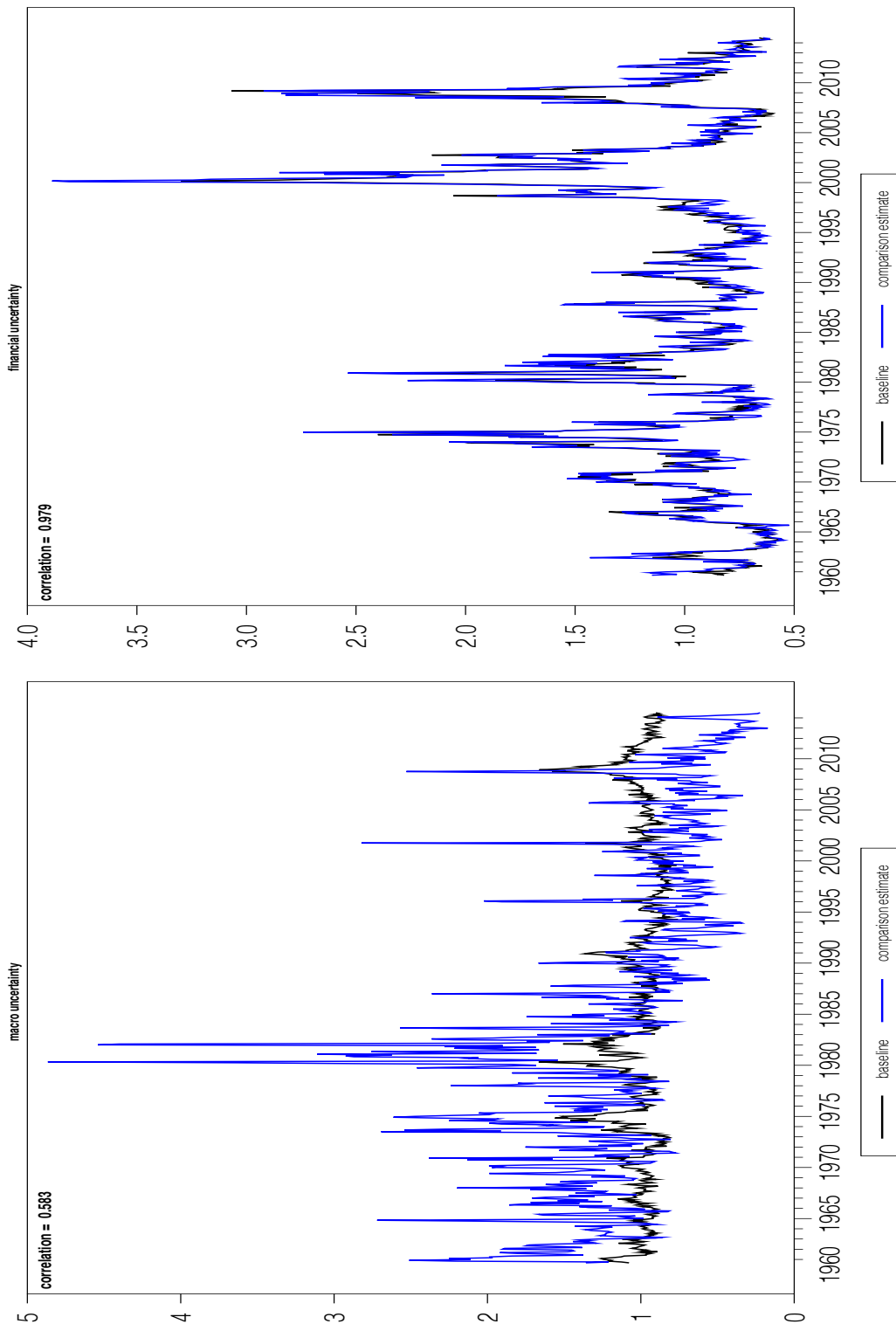


Figure A21: Uncertainty estimates from model without time variation in idiosyncratic components of volatility: baseline GFSV estimates versus estimates obtained from model in which idiosyncratic components in volatility are restricted to be constants.

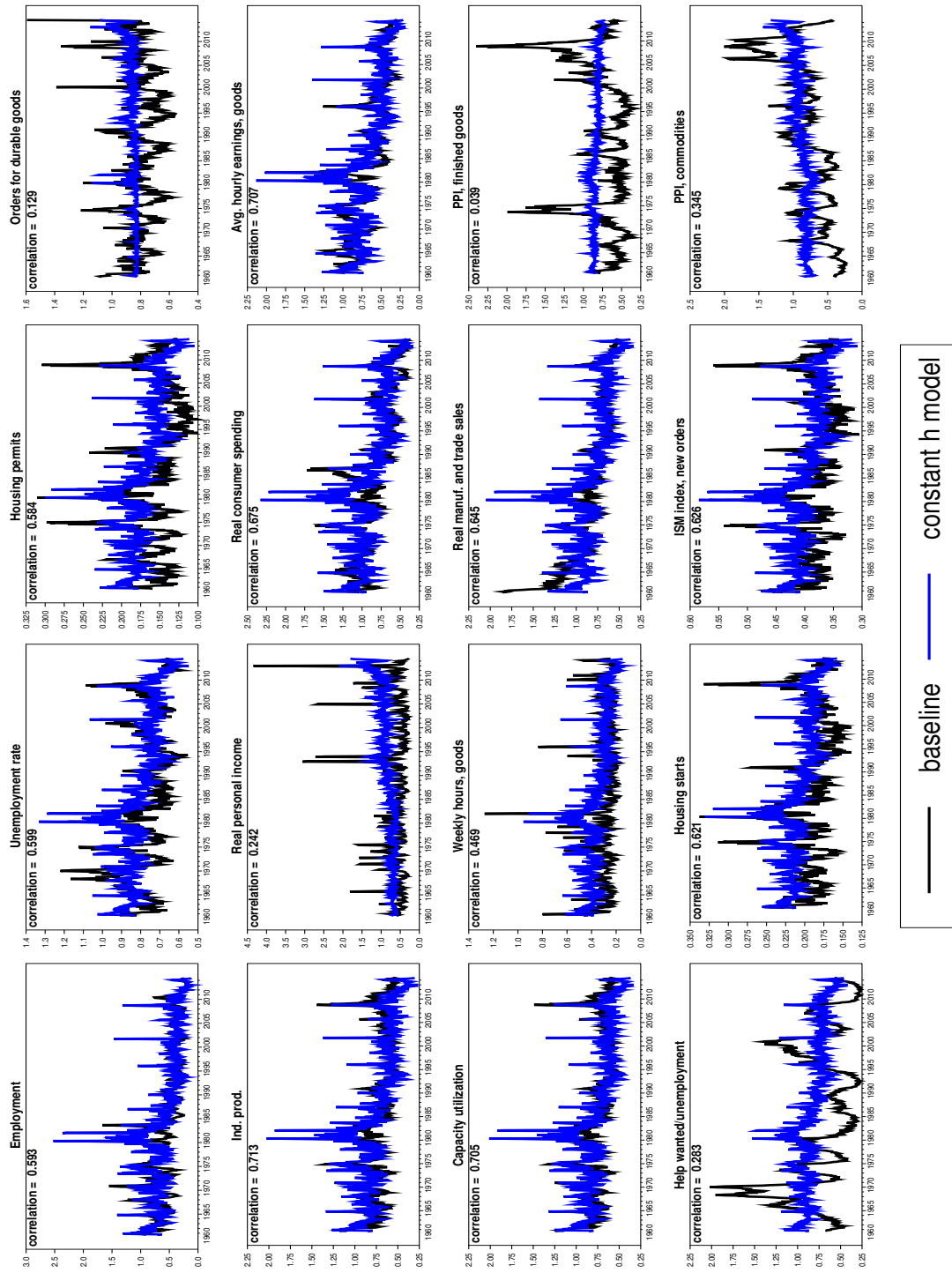


Figure A22: Reduced-form volatility estimates from model without time variation in idiosyncratic components of volatility (blue line) versus baseline estimates (black line)

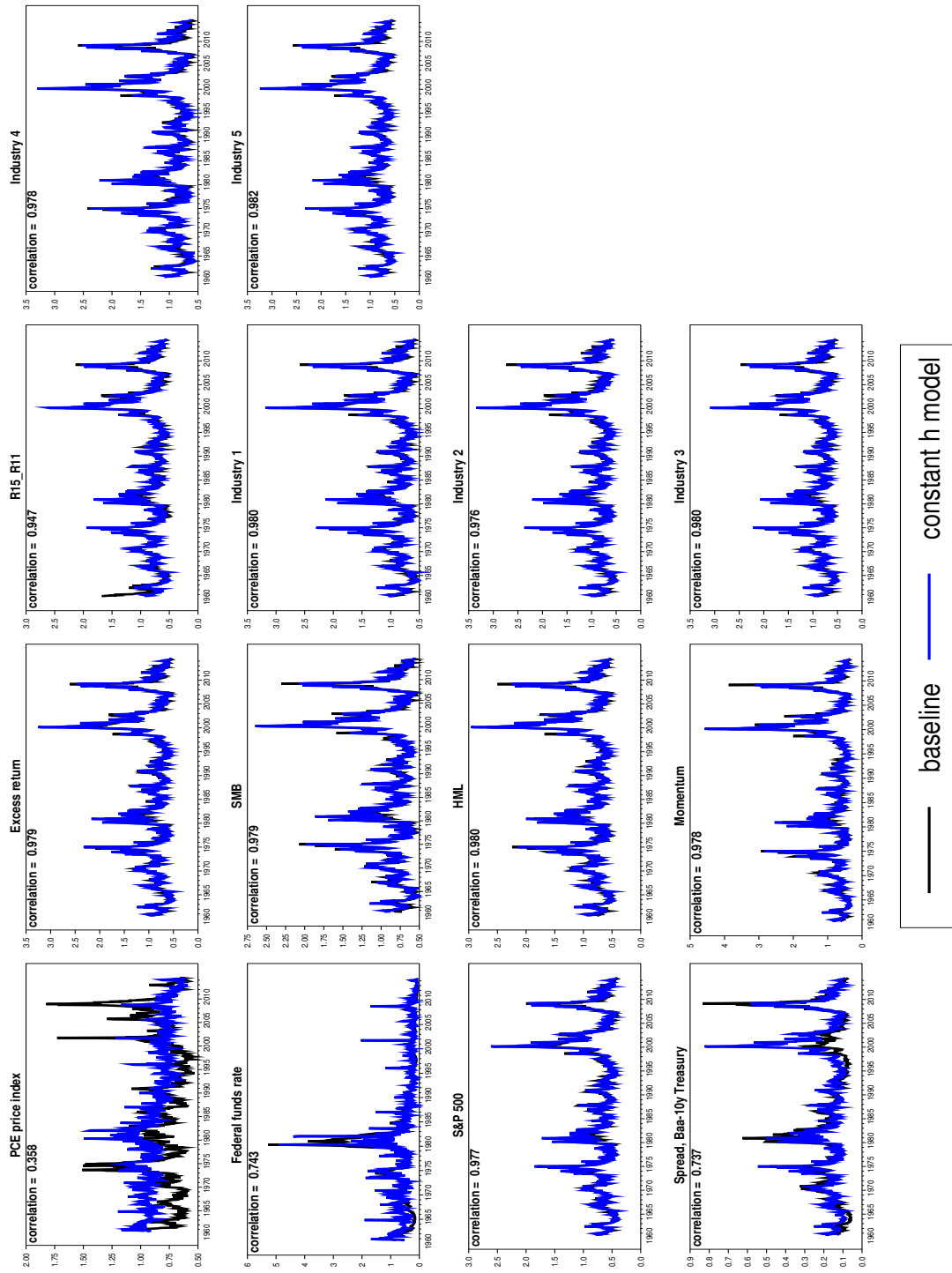


Figure A22: Reduced-form volatility estimates from model without time variation in idiosyncratic components of volatility (blue line) versus baseline estimates (black line)

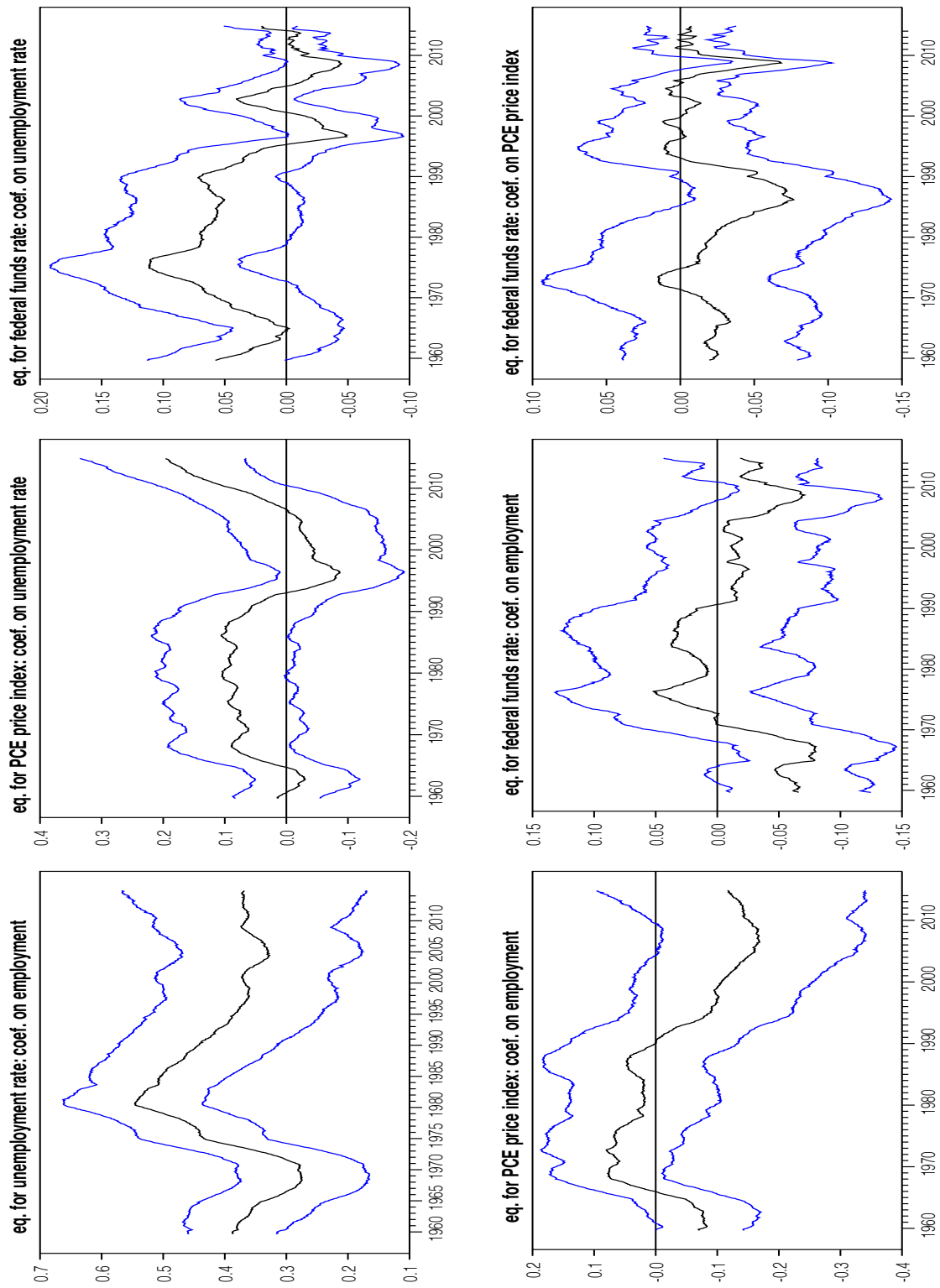


Figure A23: Estimates of a 4-variable model with one macroeconomic uncertainty factor: time-varying A coefficients, posterior median (black line) and 15%/85% quantiles (blue lines).

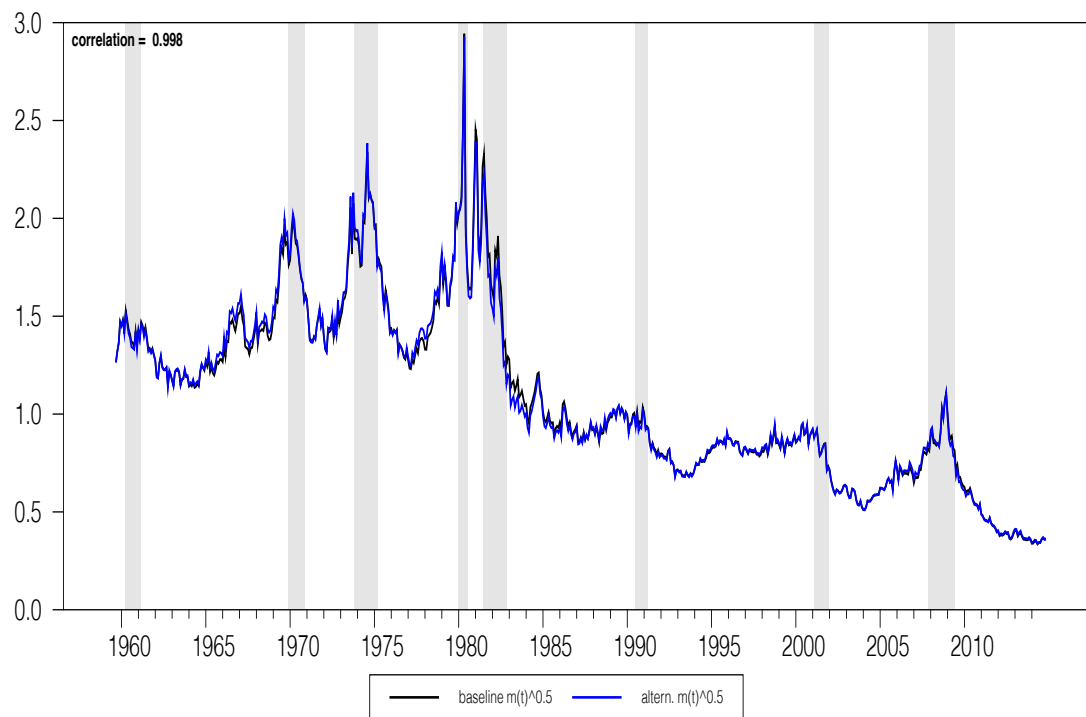


Figure A24: Estimates of a 4-variable model with one macroeconomic uncertainty factor: uncertainty factor. The black line is the posterior median of a constant parameter model. The blue line is the posterior median from a corresponding model augmented to allow the A matrix to be time-varying as in studies such as Primiceri (2005).

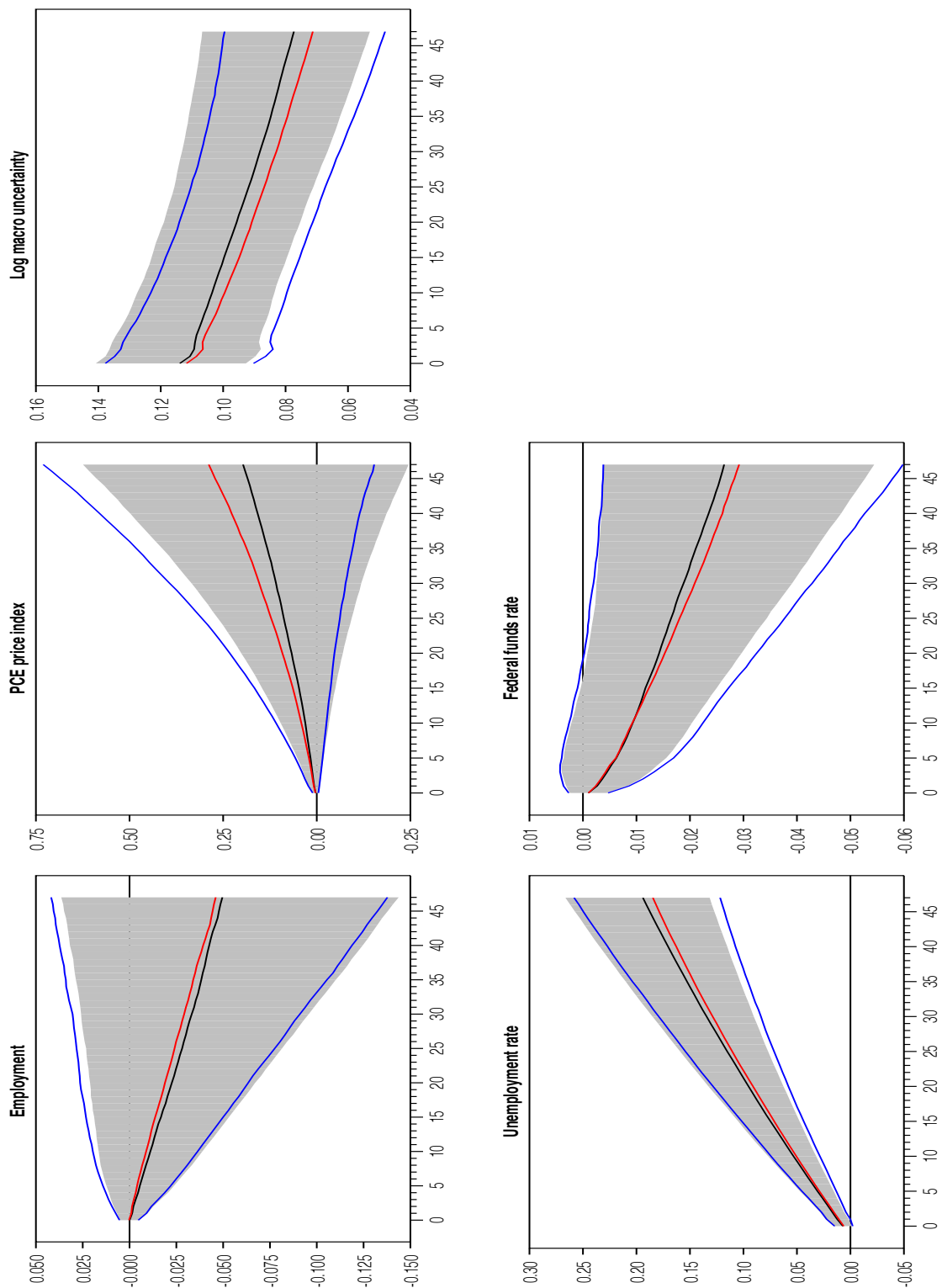


Figure A25: Estimates of a 4-variable model with one macroeconomic uncertainty factor: impulse responses from shock to macroeconomic uncertainty. Baseline model estimates: black line (median) with gray fans (15%/85% quantiles). Alternative estimates, from a corresponding model augmented to allow the A matrix to be time-varying: red line (median) with blue lines (15%/85% quantiles).

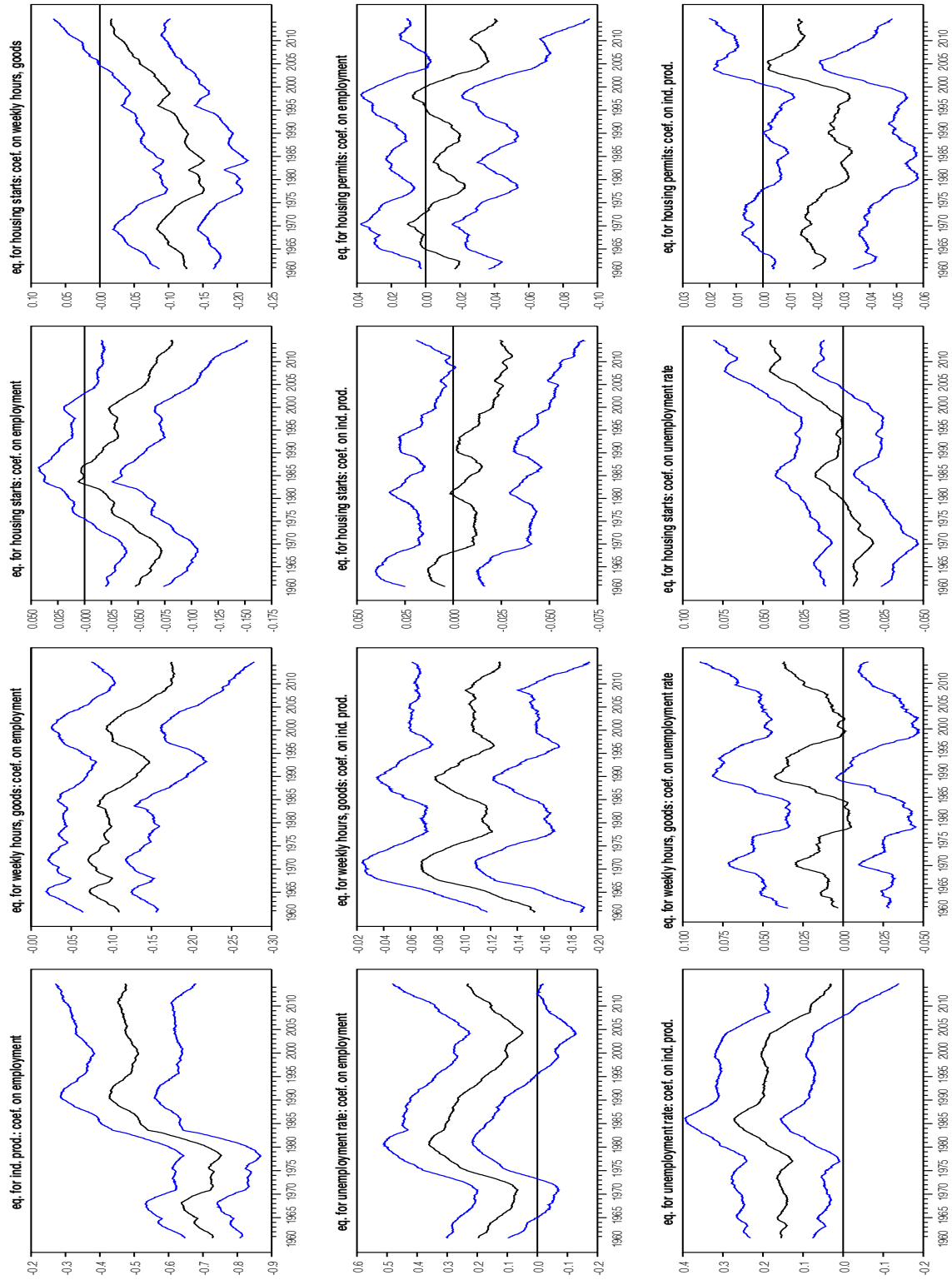


Figure A26: Estimates of a 12-variable model with one macroeconomic uncertainty factor: time-varying A coefficients, posterior median (black line) and 15%/85% quantiles (blue lines).

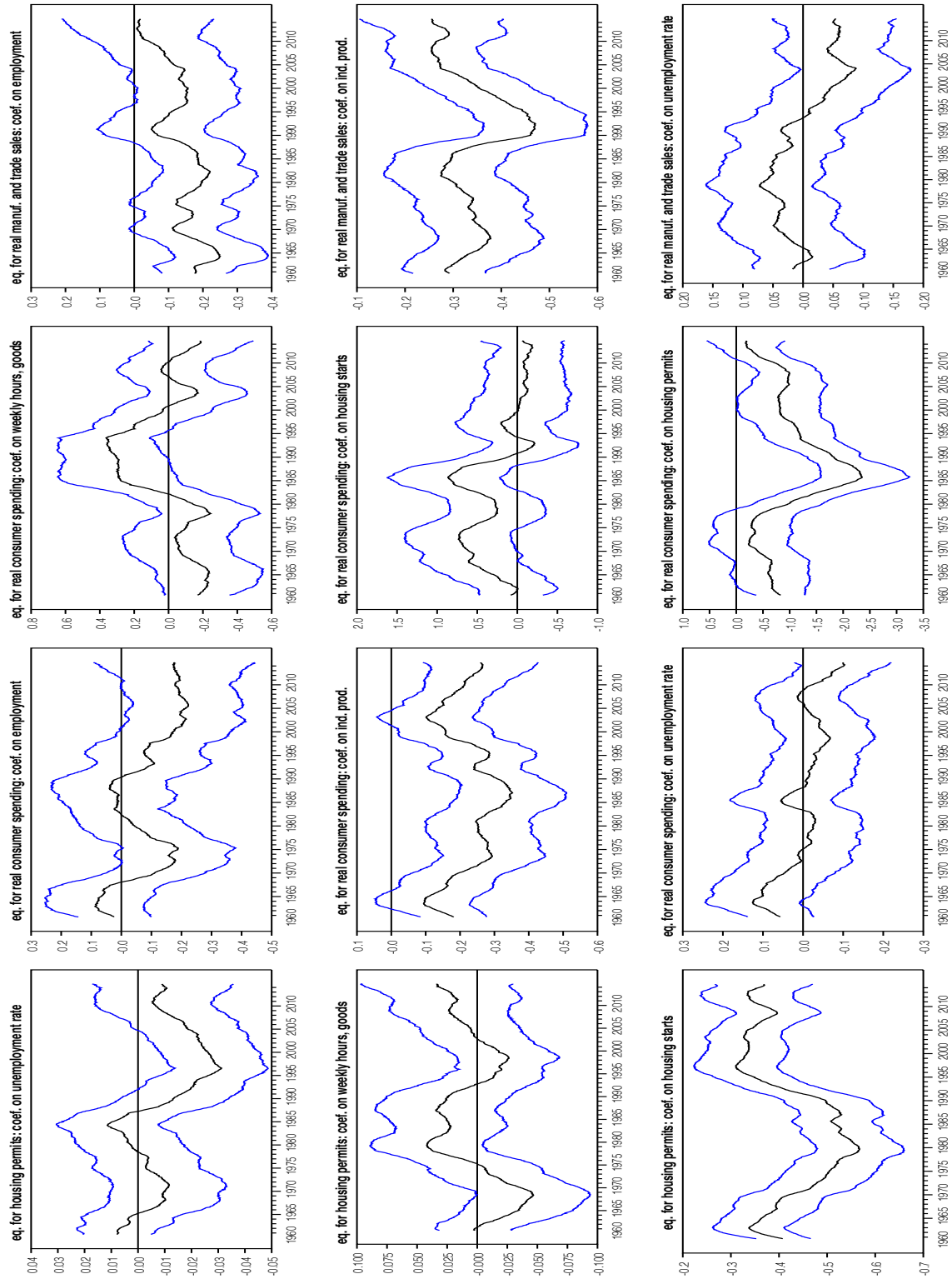


Figure A26: Continued, estimates of a 12-variable model with one macroeconomic uncertainty factor: time-varying A coefficients, posterior median (black line) and 15%/85% quantiles (blue lines).

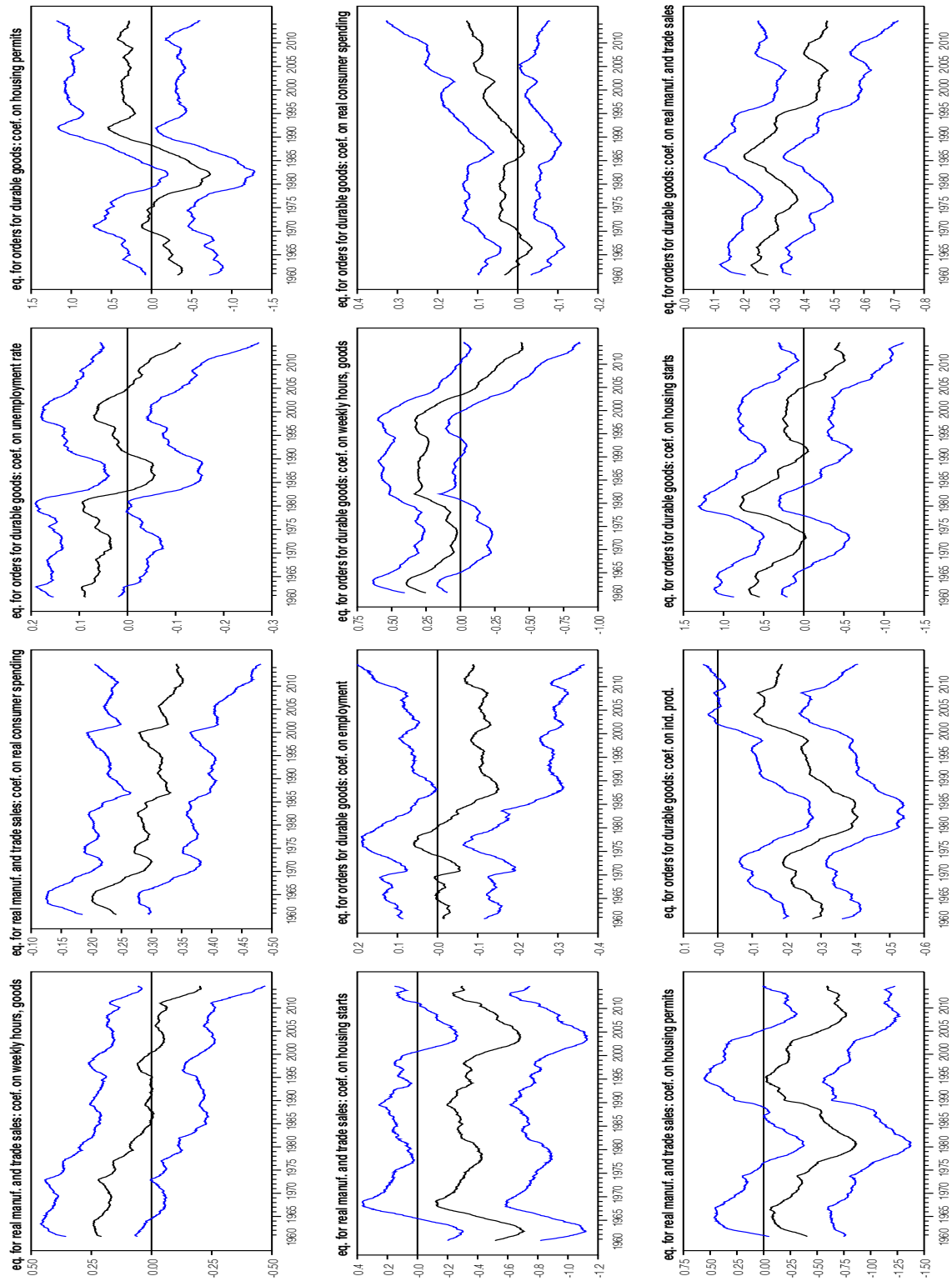


Figure A26: Continued, estimates of a 12-variable model with one macroeconomic uncertainty factor: time-varying A coefficients, posterior median (black line) and 15%/85% quantiles (blue lines).

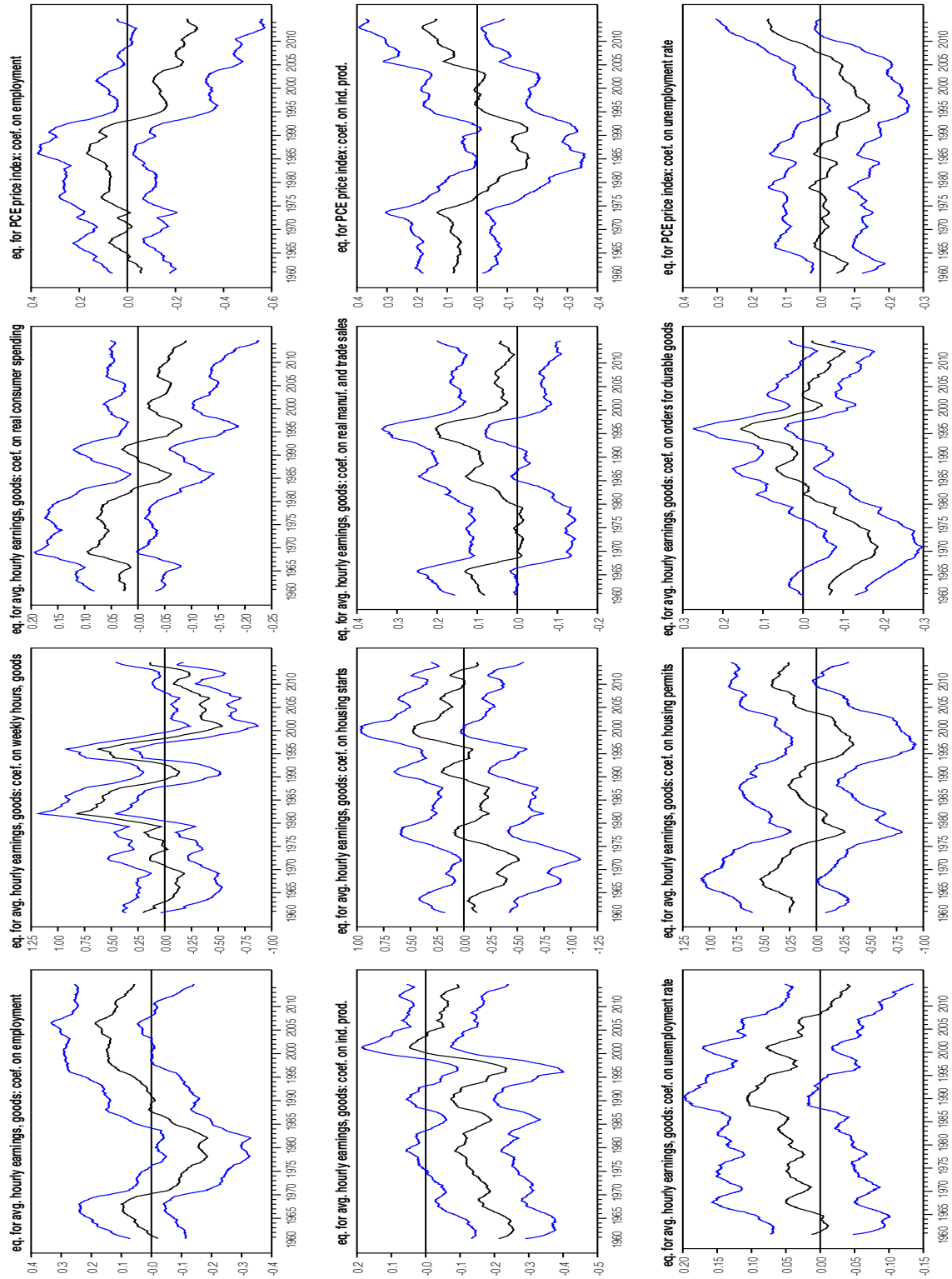


Figure A26: Continued, estimates of a 12-variable model with one macroeconomic uncertainty factor: time-varying A coefficients, posterior median (black line) and 15%/85% quantiles (blue lines).

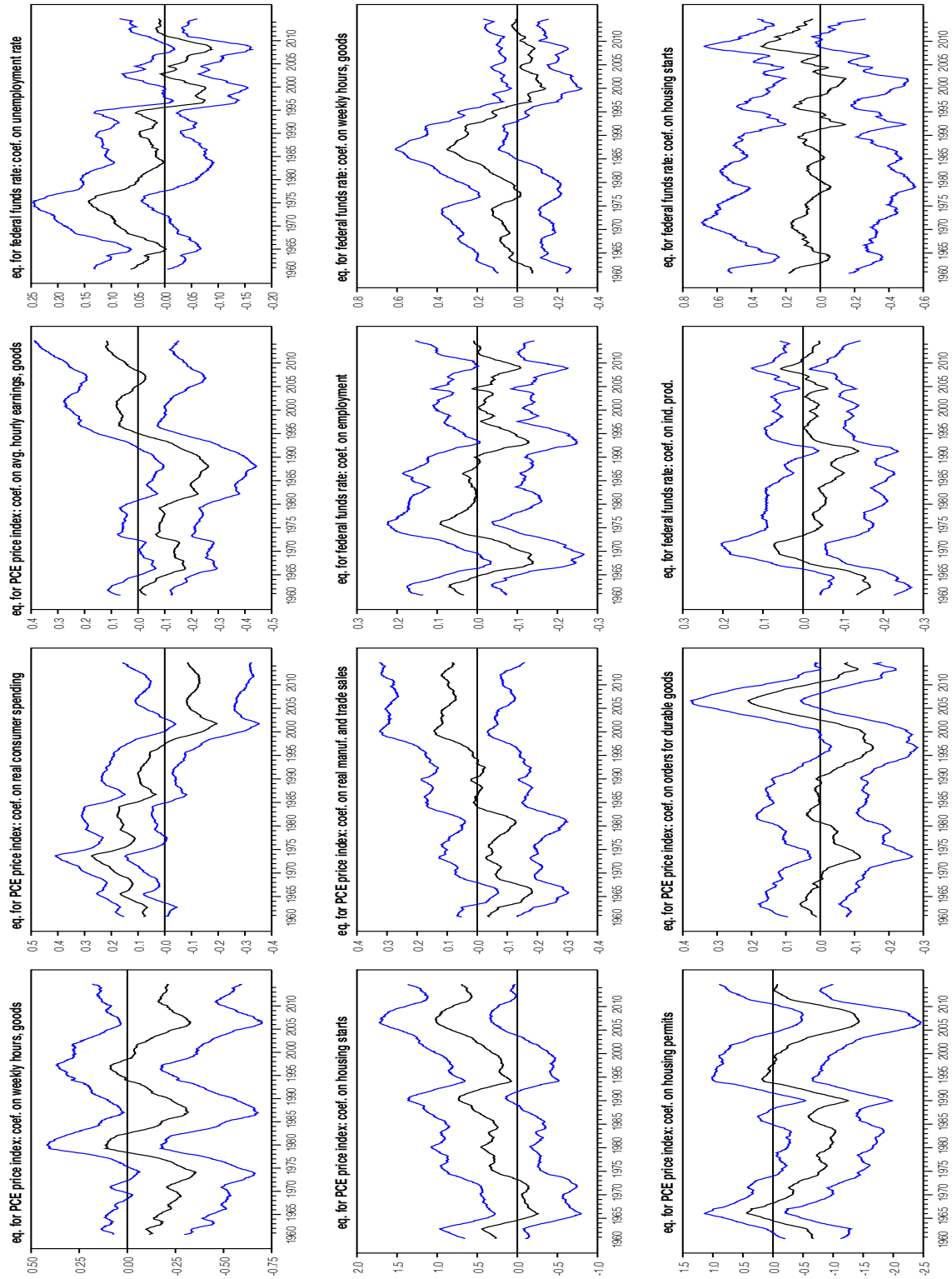


Figure A26: Continued, estimates of a 12-variable model with one macroeconomic uncertainty factor: time-varying A coefficients, posterior median (black line) and 15%/85% quantiles (blue lines).

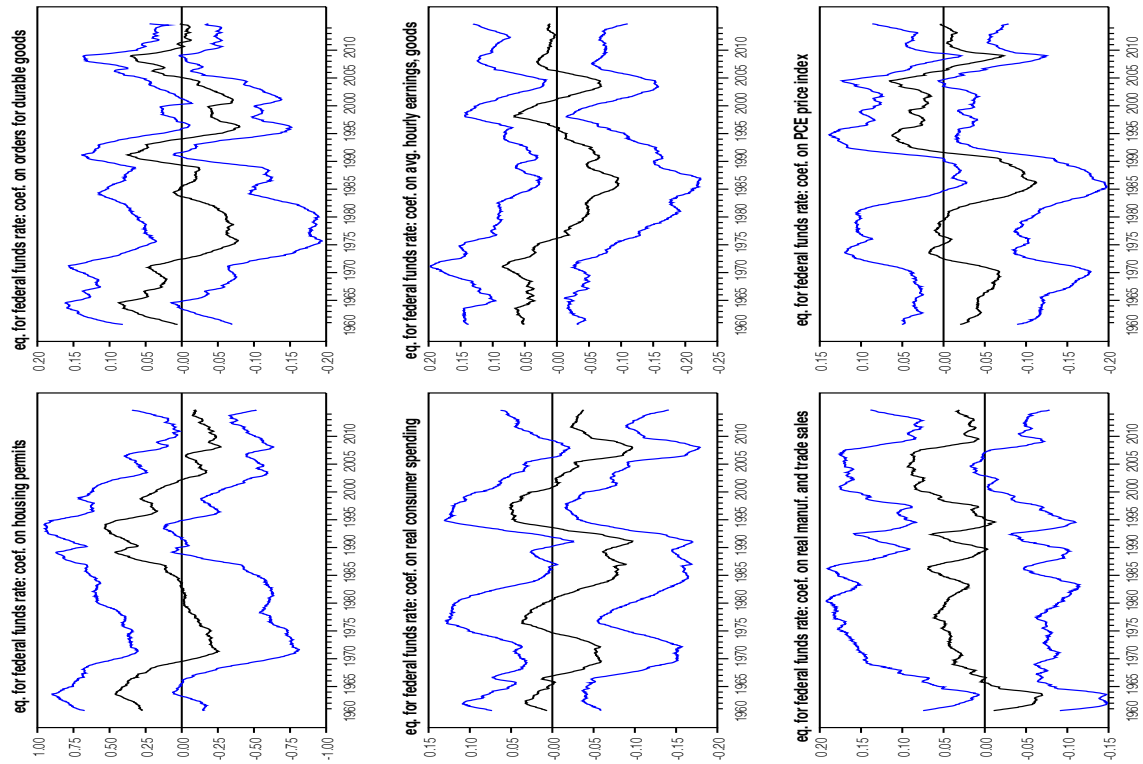


Figure A26: Continued, estimates of a 12-variable model with one macroeconomic uncertainty factor: time-varying A coefficients, posterior median (black line) and 15%/85% quantiles (blue lines).

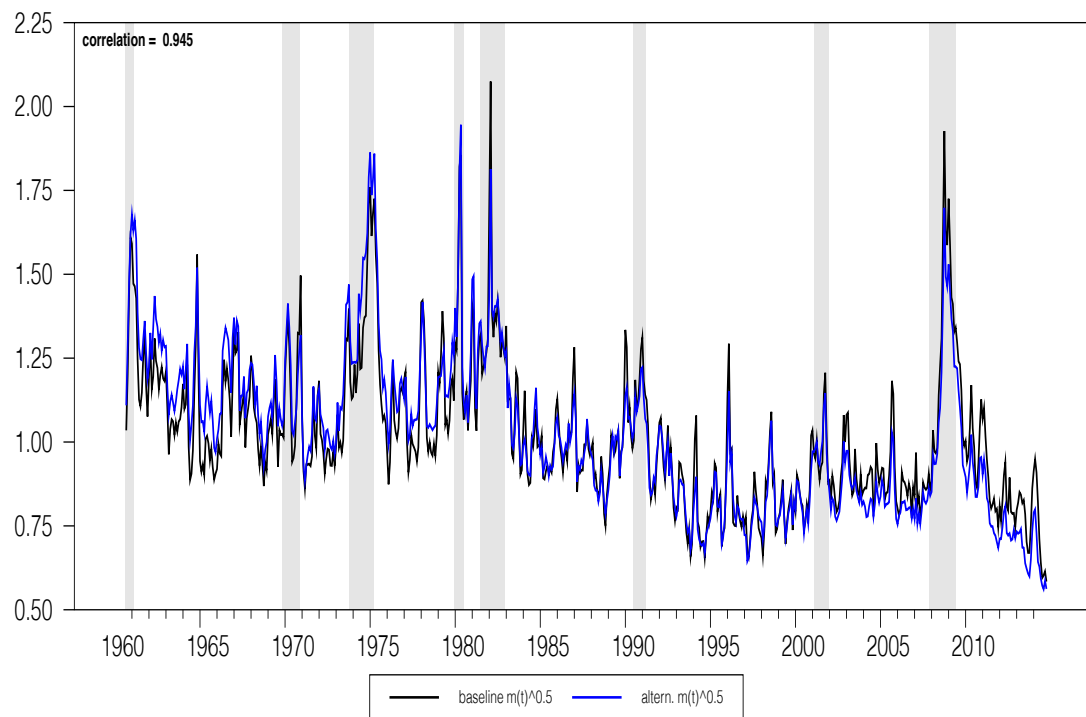


Figure A27: Estimates of a 12-variable model with one macroeconomic uncertainty factor: uncertainty factor. The black line is the posterior median of a constant parameter model. The blue line is the posterior median from a corresponding model augmented to allow the A matrix to be time-varying as in studies such as Primiceri (2005).

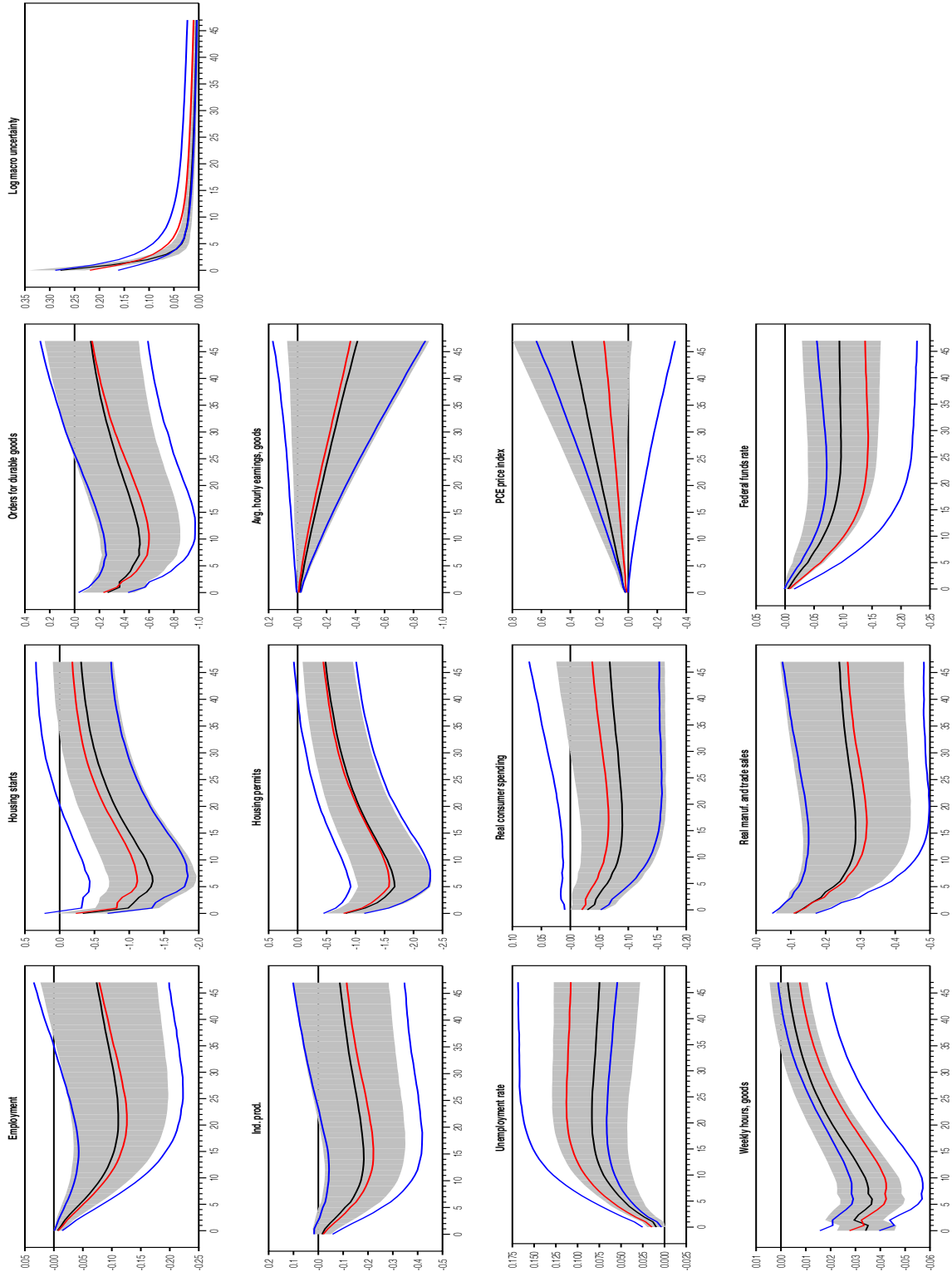


Figure A28: Estimates of a 12-variable model with one macroeconomic uncertainty factor: impulse responses from shock to macroeconomic uncertainty. Baseline model estimates: black line (median) with gray fans (15%/85% quantiles). Alternative estimates, from a corresponding model augmented to allow the A matrix to be time-varying: red line (median) with blue lines (15%/85% quantiles).

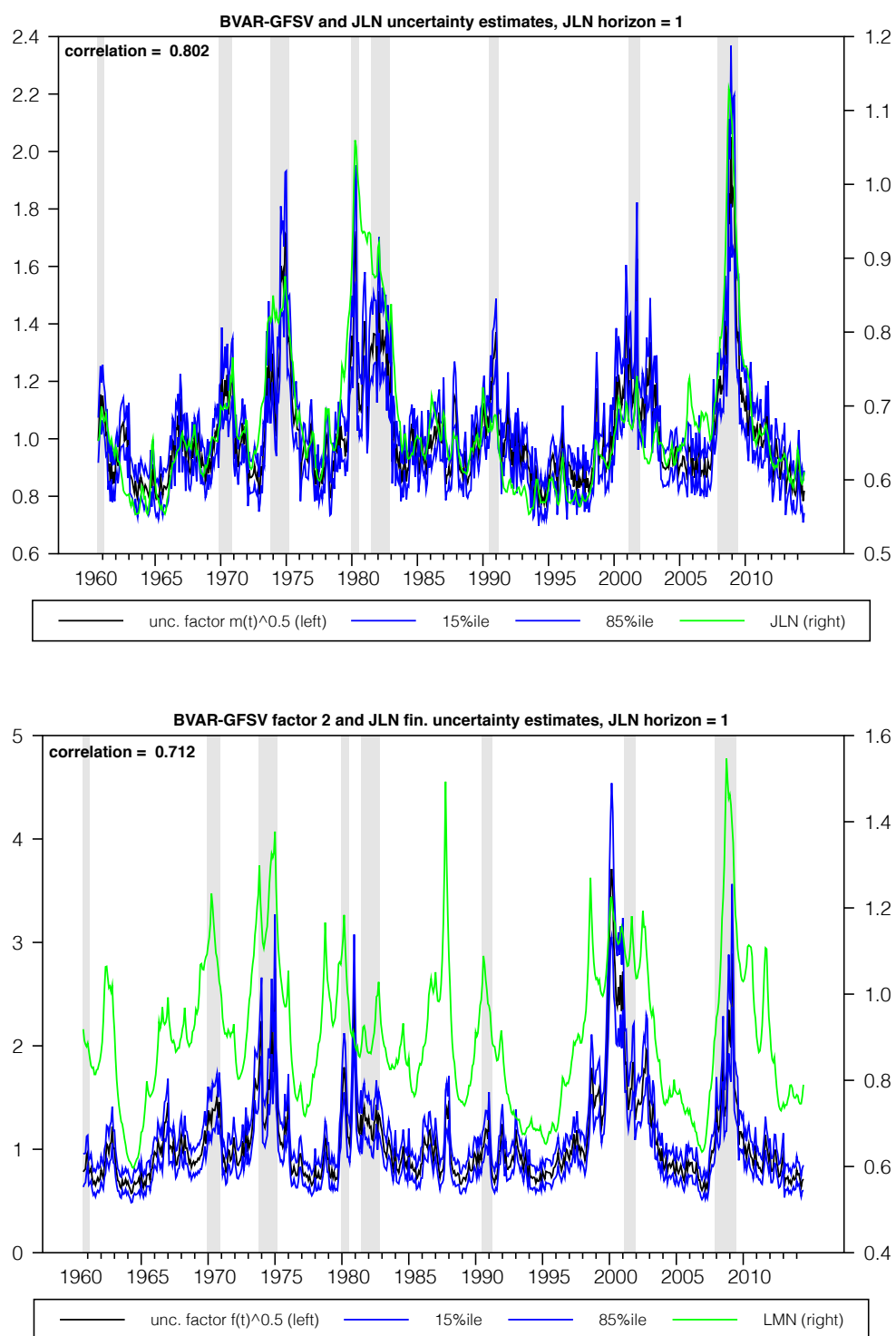


Figure A29: Results for alternative variable specification 1, with S&P500 and credit spread moved to the macro set: uncertainty estimates, macro in top panel and financial in bottom panel

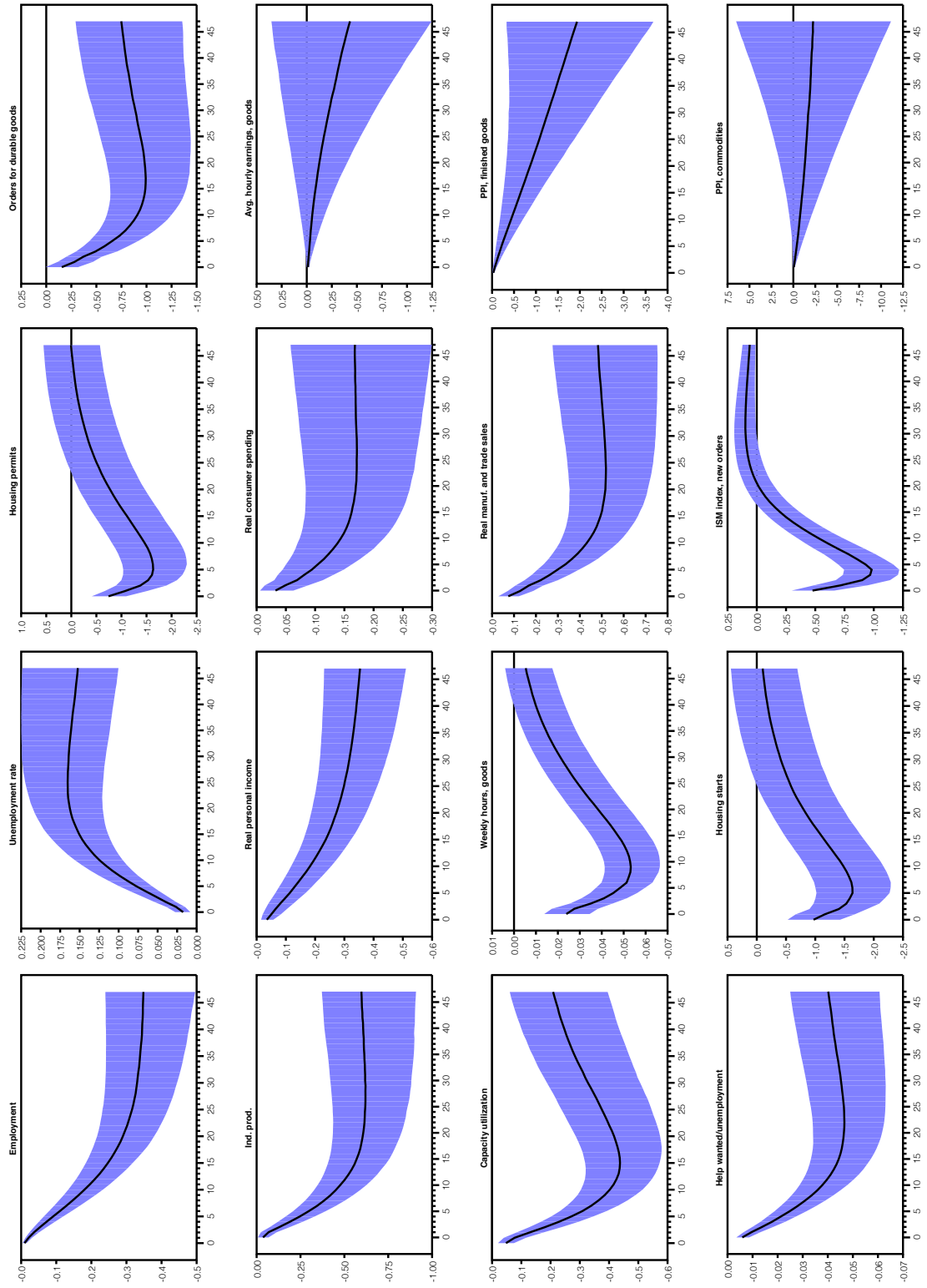


Figure A30: Results for alternative variable specification 1, with S&P500 and credit spread moved to the macro set: impulse responses, one st. dev. shock to macro uncertainty

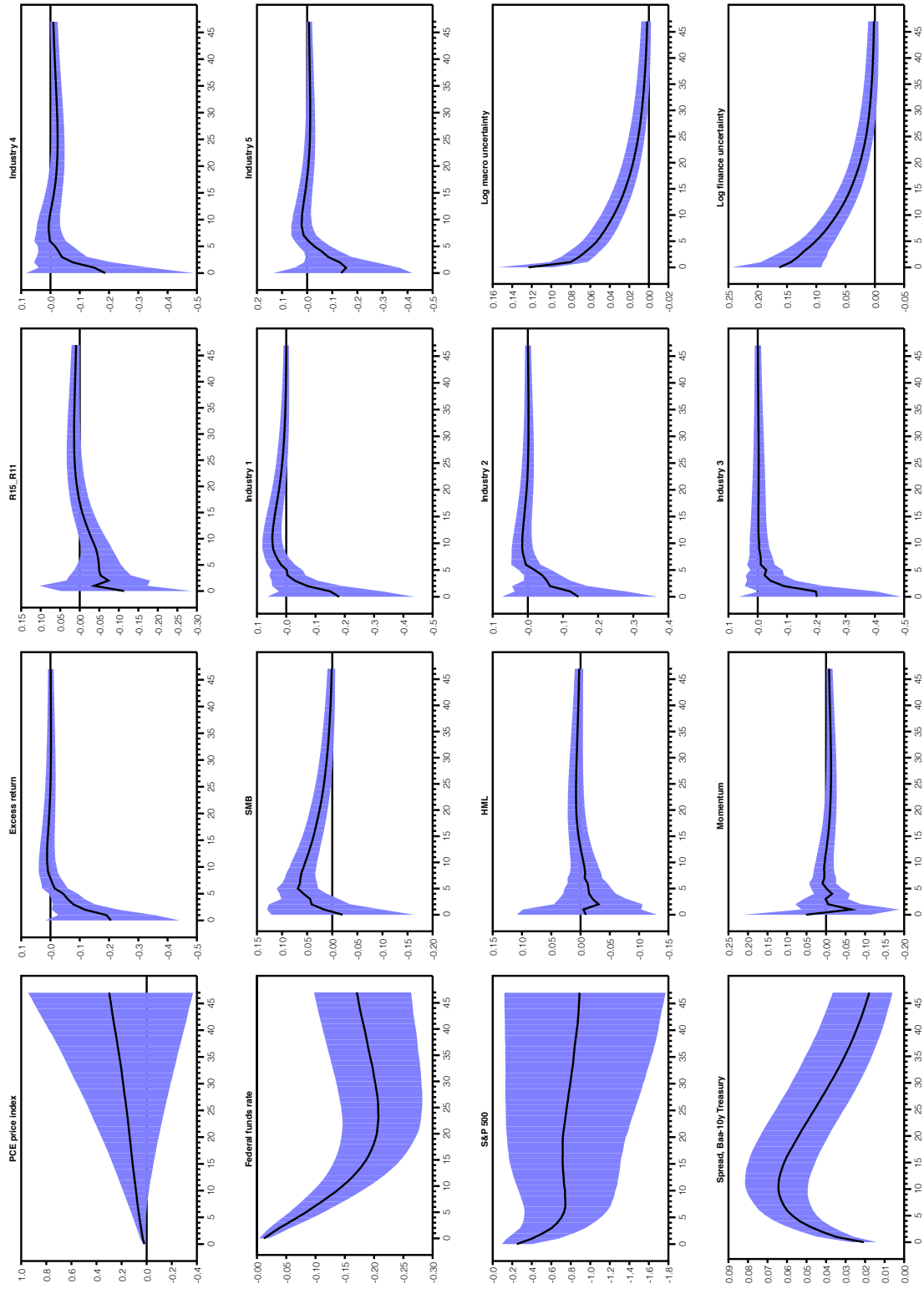


Figure A30: Continued, results for alternative variable specification 1, with S&P500 and credit spread moved to the macro set: impulse responses, one st. dev. shock to macro uncertainty

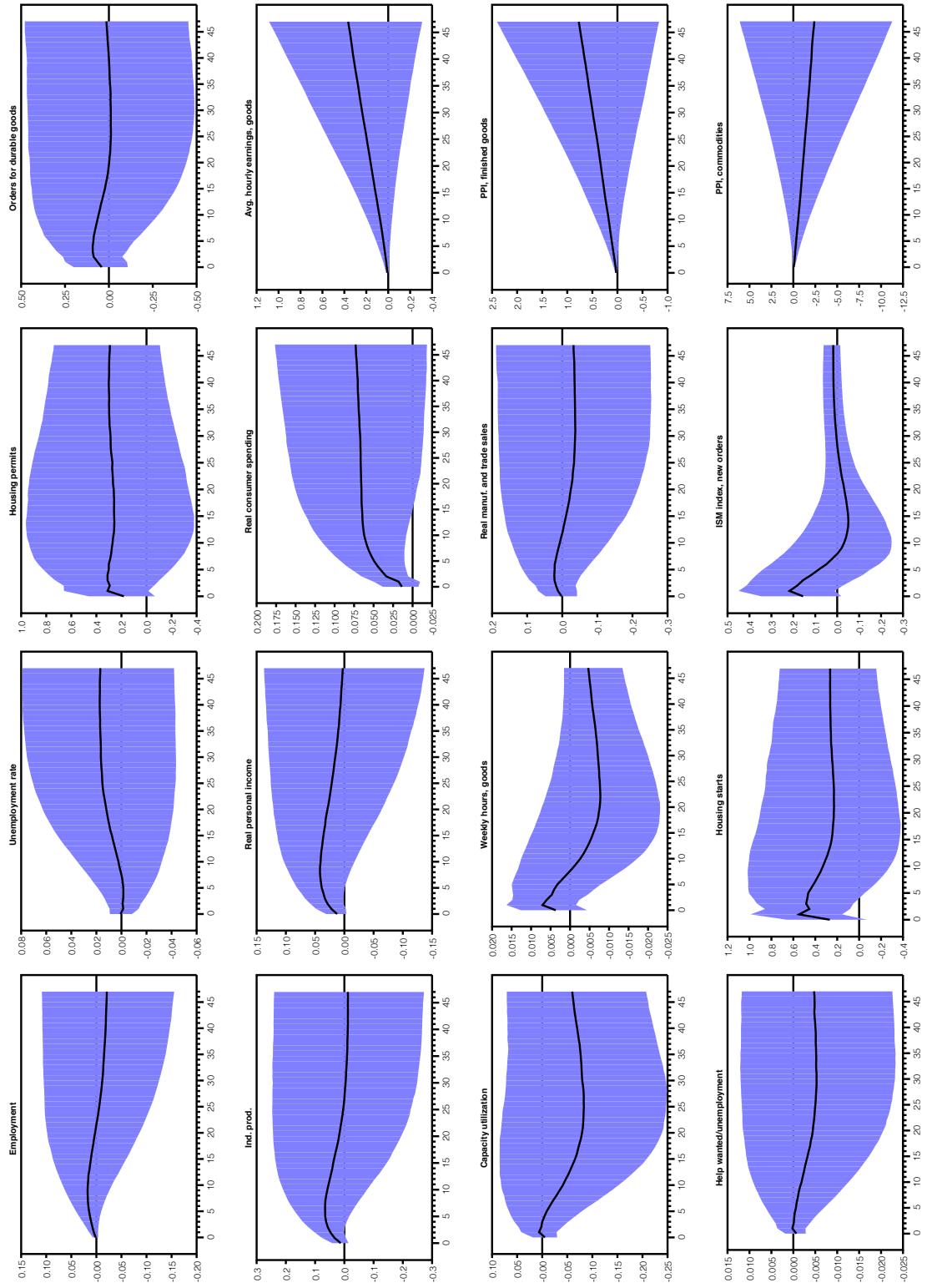


Figure A31: Results for alternative variable specification 1, with S&P500 and credit spread moved to the macro set: impulse responses, one st. dev. shock to financial uncertainty

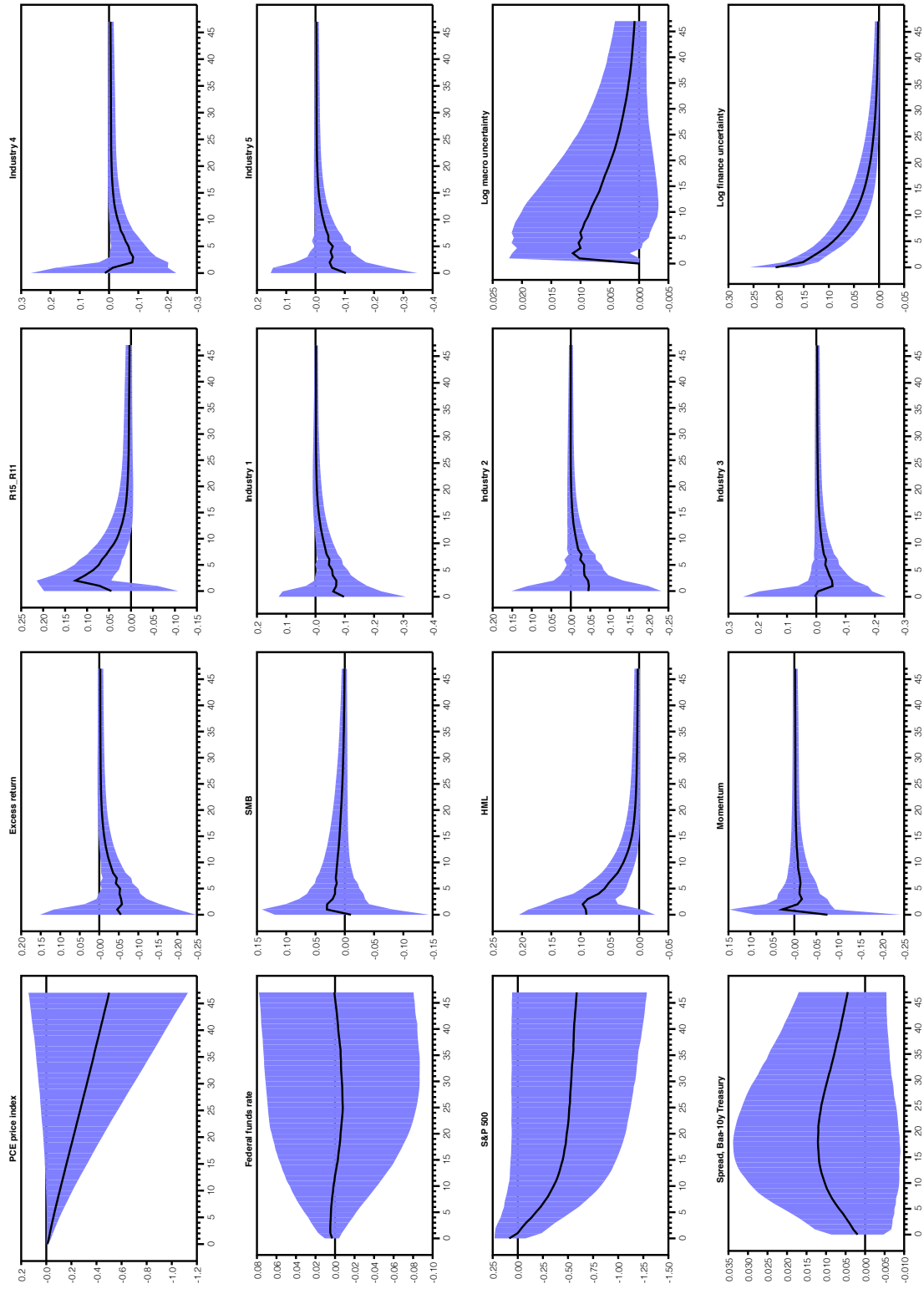


Figure A31: Continued, results for alternative variable specification 1, with S&P500 and credit spread moved to the macro set: impulse responses, one st. dev. shock to financial uncertainty

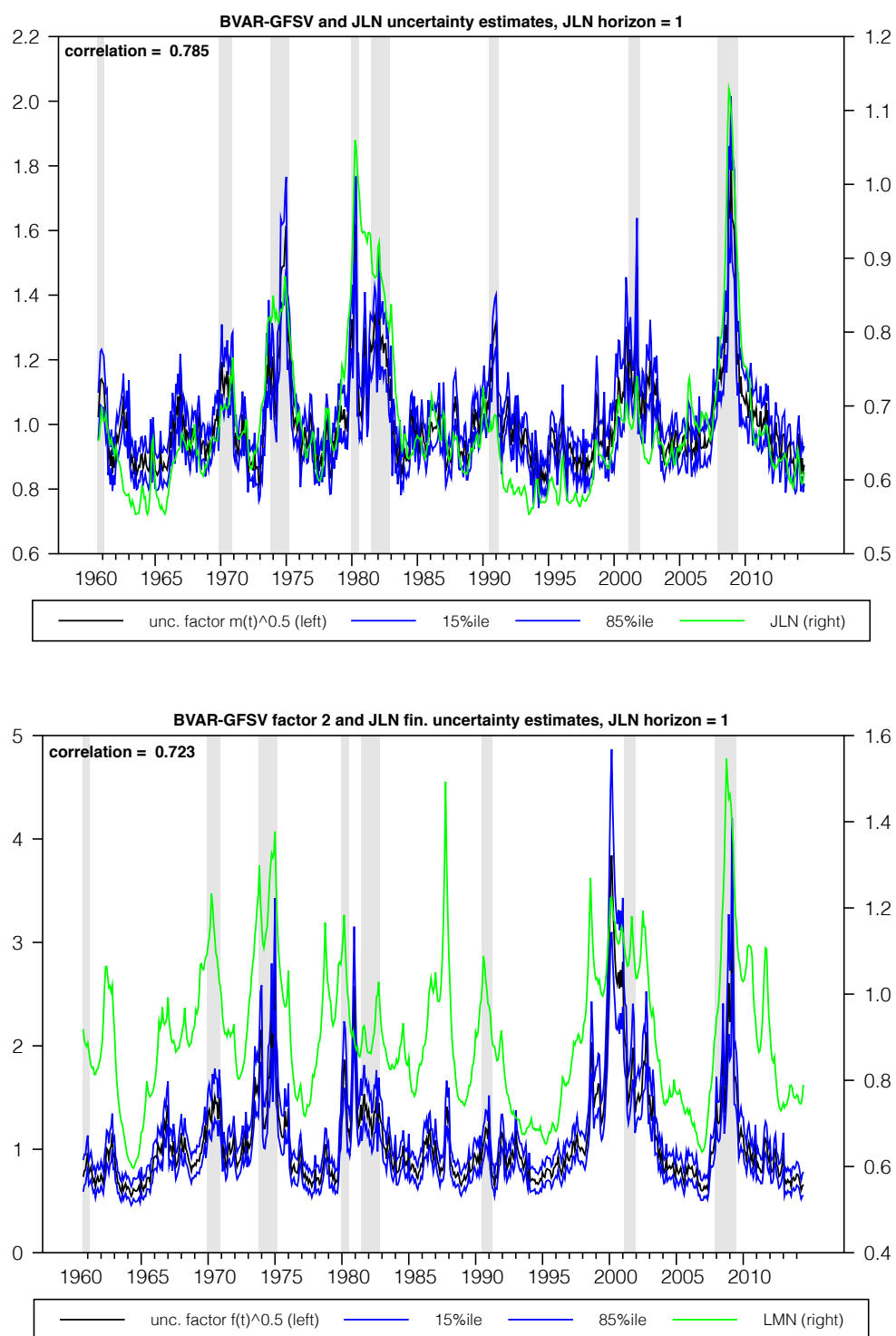


Figure A32: Results for alternative variable specification 2, with S&P500 moved to the macro set: uncertainty estimates, macro in top panel and financial in bottom panel

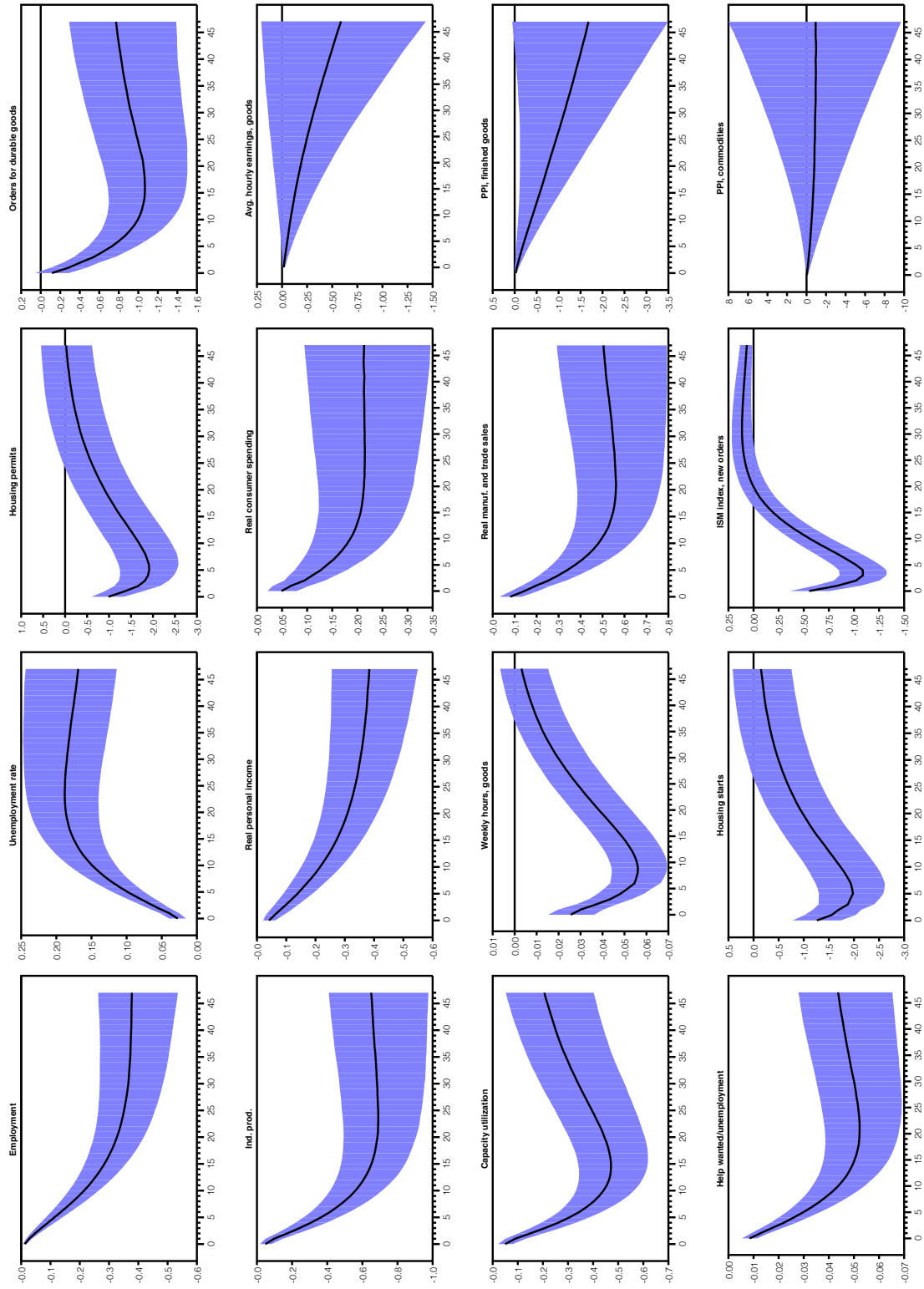


Figure A33: Results for alternative variable specification 2, with S&P500 moved to the macro set: impulse responses, one st. dev. shock to macro uncertainty

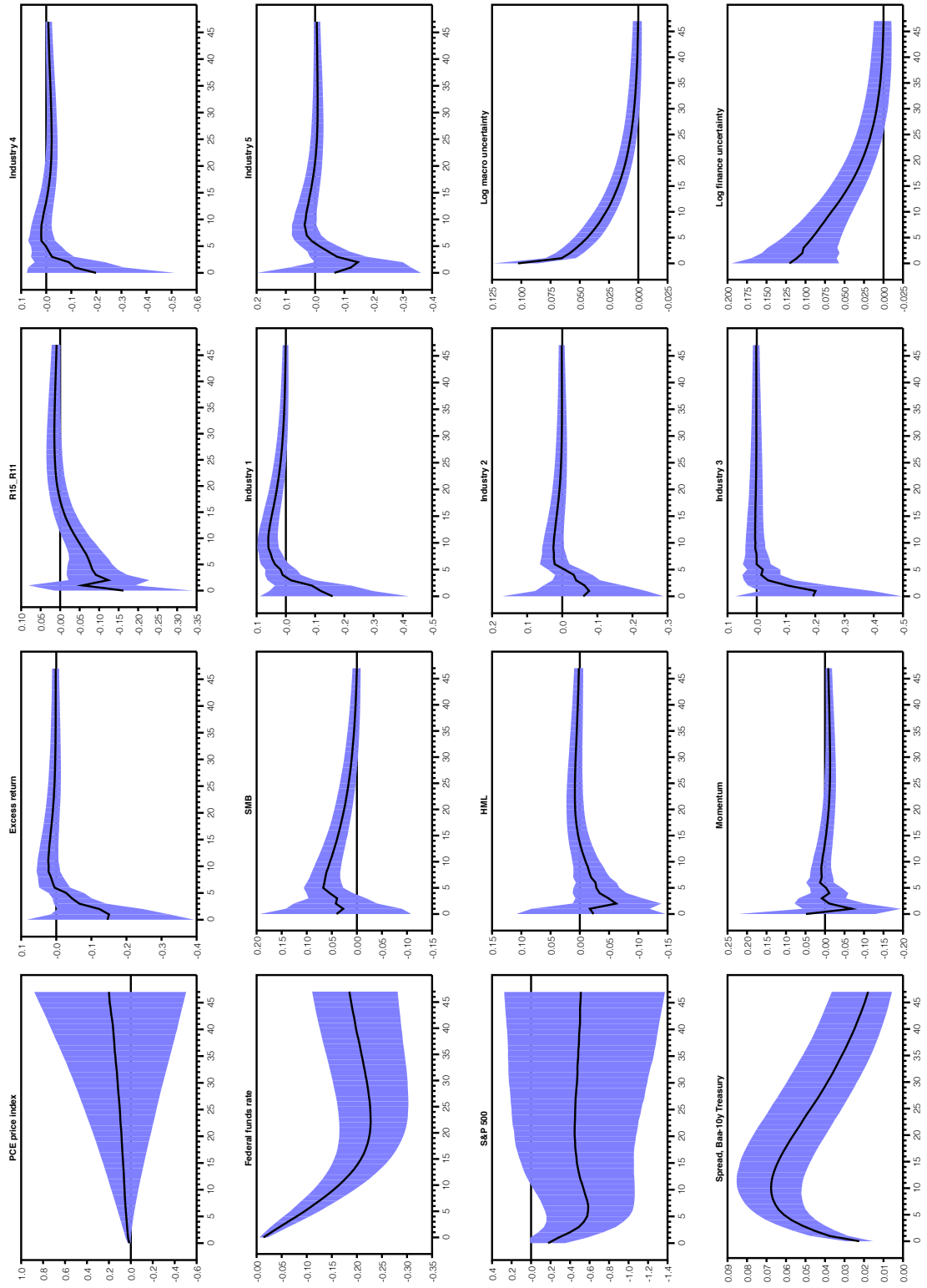


Figure A33: Continued, results for alternative variable specification 2, with S&P500 moved to the macro set: impulse responses, one st. dev. shock to macro uncertainty

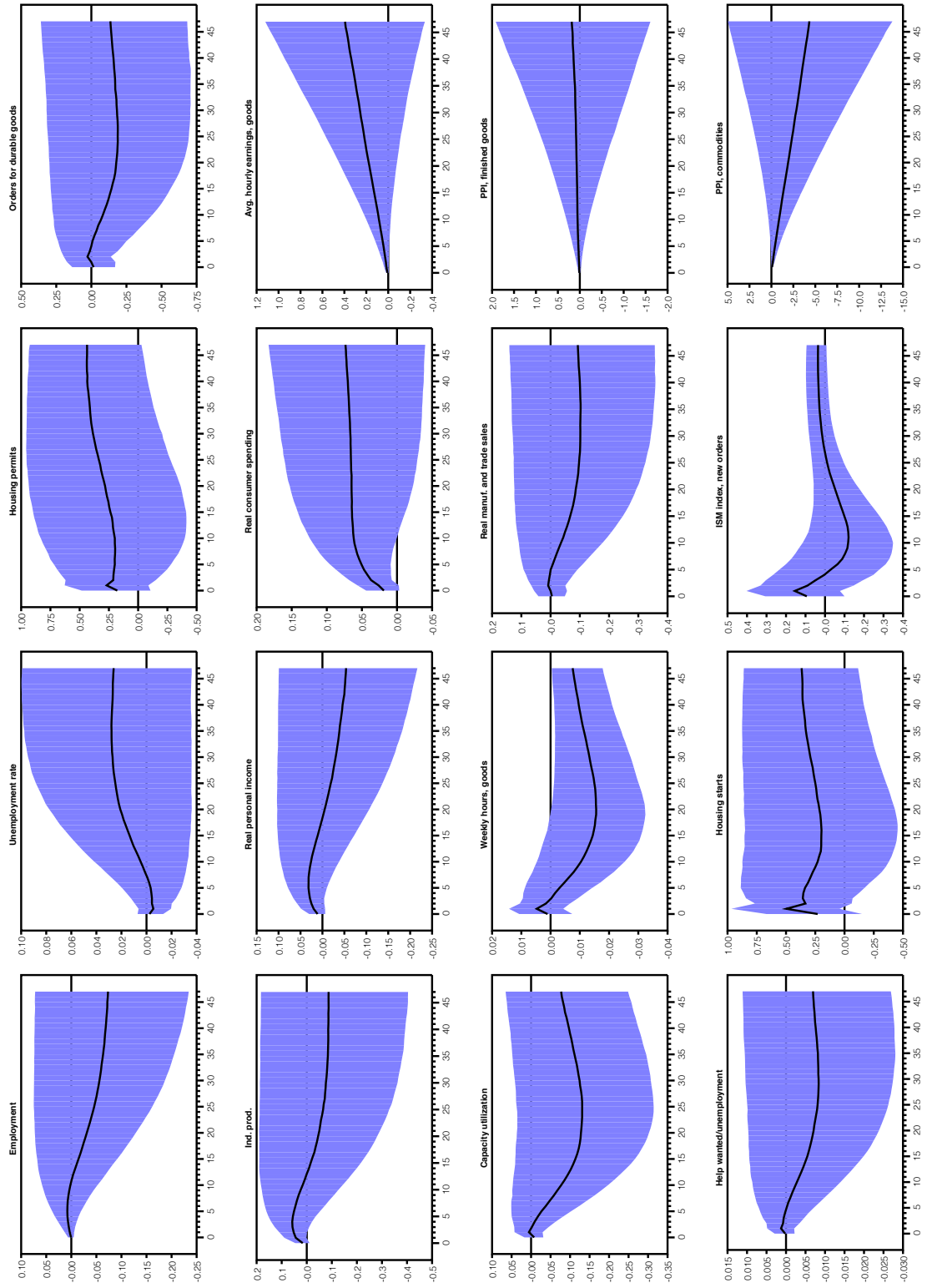


Figure A34: Results for alternative variable specification 2, with S&P500 moved to the macro set: impulse responses, one st. dev. shock to financial uncertainty

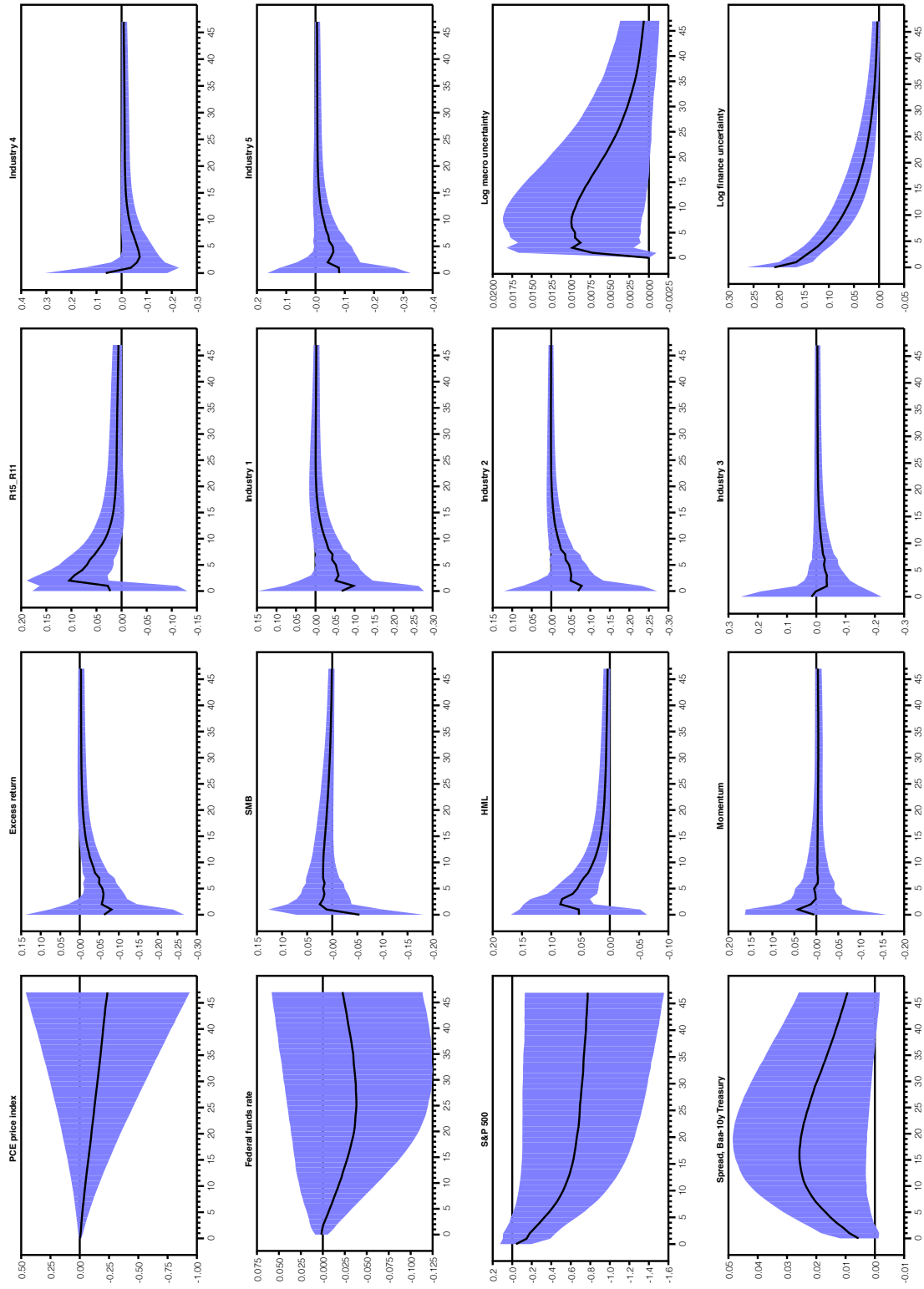


Figure A34: Continued, results for alternative variable specification 2, with S&P500 moved to the macro set: impulse responses, one st. dev. shock to financial uncertainty

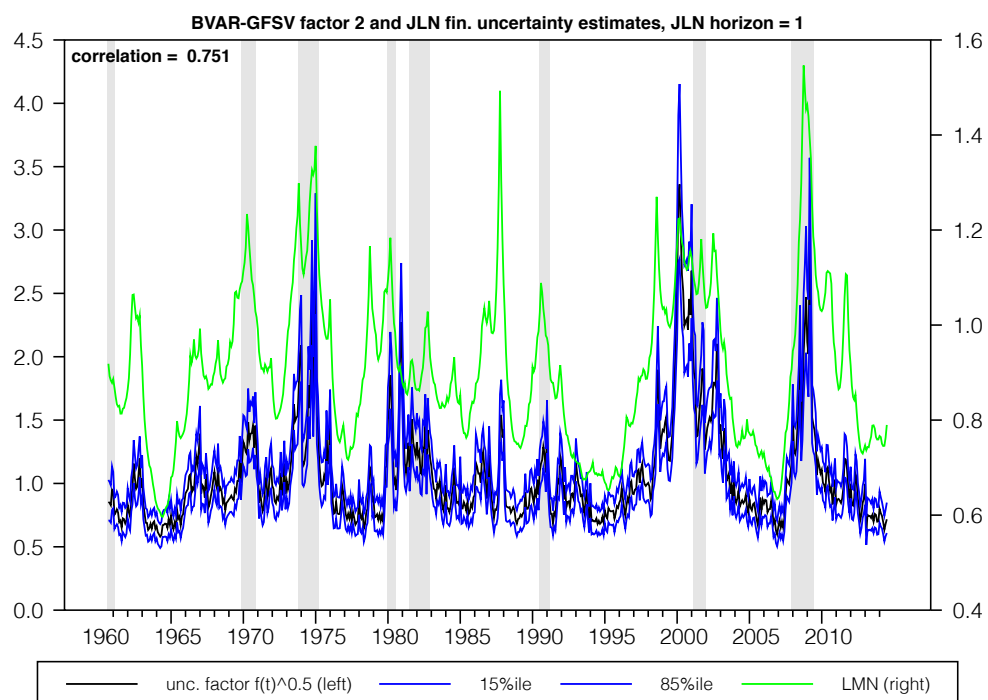
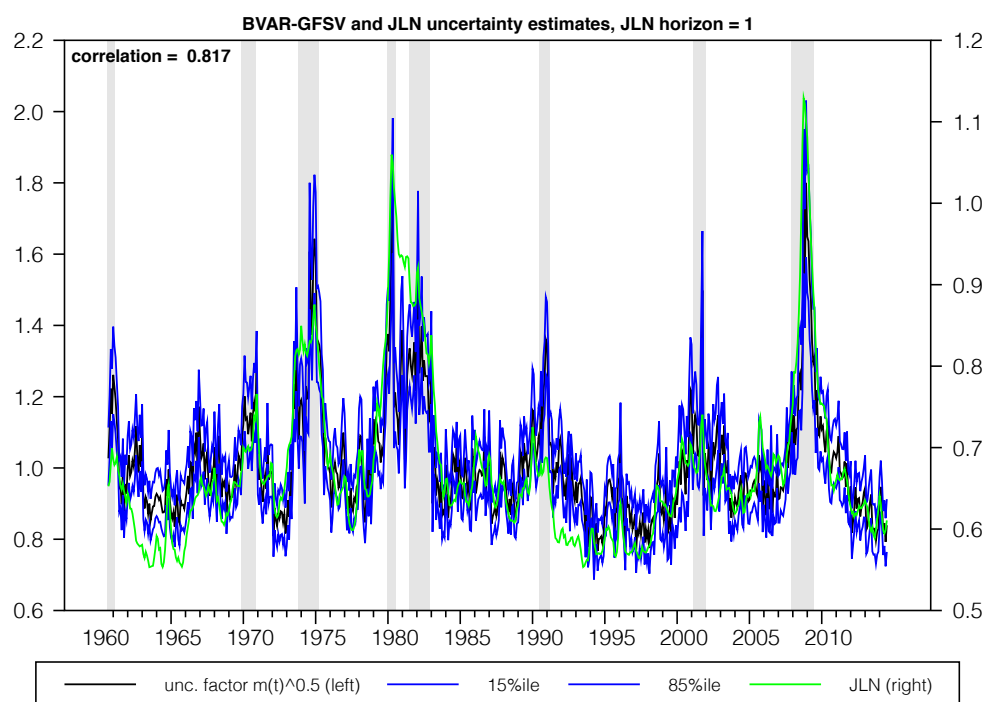


Figure A35: Results for alternative variable specification 3, with the credit spread moved to the macro set: uncertainty estimates, macro in top panel and financial in bottom panel

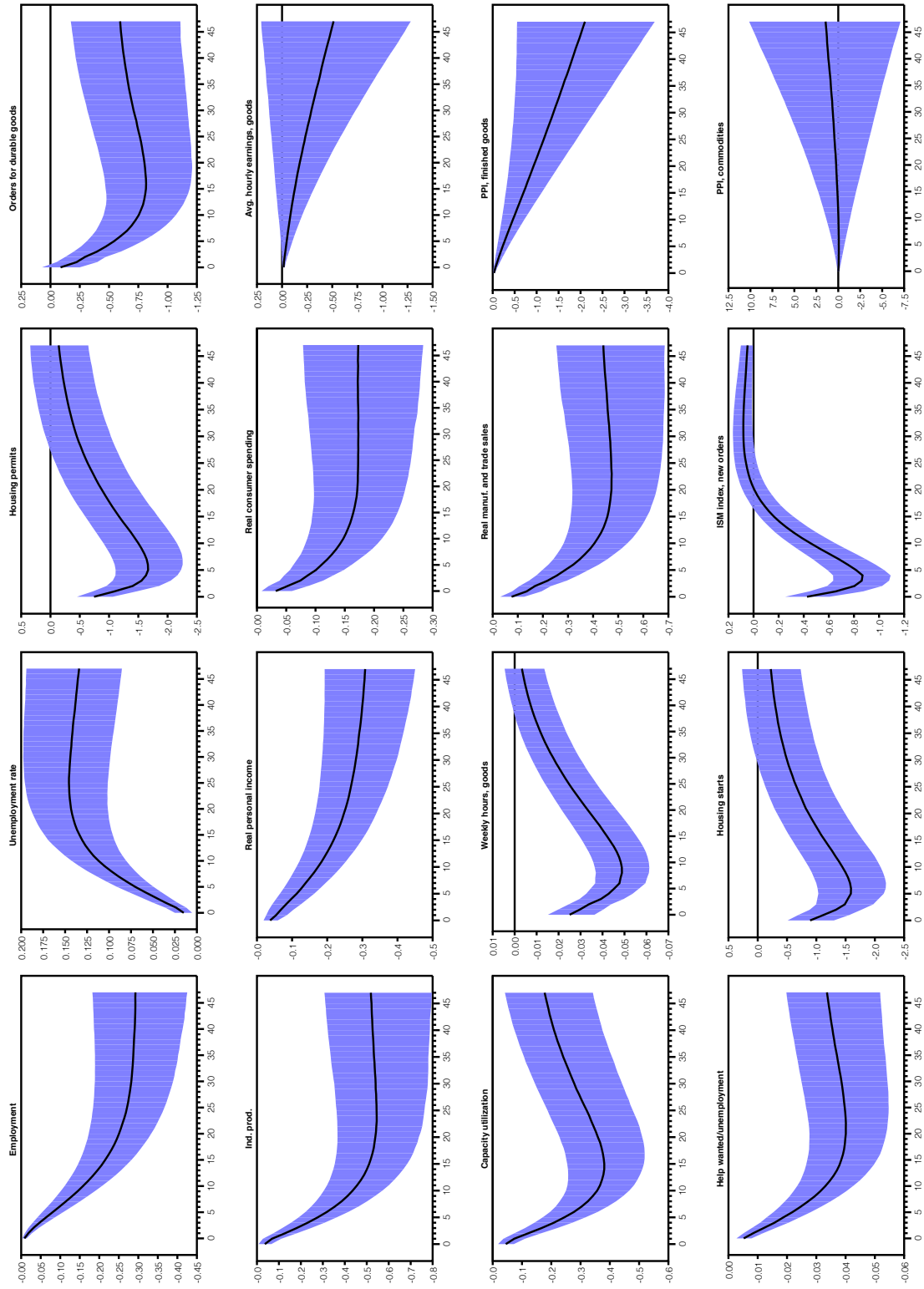


Figure A36: Results for alternative variable specification 3, with the credit spread moved to the macro set: impulse responses, one st. dev. shock to macro uncertainty

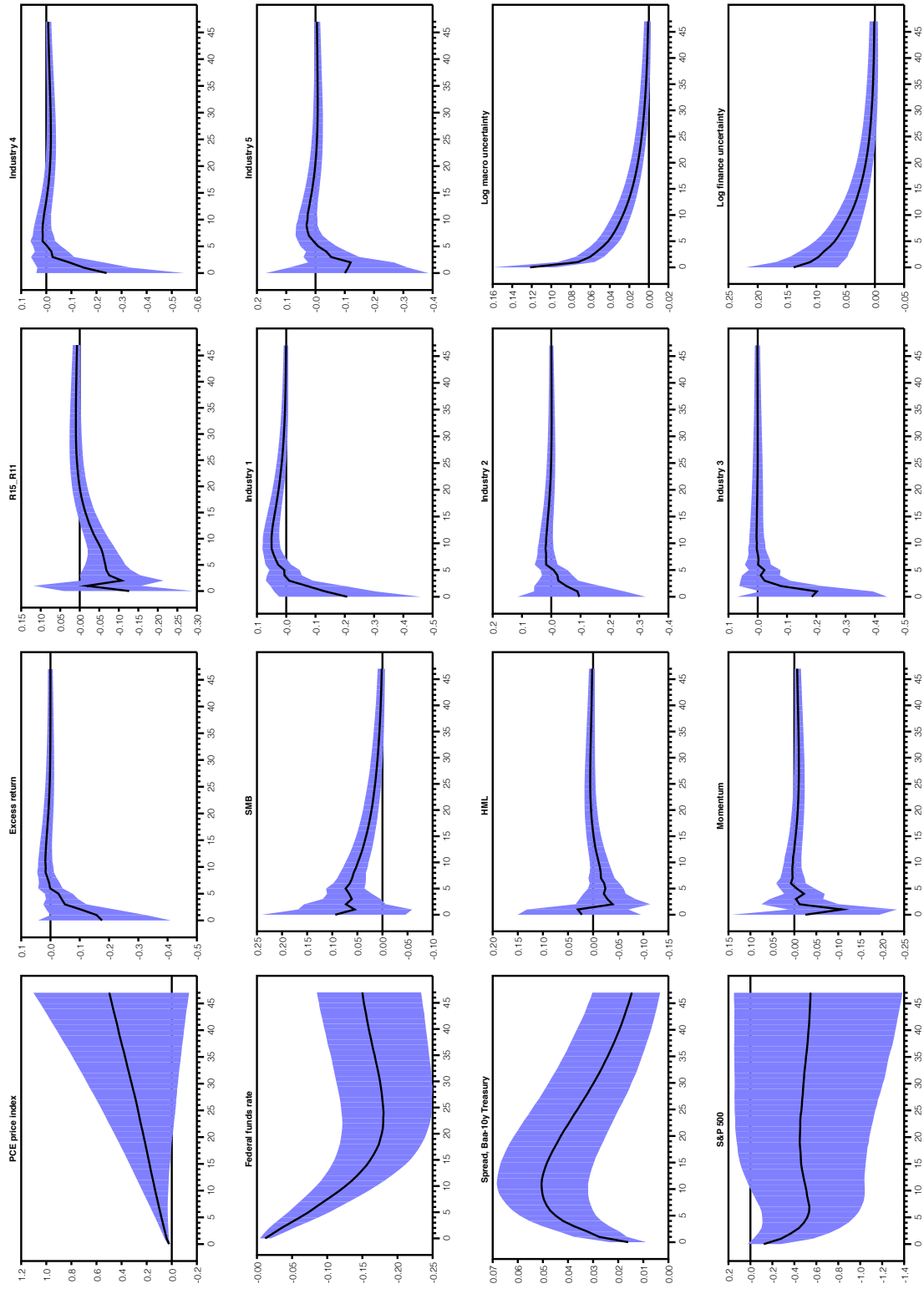


Figure A36: Continued, results for alternative variable specification 3, with the credit spread moved to the macro set: impulse responses, one st. dev. shock to macro uncertainty

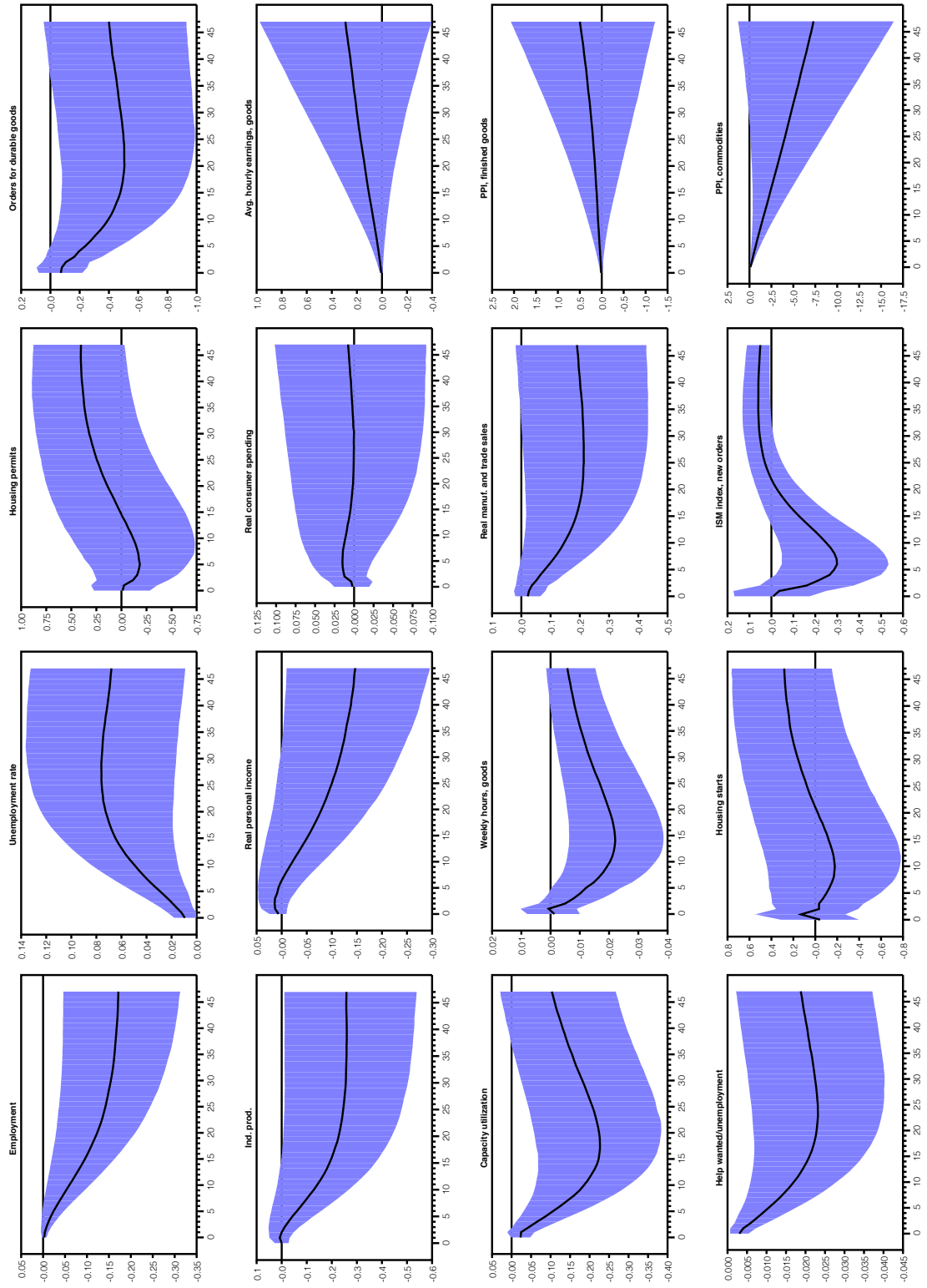


Figure A37: Results for alternative variable specification 3, with the credit spread moved to the macro set: impulse responses, one st. dev. shock to financial uncertainty

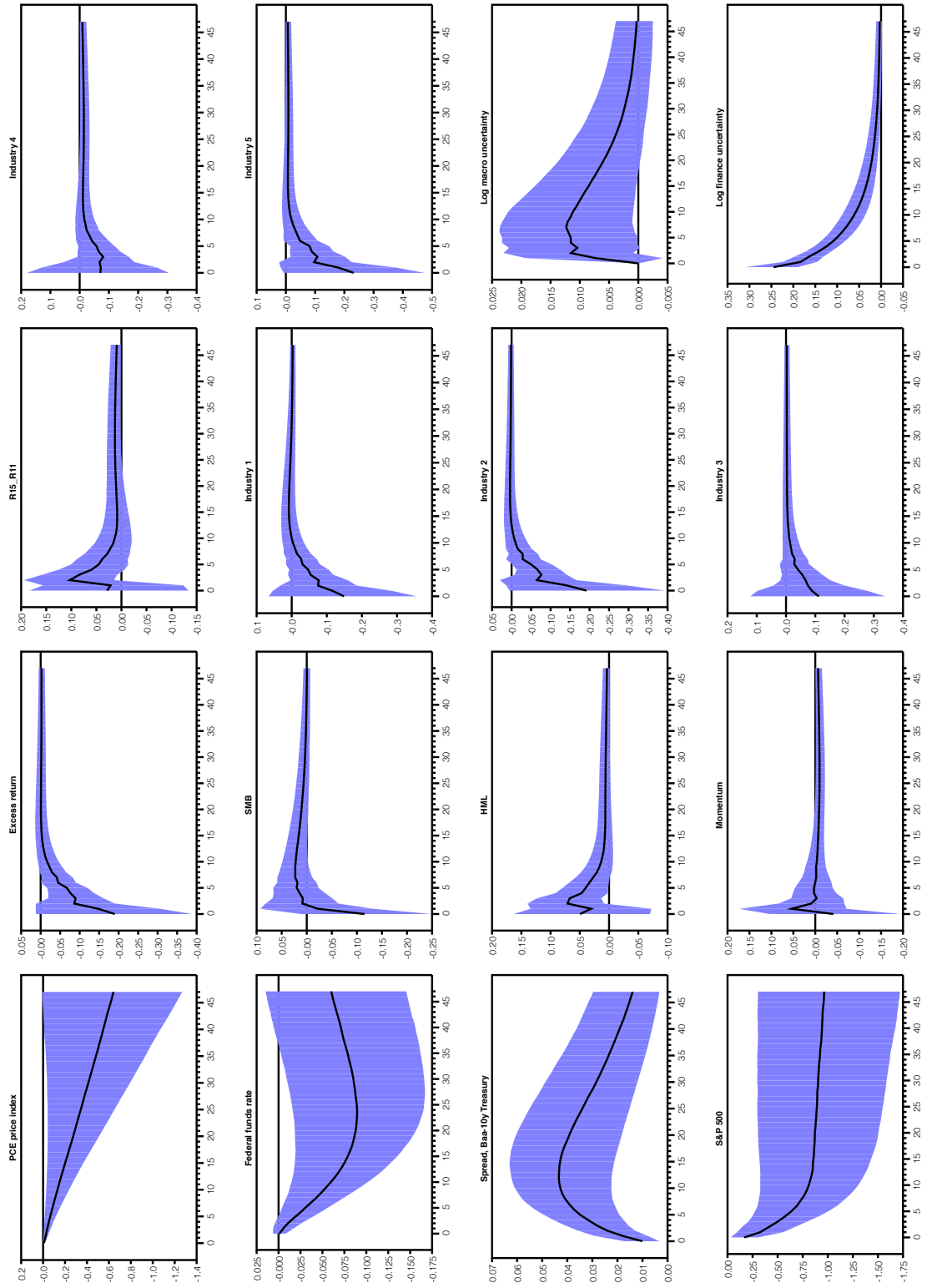


Figure A37: Continued, results for alternative variable specification 3, with the credit spread moved to the macro set: impulse responses, one st. dev. shock to financial uncertainty

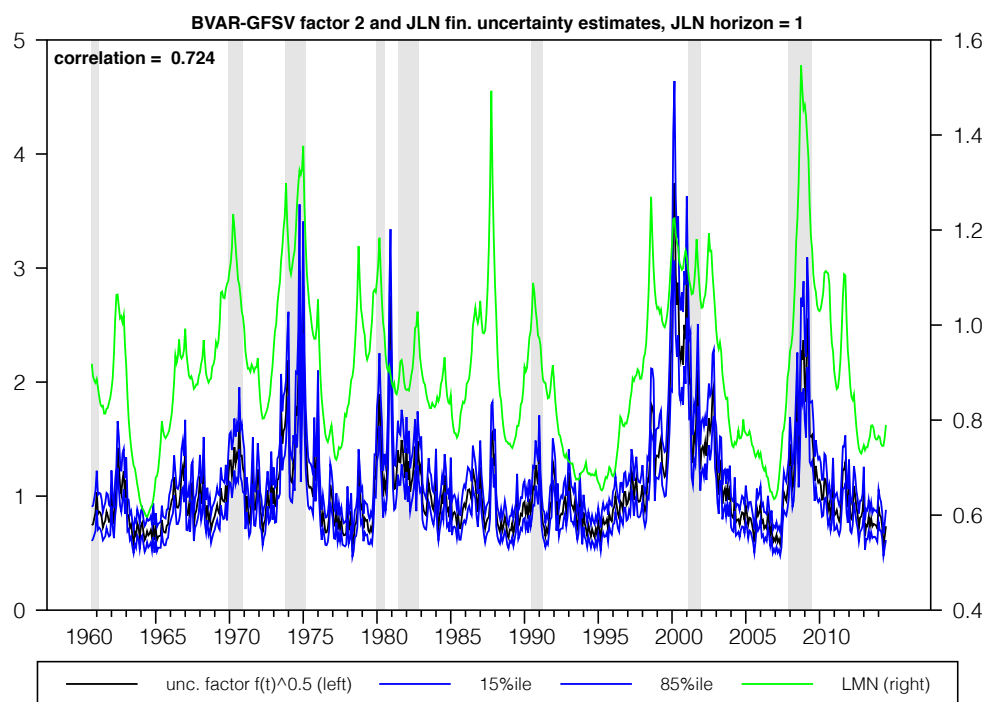
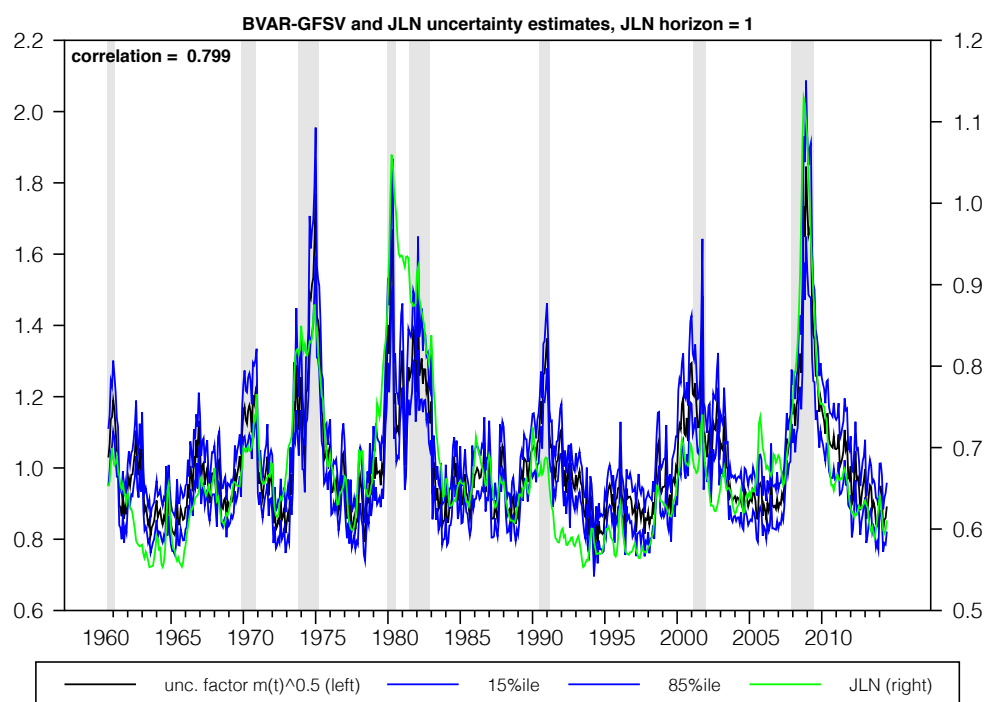


Figure A38: Results for alternative variable specification 4, with 36 variables: uncertainty estimates, macro in top panel and financial in bottom panel

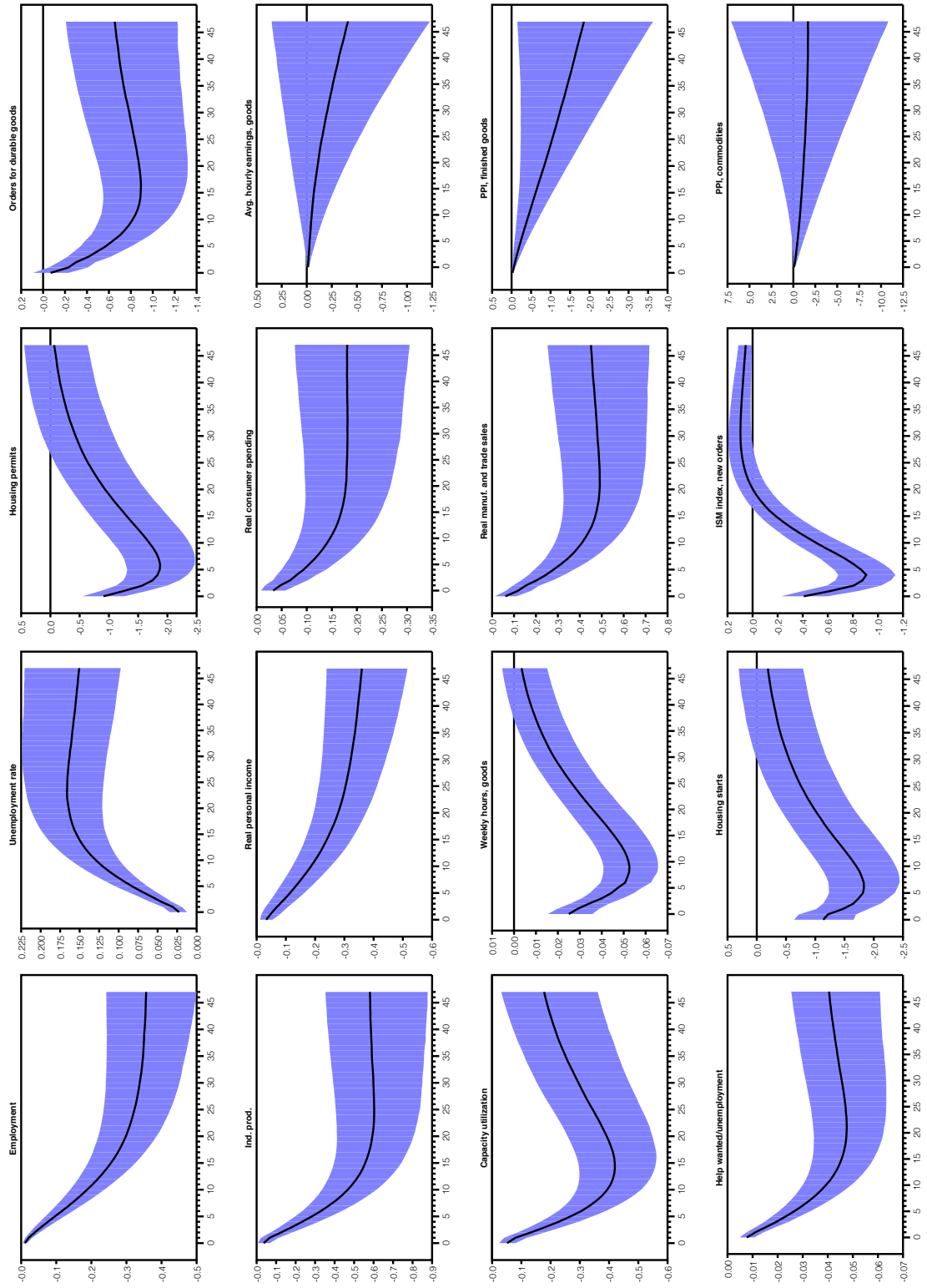


Figure A39: Results for alternative variable specification 4, with 36 variables: impulse responses, one st. dev. shock to macro uncertainty

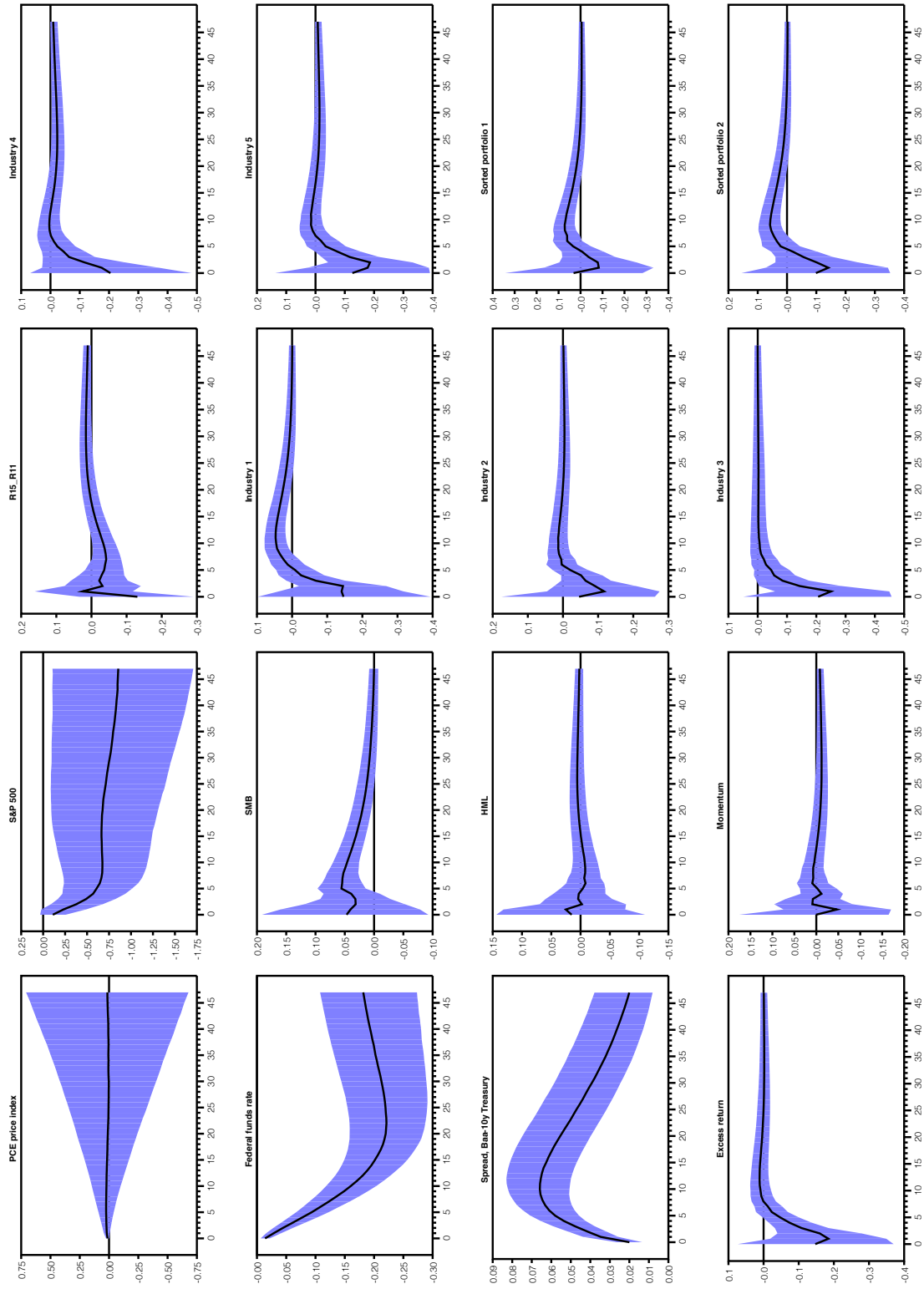


Figure A39: Continued, results for alternative variable specification 4, with 36 variables: impulse responses, one st. dev. shock to macro uncertainty

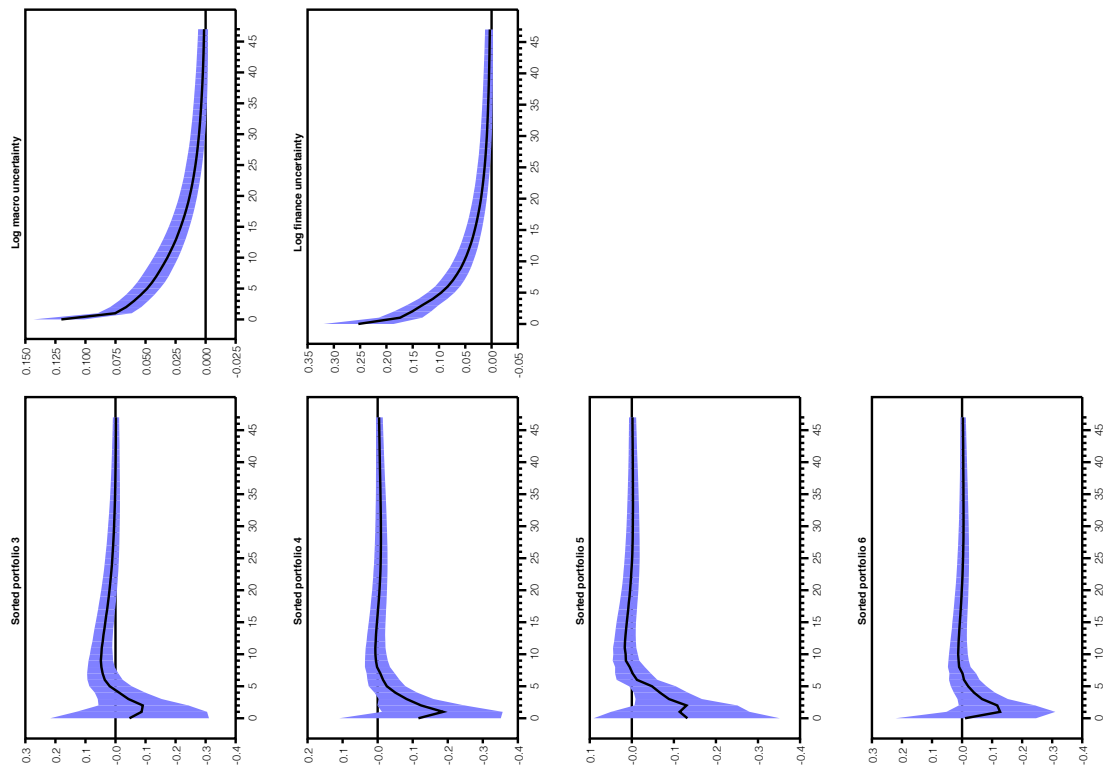


Figure A39: Continued, results for alternative variable specification 4, with 36 variables: impulse responses, one st. dev. shock to macro uncertainty

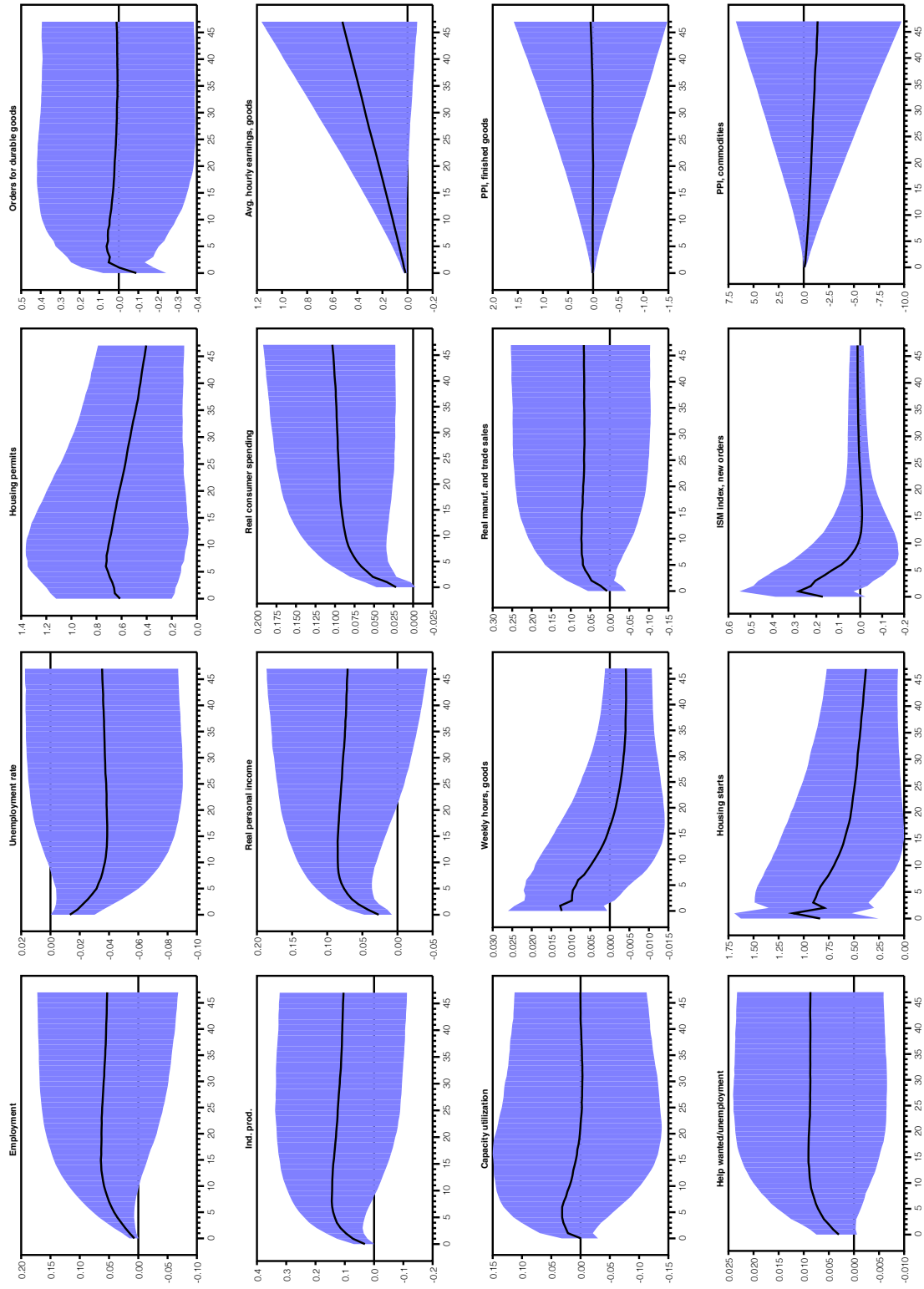


Figure A40: Results for alternative variable specification 4, with 36 variables: impulse responses, one st. dev. shock to financial uncertainty

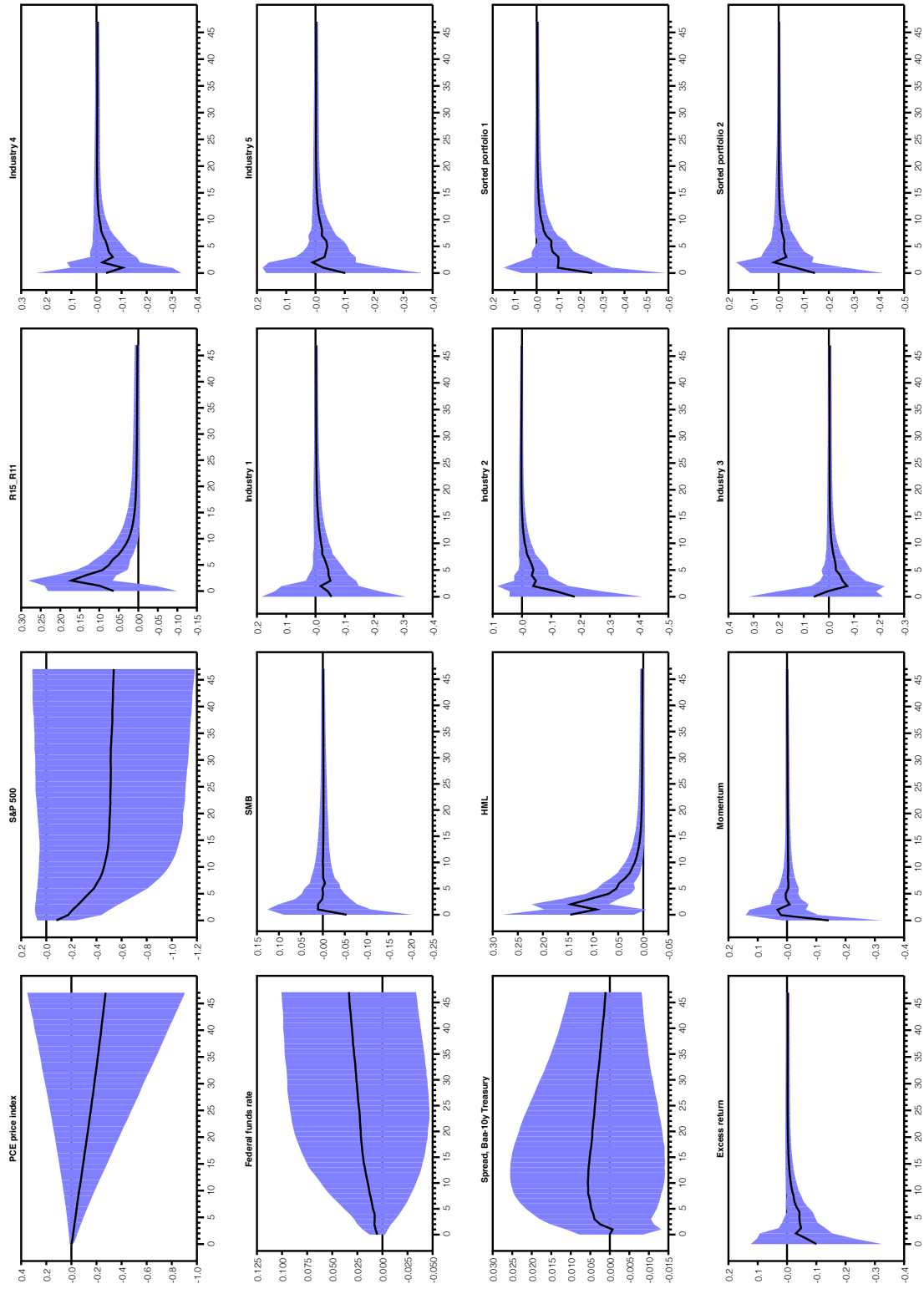


Figure A40: Continued, results for alternative variable specification 4, with 36 variables: impulse responses, one st. dev. shock to financial uncertainty

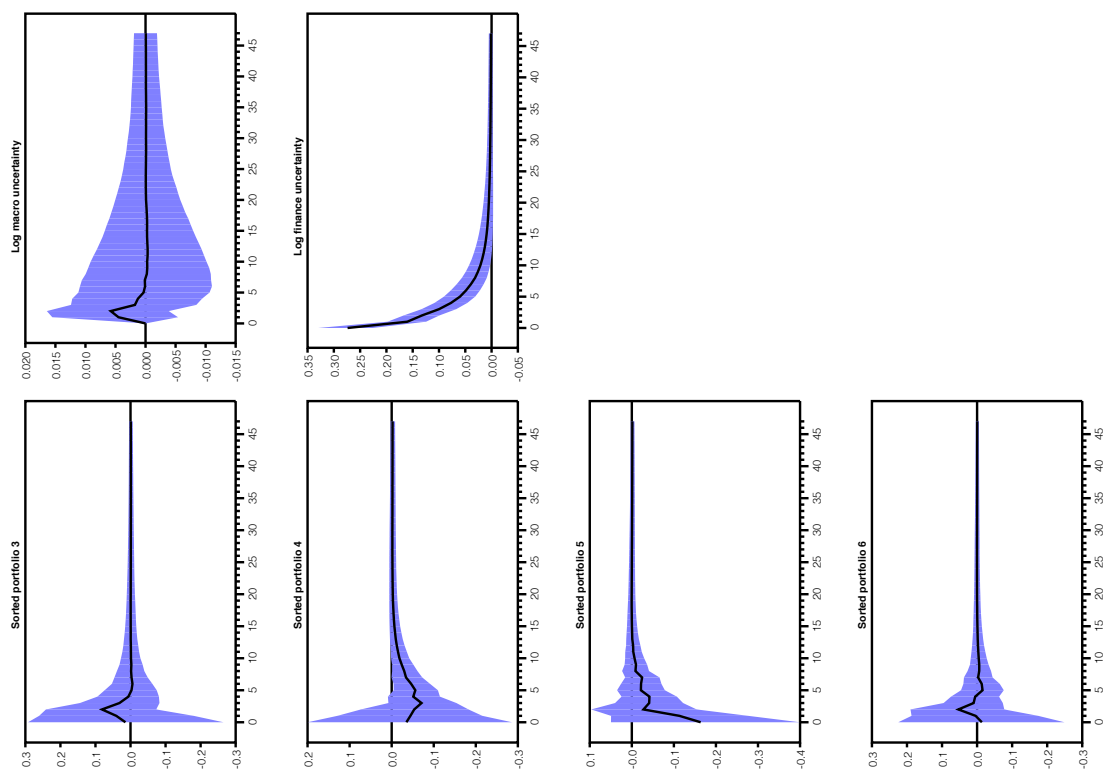


Figure A40: Continued, results for alternative variable specification 4, with 36 variables: impulse responses, one st. dev. shock to financial uncertainty

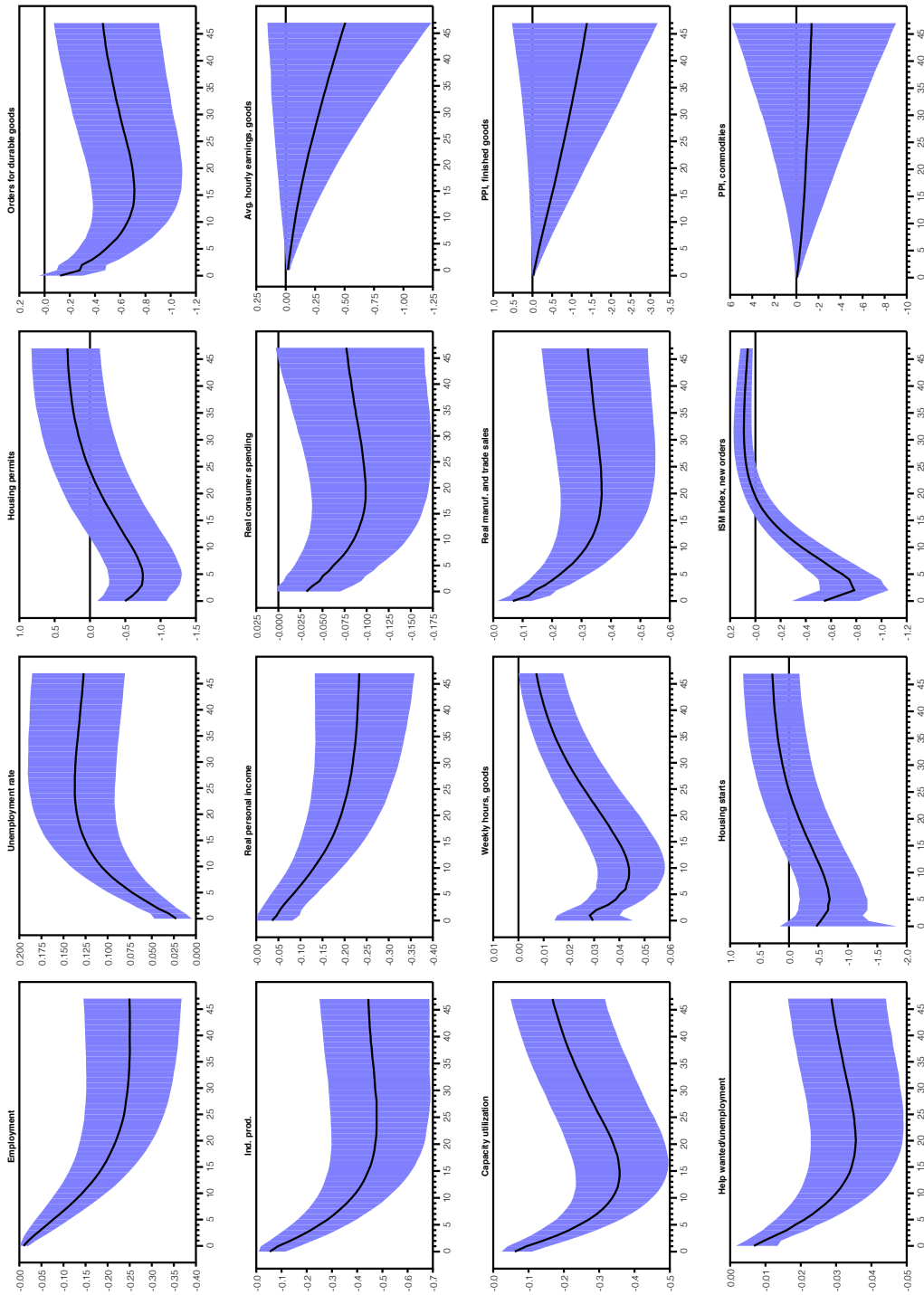


Figure A41: Results for alternative prior 1: impulse responses, one st. dev. shock to macro uncertainty

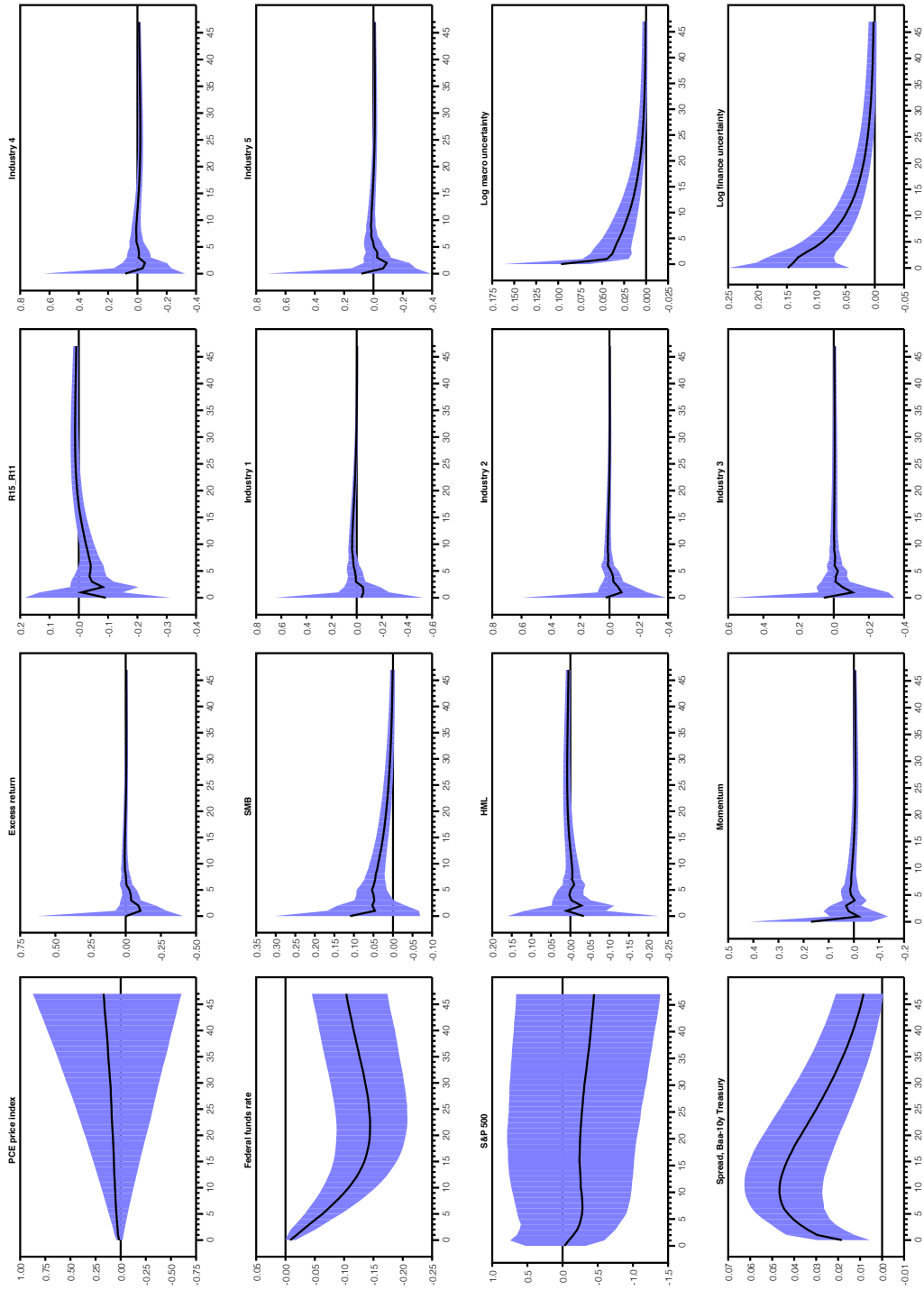


Figure A41: Continued, results for alternative prior 1: impulse responses, one st. dev. shock to macro uncertainty

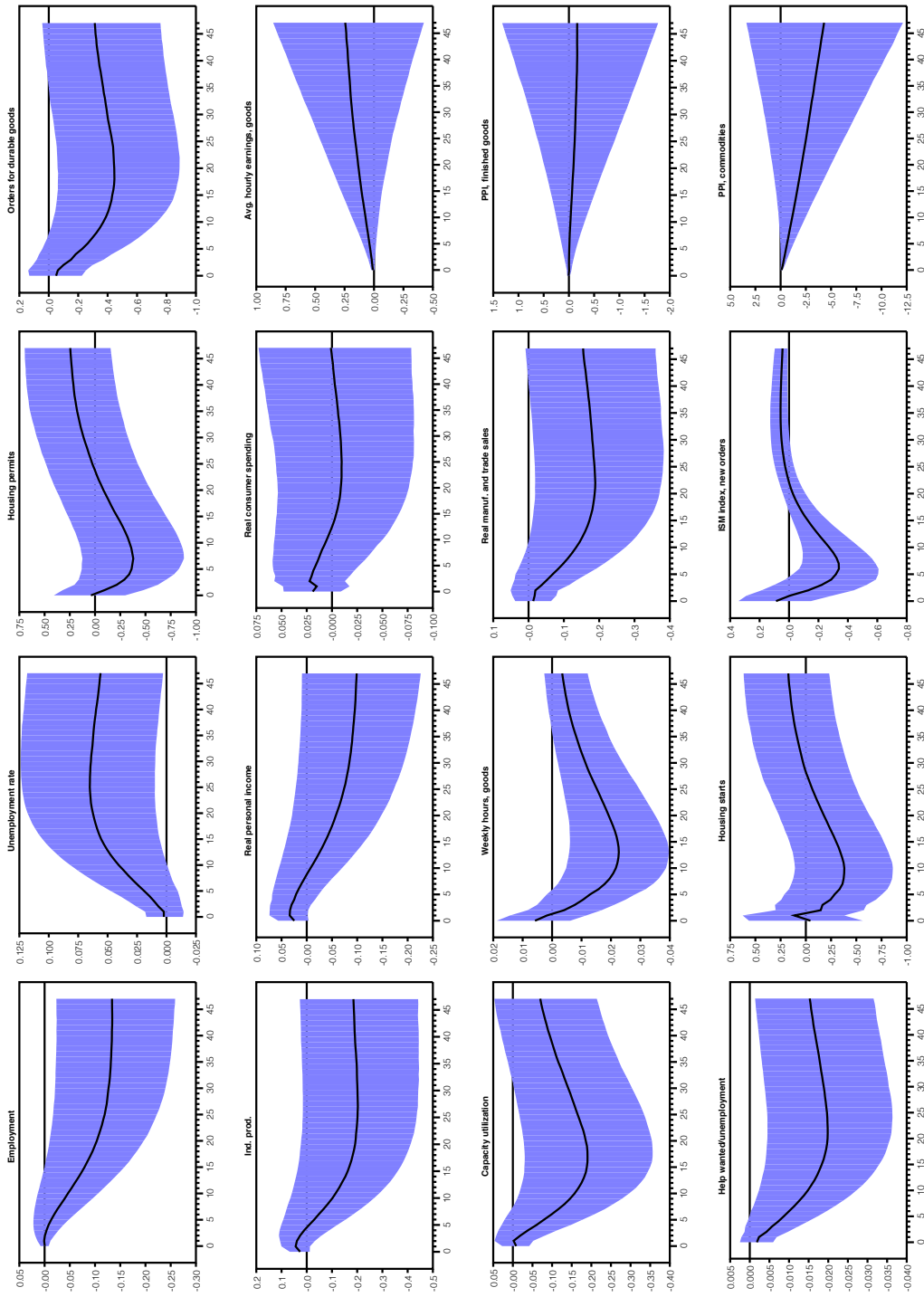


Figure A42: Results for alternative prior 1: impulse responses, one st. dev. shock to financial uncertainty

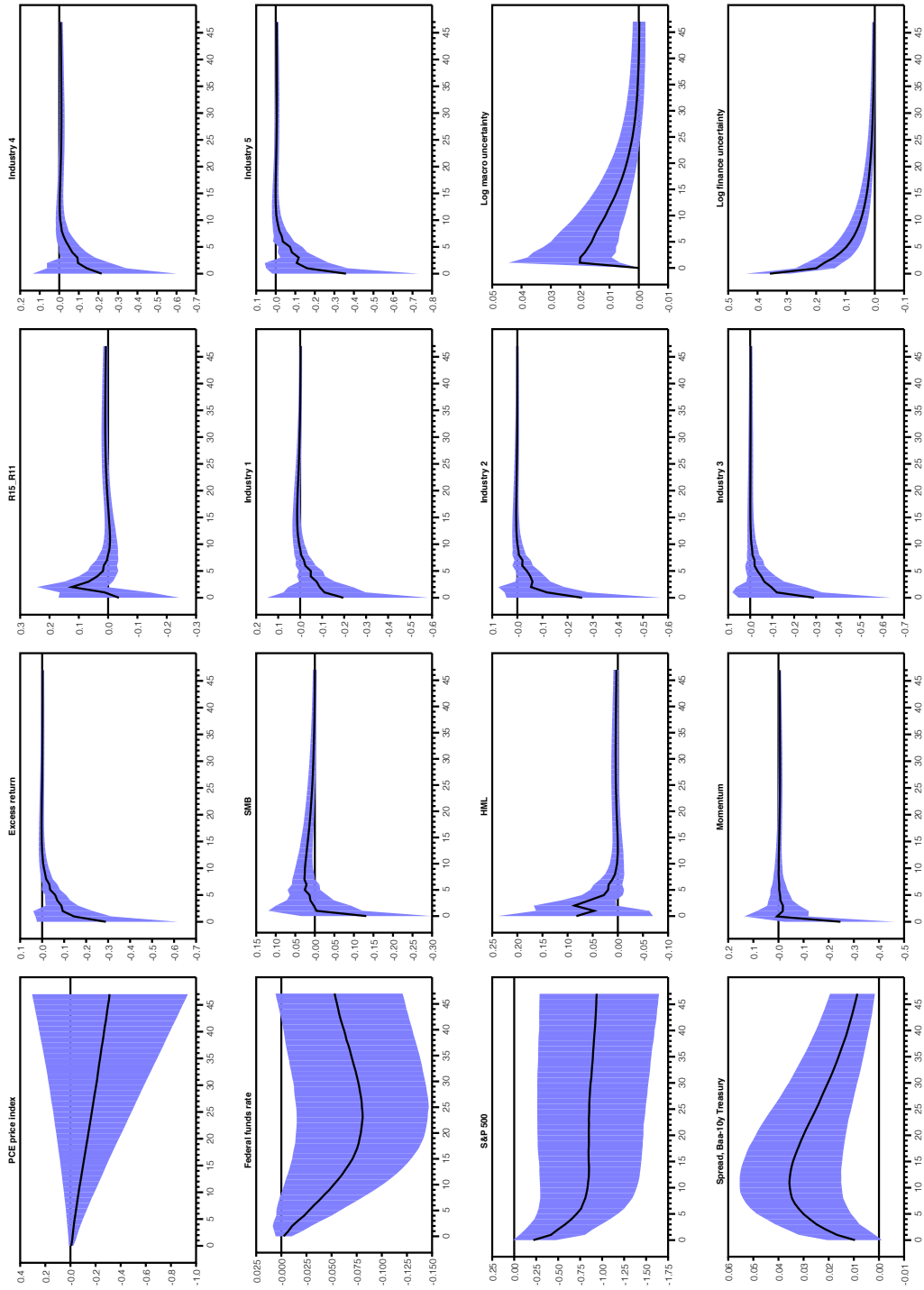


Figure A42: Continued, results for alternative prior 1: impulse responses, one st. dev. shock to financial uncertainty

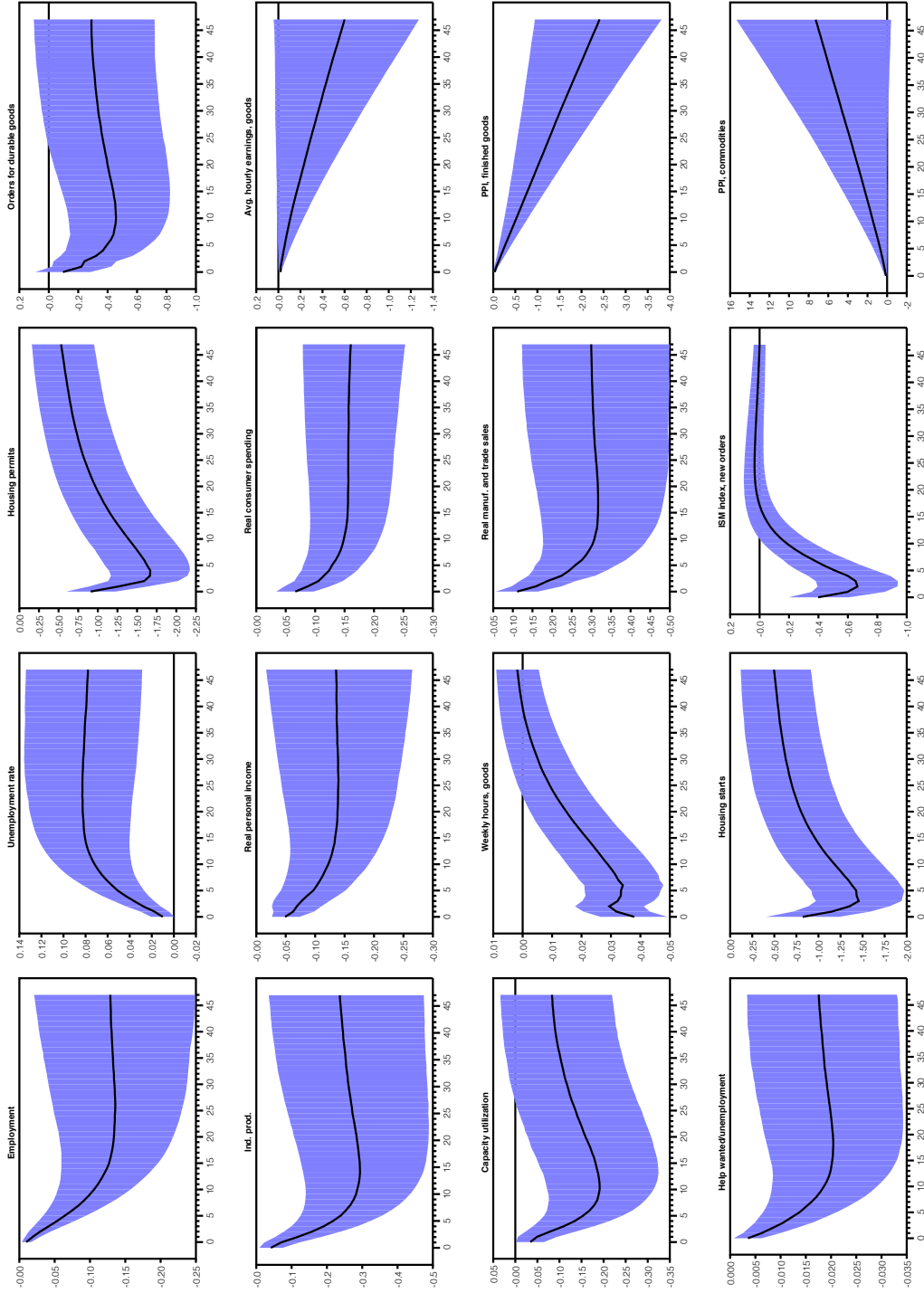


Figure A43: Results for alternative prior 2: impulse responses, one st. dev. shock to macro uncertainty

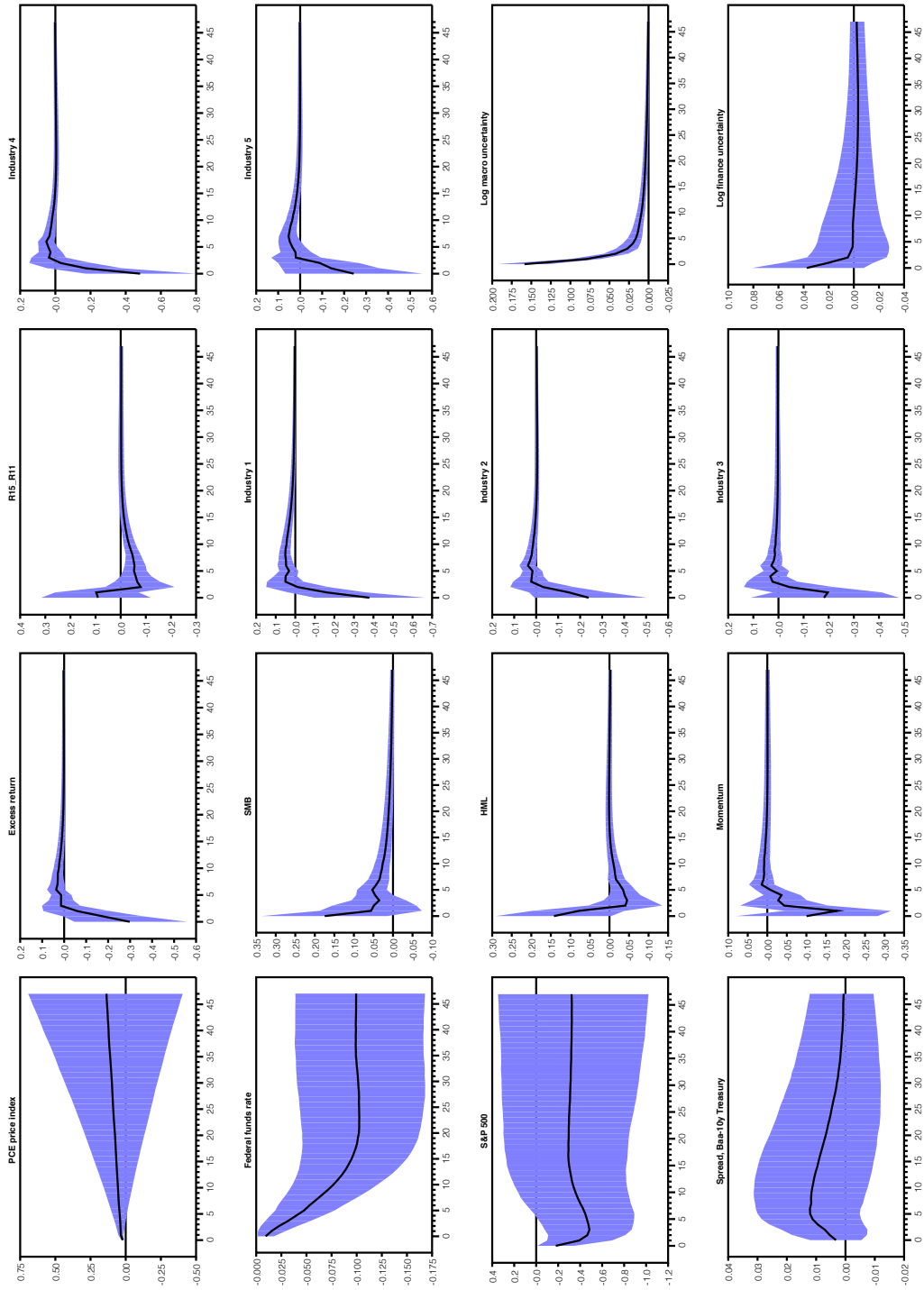


Figure A43: Continued, results for alternative prior 2: impulse responses, one st. dev. shock to macro uncertainty

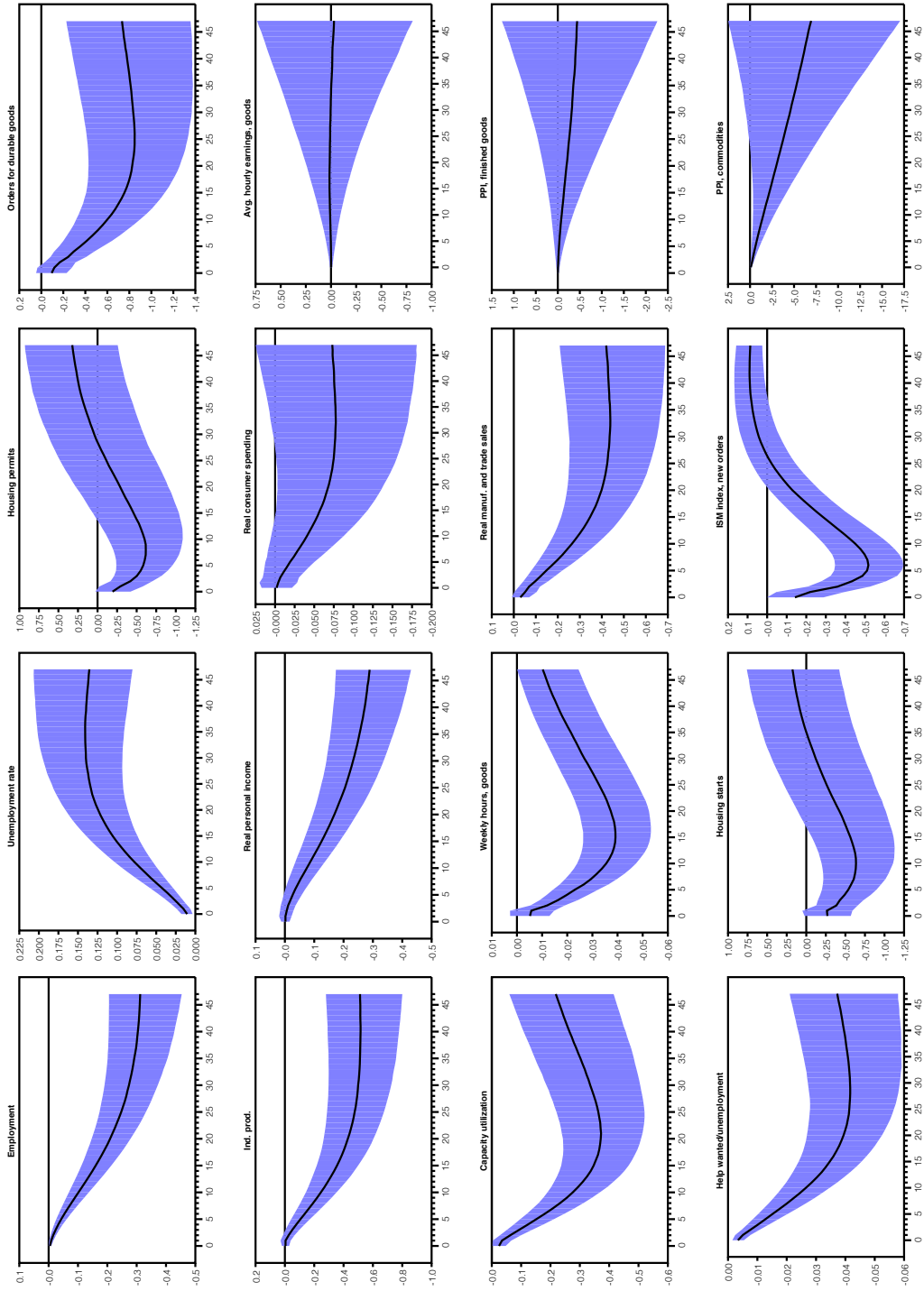


Figure A44: Results for alternative prior 2: impulse responses, one st. dev. shock to financial uncertainty

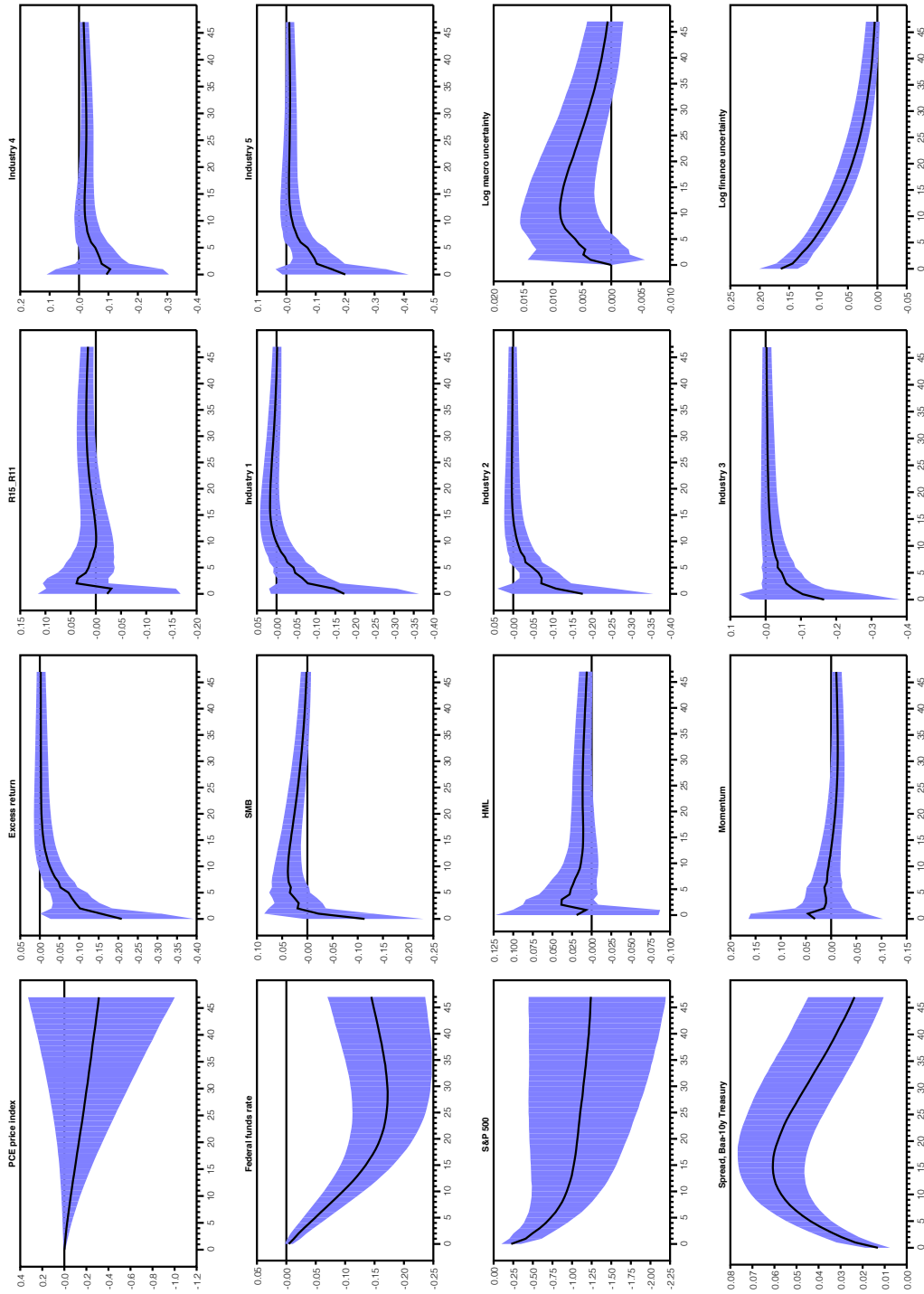


Figure A44: Continued, results for alternative prior 2: impulse responses, one st. dev. shock to financial uncertainty